Anyonic Tensor Networks

Symmetries, Categories and Circuits

Jeremias Rieser
advisors: C. Mendl and B. Decker

October 25, 2026

Contents

Introduction 2

1 Background 4
- 1.1 Literature & Related Work . . . 4
- 1.2 On Tensors . . . 5
  - 1.2.1 Tensor Networks & Lattice Models . . . 9
  - 1.2.2 Tensor Network Algorithms . . . 11

2 Symmetric Tensor Networks 12
- 2.1 Representation theory . . . 12
- 2.2 Application: The XXZ Hamiltonian . . . 15
- 2.3 Symmetric Tensors . . . 20

3 Anyons I : Introduction 23
- 3.1 Configuration Spaces & The Braid Group . . . 23

4 Categorical Networks and the Algebraic Theory of Anyons 27
- 4.1 Tensor & Fusion Categories . . . 28
- 4.2 Decompositions for Categorical Networks . . . 39
  - 4.2.1 Implementing Fusion Trees . . . 42

5 Anyons II: Spin Chains & Tensor Networks 44
- 5.1 1D Lattice Models & The Golden Chain . . . 44
- 5.2 Topological Quantum Computation . . . 45

A Appendix 50
- A.1 Some Group Theory . . . 50
- A.2 Some Algebraic Topology . . . 50
- A.3 Tensor Decompositions & Algorithms . . . 51
- A.4 Proofs for Chapter 2 . . . 51
- A.5 Supplementary categorical constructions for Chapter 4 . . . 52
  - A.5.1 TQFT . . . 52
- A.6 Overview of existing implementations . . . 52

Bibliography 54

Introduction

Tensor Networks [CWV26] allow the efficient numerical representations and computation on quantum states for large N. This includes variational methods and efficient simulation of quantum circuits [Orú14; Bia20]. Symmetric Tensor Networks [SPV10] yield a speedup over traditional methods for a subclass of tensor networks that are globally symmetric. The study of symmetric tensors is largely related to the structure of the *categories of representations* $\mathrm{Rep}(G)$ for some symmetry group G such as $\mathrm{SU}(2)$. These representation categories are foundational to the theory behind *fusion tensor categories* [Del02].

Check this one. Or find a better source.

These fusion categories also model *topological orders* .

These fusion categories also model a class of particles called *anyons* [Kit06]. These anyons are of great interest because they are *universal* for quantum computation [FKW02], and they are natively *fault tolerant* [Kit03]. This method of computation is called *topological quantum computation*. Just as for traditional quantum circuits, we can represent anyonic quantum circuits by an *anyonic tensor network*. There are also classes of interacting Hamiltonians on anyonic systems, as given by [Fei+07]. In fact, the same algorithms underlying symmetric tensor network methods can be adapted to support anyonic theories [Pfe12; Sch+20; DH25].

introduce

what even is this? a topological order?

Third, all of these concepts are embedded into a *string-diagram* formalism [**<empty citation>**]

In this thesis we give a formal overview how all these concepts fit into the same categorical framework while presenting a review of the existing literature, i.e. we give an interpretation of the results of Singh and Vidal [SV12], Weichselbaum [Wei12b], Pfeifer et al. [Pfe+10], Unfried [Unf24], and Devos and Haegeman [DH25], embed them with a rigorous categorical theory and provide missing proofs. Due to the interconnectedness with topological order, we phrase them in the language of [KZ22].

Overview This is an overview of the individual chapters and their contents.

- Chapter 1 gives an introduction to tensors and tensor networks. We provide the folklore definition of a tensor as a multilinear arrays of complex numbers, and generalize this to a definition over abstract vector spaces. We collect all the necessary operations of a tensor into the einsum notation. We also define the theory of tensor networks, and we give the most common structures and algorithms.

- Chapter 2 first introduces the concept of a symmetry, and we also give a short review of representation theory. We show how the XXZ-hamiltonian factors into a block-sparse matrix under its symmetry groups $\mathrm{U}(1), \mathrm{O}(2)$ and $\mathrm{SU}(2)$. We then begin defining symmetric tensors, so tensors that are invariant under the global application of a symmetry $g \in G$. We show that we can achieve a speedup on tensors of any shape, however the actual algorithms by which these sparse block tensors are realized are given in Chapter 4 so we do not produce redundant texts.

index the subsection

- Chapter 3 is an introduction to the theory of *anyons*. We introduce them as pointlike defects in a two-dimensional medium. We then show that their time-evolutions are governed by the structure of the *braid-group*, and its higher dimensional representations. We sketch a derivation of their fusion relations from this exchange structure.

- Chapter 4 forms the backbone of this thesis. First, we give an introduction to category theory from the ground up. Then, we begin a step-by-step derivation of the theory of *modular tensor categories*, motivated by our knowledge about representations and the physical aspects of the anyons. Each step carries along a visual interpretation in terms of standard vectors spaces, in terms of representations and in terms of anyon diagrammatics. Then, in Theorem 4.1 we prove the sparse-block decomposition of a tensor for arbitrary fusion categories. In **??**, we derive how to implement these categorical tensors

on the basis of this decomposition. We give a formal definition of a *fusion-tree* and how to represent this in a program. As above, general einsums will be implemented.

- Chapter 5 should form the last chapter . Here we explore some of the more elementary models on anyon systems in one and two-dimensions. Then we introduce the theory of *topological quantum computation*, and we show how to simulate this by our categorical tensor networks.

Does it?

In Chapter 1 we review the technical literature which is shared by the chapters (sections?). We give accounts on symmetries, representation theory, fermionic networks, anyons and categories. Theorem 4.1 presents the main theorem that is the backend of our tensor system. While we will state the first 2 sections with scarce proofs, chapter chapter 4 will be quite rigorous and supercede all statements from the previous sections.

Maybe we will present a outlook on Modular Categories

1

Background

1.1 Literature & Related Work

While we introduce this concept classically and endow it with symmetry as a start, we will start concerning ourselves with a categorical embedding of tensor networks in what is commonly reffereed to as a *fusion-category* (Lit ???).

Literature 1.1 (Symmetric Tensor Network). Originally, the idea to exploit symmetries of the hamiltonian for a speedup was proposed to be used in dmrg by [MG02] (What is [SM98]? This sounds like it as well.). More general theory for *globally invariant tensor networks*, i.e. networks that are built of tensors that are globally invariant is given by Singh, Pfeifer, and Vidal [SPV10] preceeding more detailed accounts on abelian symmetries [SPV11] and nonabelian (SU(2)) symmetries in [SV12]. A similar approach to (non)abelian symmetries is given by Weichselbaum [Wei12a; Wei20]. These rely on the general idea, that under representations of some compact group G, there are on site symmetries on each tensor, i.e. $[A, U_g^{\otimes k}] = 0$ there is a generalized *wigner-eckart-theorem* for tensors with n legs of the form

$$T \in \bigoplus_a D_a \otimes V_a$$

where D_a is a matrix and V_a is a *fusion-tree*, i.e. a clebsch-gordan tensor. Reviews and specific algorithms in the SU(2) case are given by [SO20; Sch20; Sch+20]. [Sch+20, § 8] proposes that this method is appropriate for Anyonic simulation as well (although this is done earlier by singh & pfeiffer.). Symmetries as categorified networks, as we will present in **??**, are explained in [Unf24] and [DH25] and in this work in chapter 5.

For a list of software implementing symmetries see **??**.

Literature 1.2 (Anyons). *Anyons* as quasiparticles with exotic exchange statistics were originally propsed by [Wil90] [LM77]. Anyons are usually described as *defects* in a (topologically orderd?) medium of dimension two (technically codimension after [**SS23AnyonsInTedK**]). The exchange statistics of N particles on this medium M is goverend by the *configuration space*

$$\mathrm{Conf}_N(M) = \{(x_1, \dots, x_N) \in M^N : x_i \neq x_j\} \setminus S_N.$$

The statistics of the particle exchange is given by the representation of $\pi_1(\mathrm{Conf}_N(M))$ (the fundamental (homology?) group). If $M = \mathbb{R}^2$, this corresponds to the *braid group* on N strands B_N. If the dimension of $M \geq 3$ this corresponds to the symmetric group S_N. If these representations are unitary and the (ground state?) is degenerate, this allows for *topological quantum computation* (cf. Literature 1.6).

Literature 1.3 (Anyonic Spin Chains and Tensor Networks)**.** (Check if this is the first). A first appearance of *Anyonic Tensor Networks* in terms of spin chains is given by Feiguin et al. [Fei+07] as the *golden chain*, which realizes a interacting 2-local hamiltonian on a n Fib-anyon lattice. This is reviewed in [Tre+08] with generalizations to 3-local and longer range hamiltonians. A general account outside of the Fib-model can be found in [Gil+13] and [BG17] (Check these again.). We can consider these approaches as a sort of manualization of the tensor network approach. Anyonic tensor networks are first introduced by [Pfe+10] and [Pfe12], developed out of the symmetry methods from chapter 2 and both the *isotopy* rules and anyonic operators defined in [BSS08]. Furthermore, stuctures such as an anyonic MERA are introduced.

We also need to discuss the topic of *anyonic entanglement entropy*, as given in [Pfe14; BKP17; Pfe12].

Literature 1.4 (Simulation of Anyons and String Nets)**.** This should probably go in the previous section. [Liu+22] [Kir+23]

Literature 1.5 (Anyons as Categories)**.** The correspondence between the *physical* treatment of anyons and the *categorical treatment* of anyons as a *braided fusion category* (modular functor?) was originally descibed by Kitaev [Kit06, § E, § 8] based on previous work in quantum field theory already making use of modular categories and their diagrammatic calculus such as [RT91]. Further treatments are found in [**wang**] and [**simon-nayak-das-sarma**]. A modern treatment of the realization through the Configuration space of n-anyons if given by [**90eb98357acd4634abd18d906cc3811d**] and in terms of modular functors by [Bur16].

Literature 1.6 (**Topological Quantum Computation**)**.** Fault tolerant *Topological Quantum Computation (TQC)* using anabelian anyons was originally proposed by Kitaev [Kit03, § 7] (Originally published in 1997). Also [Fre98] and .... ??? has shown that there are nonabelian anyon models that are universal [DN05] [Moc03] [Moc04].

[MSS24] gives nicer literature. [Sim23] is a great textbook account from one of the original authors on quantum compiling and computing. [Nay+08] is also one of the original accounts. This does not need to be any longer. Maybe mention categorical computation in general? but idk what htis is yet [KZ23].

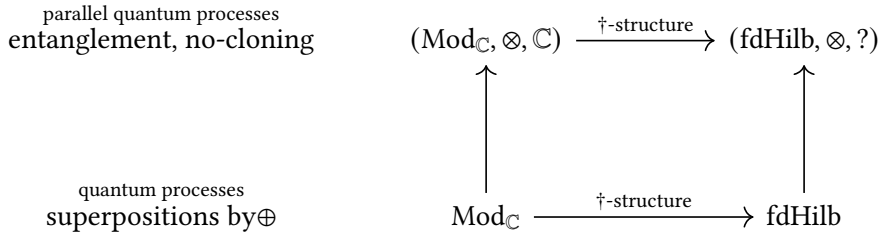
Literature 1.7 (**Quantum Compiling**)**.** Some (Many?) anyon theories are universal for quantum computation. The *Solovay-Kitaev theorem* [Kit97][DN05] yields an efficient algorithm for approximating any quantum gate using only basic gates from a universal set. Here we can consider 1.) The representation of braid circuits as a tensor nework operation and 2). A natural embedding of injection weaves as proposed by Simon et al. [Sim+06] where 2 anyons are braided in their composite fusion channel by realizing this operation as a tensor reshape.

Literature 1.8 (Categorified Quantum Mechanics)**.** There are multiple approach to a diagramatic and categorified approach to quantum mechanics. Generally one wants both spaces and maps represented by a framework that allows for parallel and entangled operations, i.e. has a tensor product structure. [Sel11] gives a review of this [VVW18] does for tensor networks specifically.

TODO: find the XZ-calculus literature by Kissinger? Specific examples such as XZ-calculus in the context of TQC is handled by [AK23].

TODO: How to mention that this results in or from $\mathrm{Hilb}_{\mathrm{fd}}$? Does it? The work from selinger et.al does not yet include anyons. For this, one needs Literature 1.5.

Also, take [SS26, § 1] for a description of categorified mechanics. This stats that structure for *entangled* processes is given by the monoidal category of complex vector spaces.



The categorical language specifically yield interpretations of theorems such as no-cloning and no deleting [Abr12].

Additionally, many of the concepts of topological order are governed by category theory [KZ22], as well as the topic of higher symmetries (should this be its own Lit sect if relevant? It is for MPO symmetries [DT24] [Kon+20]).

1.2 On Tensors

Throughout this text, we build tensors over the field $\mathbb{C}$, note however that all of this will work for any algebraically closed field k.

A *tensor* is an element of a tensor product of finite-dimensional vector spaces,

$$T \in \bigotimes_{j \in J} V_j,$$

and we call T a rank $|J|$ tensor. The *dimension* of the tensor is then $\dim(T) = \prod_{j\in J} \dim(V_j)$.

The *dual space* of a vector space V is $V^* = \mathrm{Hom}(V, k)$, the space of linear functionals $f : V \to k, x \mapsto f(x)$. In Dirac notation we write $|v\rangle \in V$ for vectors (kets) and $\langle f| \in V^*$ for linear functionals (bras), so that $\langle f|v\rangle = f(v)$.

Since all spaces are finite-dimensional, we have the canonical isomorphisms

$$V^{**} \cong V, \qquad (V \otimes W)^* \cong V^* \otimes W^*, \qquad W \otimes V^* \cong \mathrm{Hom}(V, W),$$

where the last is given on simple tensors by

$$|w\rangle \otimes \langle f| \;\longmapsto\; \big(v \mapsto \langle f|v\rangle \; |w\rangle\big),$$

i.e. $|w\rangle\langle f|$ is the outer product.

Because $V^{**} \cong V$, any tensor $T \in V_1 \otimes \cdots \otimes V_n$ can be regarded as a linear map after choosing a split of its legs at some $0 \le m \le n$:

$$T : V_1^* \otimes \cdots \otimes V_m^* \longrightarrow V_{m+1} \otimes \cdots \otimes V_n.$$

We can also make the default definition a linear map if we do not want to have a split into domain and codomain by equating it to the map

$$\mathbb{C} \to V_1 \otimes \cdots \otimes V_n \cong V_1 \otimes \cdots \otimes V_n, \quad f \mapsto f(1).$$

Note that this does not depend on the tensor product structure, we could also have defined $X = V_1 \otimes \cdots \otimes V_n$ as its own vector space.

Example 1.1. Let $t \in V_1 \otimes V_2 \otimes W_2^* \otimes W_1^*$. Choosing the split after V_2, t is an operator $W_1 \otimes W_2 \to V_1 \otimes V_2$, which in Dirac notation reads

$$t = \sum_{i_1,i_2,j_1,j_2} T^{i_1 i_2}_{j_1 j_2} \; \left|v^1_{i_1}\right\rangle \left|v^2_{i_2}\right\rangle \left\langle w^2_{j_2}\right| \left\langle w^1_{j_1}\right|,$$

where $\{\left|v^a_i\right\rangle\}$ is a basis of V_a and $\{\left\langle w^b_j\right|\}$ is a basis of W_b^* (the dual basis of a basis of W_b).

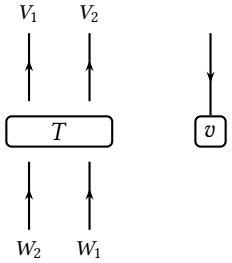
Example 1.2. Note that while we gave a definition of a vector as $|v\rangle \in V$, we can alternatively define a vector as a linear map

$$v : \mathbb{C} \to V, \lambda \mapsto \lambda \, |v\rangle \, .$$

This means we can effectively let the space of vectors be the space $\mathrm{Hom}(\mathbb{C}, V)$.

Example 1.3. Since we have given tensors as linear maps, we can define some trivial contractions as *compositions* by $circ$. We can compose two maps f, g if the domain of f agrees with the codomain of g. So we can define a matmul between $f : V \to W, g : X \to V \longrightarrow (fg) : X \to W$.

Graphical Notation 1.1. Graphically, a tensor $T : W_n \otimes \cdots \otimes W_1 \to V_1 \otimes \cdots \otimes V_m$ is a coupon with a leg for each factor of its spaces: legs for the codomain are outgoing from the top of the coupon, legs for the domain point inwards towards the bottom of the diagram. Arrows are oriented so that a leg carrying V points up and a leg carrying V^* points down.



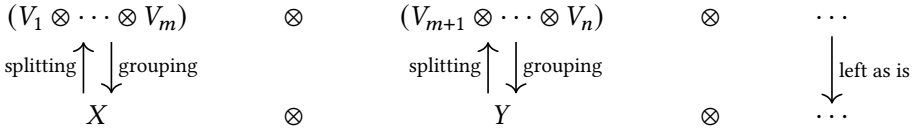
Now we want to introduce the common methods of *reshapes* and *contractions* onto a tensor. Reshapes come in ?|?? flavors:

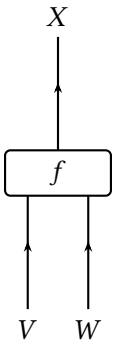
1. Grouping and splitting indices: $X \cong V \otimes W$

2. Transpositions: $\mathrm{Hom}(V \otimes W, X) \to \mathrm{Hom}(V, X \otimes W^*)$

3. Permutations: $\mathrm{Hom}(V \otimes W, X \otimes Y) \to \mathrm{Hom}(W \otimes V, Y \otimes X)$

For grouping legs, we can act as above and take the isomorphic vector space to be X. Equivalently we can also consider a bracketing of the vector spaces





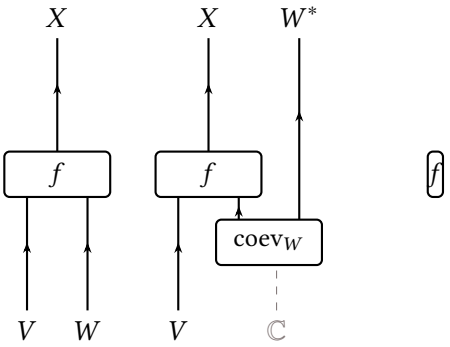
$f \in \mathrm{Hom}(V \otimes W, X)$

For the other operations, we need a relation between a vector space and its dual space. For this we use the notion of *evaluation* and *coevaluation* maps.

$$\begin{aligned} \mathrm{ev}_V &: V^* \otimes V \to k, & f \otimes v &\mapsto f(v), \\ \mathrm{coev}_V &: k \to V \otimes V^*, & \lambda &\mapsto \lambda \sum_i v_i \otimes f_i. \end{aligned}$$

Take a arbitrary tensor $\mathrm{Hom}(V \otimes W, X)$. Now we can do a reshape by composing these maps. The transposition here would then be the isomorphism

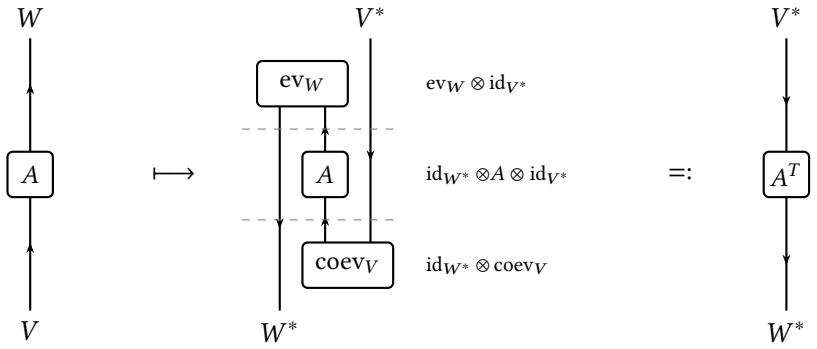
$$\mathrm{Hom}(V \otimes W, X) \to \mathrm{Hom}(V, X \otimes W^*), \quad f \mapsto (f \otimes id_{W^*}) \circ (\mathrm{id}_V \otimes \mathrm{coev}_W).$$



$f \in \mathrm{Hom}(V \otimes W, X)$

where we have omitted the extra leg carrying $\mathbb{C}$ as this is a trivial leg of dimension one.

If we have a matrix operator $A : V \to W$ we can execute a transpose by application of ev and coev:



$A \in \mathrm{Hom}(V, W)$ $\quad (\mathrm{ev}_W \otimes \mathrm{id}_{V^*}) \circ (\mathrm{id}_{W^*} \otimes A \otimes \mathrm{id}_{V^*}) \circ (\mathrm{id}_{W^*} \otimes \mathrm{coev}_V)$ $\quad A^T \in \mathrm{Hom}(W^*, V^*)$

Next, we want to realize *permutations*. Recall that a permutation is a rearrangement on a set of labels $(a_1, a_2, \ldots, a_n) \mapsto (s_1(a_1), s_2(a_n), \ldots, s_n(a_n)$. It is known that an arbitrary permutation can be realized by swapping repeatedly applying *swap operations* between two adjacent symbols, these operators $s_i : (a, b) \to (b, a)$ are the generators of the symmetric group of order n, S_n (cf. appendix A.1). So it is enough for us to implement these simple swaps.

where $\{v_i\}$ is a basis of V and $\{f_i\}$ its dual basis. A *contraction* is a composite of identities and evaluation maps that removes the paired legs from the tensor product. For example, for operators $A \in U \otimes V^*$ (a map $V \to U$) and $B \in V \otimes W^*$ (a map $W \to V$), the product $AB \in U \otimes W^*$ is

$$AB = (\mathrm{id}_U \otimes \mathrm{ev}_V \otimes \mathrm{id}_{W^*})(A \otimes B).$$

Equip U, V, W with bases $\{u_a\}, \{v_i\}$ and w_b respectively. We can now write

$$A \otimes B = \sum_{a,i,j,b} A_{ai} B_{jb} \, |u_a\rangle \otimes \langle v_i| \otimes |v_i\rangle \otimes \langle w_b|$$

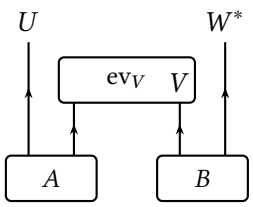
and by evaluation $\mathrm{ev}_V(\langle v_j| \otimes |v_i\rangle) = \langle v_j | v_i \rangle = \delta^i_j$ we can obtain

$$AB = \sum_{a,i,b} A_{ai} B_{jb} \, |u_a\rangle \otimes \langle w_b|$$

with coordinates $(AB)^a_b = \sum_i A^a_i B^i_b$ which is precisely the well known matrix-multiplication formula.

If we had taken them both as maps instead, we can still do the contraction. However now under application of both evaluation maps.

SHow this. But there should be a deeper reasoning, i.e this is probably just the fact that ev itself is the isomorphisms tbetween the presentations.



$A \in \mathrm{Hom}(V, W)$

Remark 1.1. In this formulation, we hvae generally defined our tensors to work without choosing a basis. Even if we do, our bases do not need to be coordinatized. We will later see the use of categories (cf. Chapter 4) where we have vector spaces that only admit a formal basis of symbols.

Traditionally however, we are developing tensors at least over the field $\mathbb{C}$. Furthermore, all our vector spaces are indexable by $\mathbb{C}^n$.

rephrase!

It should also be mentioned that on the implementation level, we also have to consider how we store these tensors. If we have real tensors we may use datatypes such as `float64`, if we have complex tensors we can use `complex128`. Generally, we store these tensors as arrays (vectors?) in contiguous memory.

In such a memory-layouted representation of a tensor, the underlying data is never changed, but the layout of its storage is. For example, if we index vector space V by index $i : \texttt{int}$ and W by $j : \texttt{int}$, we can choose to index $V \otimes W$ as $(i, j) : \texttt{int} \times \texttt{int}$. We can however due to the layouting designate another index $k = i + n \cdot j : \texttt{int}$ (column-major) or $k = j + m \cdot i : \texttt{int}$ (row-major). where $n = \dim(V)$.

The semantics of the reshape operations are captured by operations called *einsums*[1] (Einstein-sums) [Wen+25]. The matrix-matrix product $A \cdot B$ is here written as

how much? consistently?

$$einsum(ij, jk \to ik, A, B)$$

1.2.1 Tensor Networks & Lattice Models

Literature 1.9 (Tensor Networks). The modern theory of *tensor networks* shaped out of the study of *renormalization* and quantum many-body systems. A common application is the usage of numerical methods to simulate a quantum system. White [Whi92; Whi93] introduced a variational method for finding the ground state of a one-dimensional quantum system called *density matrix renormalization ansatz (DMRG)*, and this method was later adapted to efficiently simulate large systems by use of *matrix product state (MPS)*by Ostlund and Rommer [OR95]. . This was further generalized to two-dimensional systems with the introduction of *projected entangled-pair state (PEPS)* [VC04]. *multiscale entanglement renormalization ansatz (MERA)* [Vid08] can be considered as an extension of MPS allowing for long-range entanglement (this is probably wrong). Whats missing? Holography [SMB18] but there are earlier ones.

Maybe this is actually due to [PNW92]

Tensor networks can also be used to simulate quantum computations [MS08] efficiently.

In the purely mathematical view as a network of tensor contractions however, both the concept and the string notation we use today were introduced by [Pen71] under the name of *tensor-systems*.

Often times we will study quantum states and systems on a lattice. For 1-dimensional systems, we can consider a lattice of n *sites* where each holds a *local site* $|i\rangle$. We can give an arbitrary state on this lattice as

$$|\Psi\rangle = \sum_{i_1 \dots i_n} c_{i_1 \cdots i_n} |i_1 i_2 \cdots i_n\rangle .$$

Storing this state requires representing the exponentially many coefficient $c_{i_1 \cdots i_n}$. As an example, we may have two sites of spin $1/2$, so we have a total state space of dimension 4 spanned by $|i_1 i_2\rangle \cong \mathbb{C}^4$ with $i_1, i_2 \in \{\uparrow, \downarrow\}$. Then, we can give the coefficients to obtain the state

$$\Psi^+ = \frac{1}{\sqrt{2}} (|\uparrow\uparrow\rangle + |\downarrow\downarrow\rangle)$$

and we can store this state as a 4×1 matrix (a statevector of dimension 4). If we keep the spin basis on the lattice sites, a matrix on n sites is thus representable by a statevector $|\Psi\rangle \in \mathbb{C}^{2^n}$.

Now, we are allowed to apply a reshape operation and change $\mathbb{C}^{2^n}$ into $\mathbb{C}^{2 \times 2^{n-1}}$ which corresponds to a tensor of order 2, i.e. a matrix with 2 columns and 2^{n-1} rows. Now that we have a matrix, we can apply the singular value decomposition (SVD) yielding a decomposition

$$c_{i_1,(i_2 \dots i_n)} = \sum_{j_1} U_{i1,j1} S_{j_1,j_1} (V^\dagger)_{j1,(i_2 \dots i_n)}$$

Recall that we can choose the rank of the SVD, so that the matrix U that is produced is of dimension $r \leq i_1$. We can absorb the factors S and V into the remaining state and again reshape to a state, and we get the expression

NOTATION!

$$c_{i_1,(i_2 \dots i_n)} = \sum_{j_1} U_{i_1,j_1} c_{j_1,i_2 \dots i_n}$$

[1] Einsum landing page: `https://einsum.org`

We now reshape the new c again to a tensor of shape $[(j_1 i_2, i_3 \dots i_n)]$.

$$c_{i_1,i_2,\dots} = \sum_{j_1,j_2} U^{i_1,j_1} U_{(j_1 i_2),j_2} c_{j_2,(i_3,\dots,i_n)}$$

and we can reshape the expression $U_{(j_1 i_1),j_2}$ to a trivalent tensor having an extra leg for the physical dimension, i.e. we want ot build the tensor

$$c_{i_1,i_2,\dots} = \sum_{j_1,j_2} U^{i_1}_{j_1} U^{i_2}_{j_1,j_2} c_{j_2,(i_3,\dots,i_n)}$$

We can apply this procedure recursively to the new coefficient, and in the end we will get an expression for our state

$$c_{i_1,\dots,i_n} = \sum_{j_1,\dots,j_{n-1}} U_{i_1,j_1}$$

Definition 1.1. A *Matrix Product State* is a quantum state $|\psi\rangle$ with n legs written as the contraction of n trivalent tensors $\{A^{[i]}\}_{i\in[n]}$. Formally this reads

$$|\psi[A]\rangle = \sum_{i_1,\dots,i_n} \mathrm{Tr}\left(A^{i_1} A^{i_2} \cdots A^{i_n}\right) |i_1 i_2 \cdots i_n\rangle ,$$

where each A is of the form

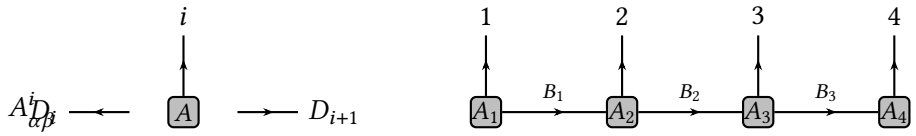
$$A^{[i]} = \sum_{i\delta_i\delta_{i+1}} A^i_{\delta_i\delta_{i+1}} |i\rangle \otimes \langle\delta_i| \otimes \langle\delta_{i+1}|$$

where $A^{[i]}_{i_k} \in \mathbb{C}^{D_i \times D_{i+1}}$. The elements of the collection $\{D_i\}_{i\in[n]}$ is referred to as the *bond-dimensions*.

Note that this is a state with closed boundaries. If we want open boundaries we have to replace the trace with appropriate end vectors (nice source?).

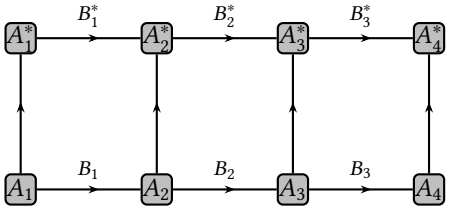
Example 1.4. We can represent each product state $|\Psi\rangle = |\psi\rangle^{\otimes n}$ as a MPS with bond dimension 1.

Graphical Notation 1.2. We represent a MPS tensor to the left and a full MPS with $N = 4$ to the right.



Note that we can easily define an inner product on an MPS (as it represents a state) and we can view this as

$\langle\Psi[A]|\Psi[A]\rangle =$



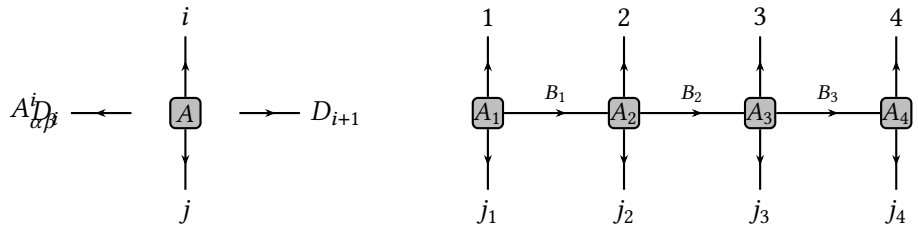
Definition 1.2. A *matrix product operator (MPO)* is an operator on a state with n legs on each side written as the contraction of 4 legged tensors $W^{[}i]$....

$$\sum_{i_1,..,i_n j_1,\dots,j_n} \mathrm{tr}(M^{i_1 j_1} M^{i_2 j_2} \cdots M^{i_n j_n}) |i_1 \cdots i_n\rangle \langle j_1 \cdots j_n|$$

We again use the convention of denoting the bonds by D, and we can regard each mps tensor as a mapping

$$\mathrm{Hom}(P_i \otimes D_i, P_j \otimes D_{i+1})$$

Graphical Notation 1.3. We represent a MPO tensor to the left and a full MPO with $N = 4$ to the right.

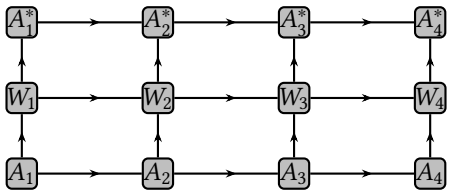


Constructing these operators can be done by the use of a *finite state machine* [CB08]. As an example, the XXZ-Hamiltonian is represented by the MPO tensors

$$W_{ij} = \begin{pmatrix} 1 & 0 & 0 & 0 & 0 \\ S^x & 0 & 0 & 0 & 0 \\ S^y & 0 & 0 & 0 & 0 \\ S^z & 0 & 0 & 0 & 0 \\ 0 & S^x & S^y & S^z & 1 \end{pmatrix}_{ij} \tag{1}$$

with bond dimension 5, note that this acts on the physical states as such.

As an application, we can immediately represent the expectation value of a state $\langle\Psi| H |\Psi\rangle$ by using MPSs and MPOs:



1.2.2 Tensor Network Algorithms

2

Symmetric Tensor Networks

This section develops the theory underlying *symmetric tensor networks*, specifically for non-abelian groups. We give a quick overview over the necessary representation theoretic concepts in Section 2.1. We employ the example of the XXZ-Hamiltonian for both abelian and non-abelian symmetries and give an exact calculation of its conserved quantities for lattices of size 2 and 3. Here we are mostly interested in the *speedup* obtained by exploiting the emergent block-structure of symmetric operators. Most of the proofs and formal definitions required for this formalism are delegated to Chapter 4, however there will still be repetitions of some of the theory.

2.1 Representation theory

This section gives a quick review of representation theory. We consider all vector spaces and their representations here to be finite, the groups may only be compact. For some of the more basic definitions as those of a group and some of the matrix groups we are using here, see Appendix A.1. We follow the books by Ibort and Rodriguez [IR19], Serre [Ser77], **woit** as well as Etingof et al. [Eti+11].

cite woit book

Definition 2.1. A *representation* of a group $(G, \cdot)$ which is just G here over a vector space V is a group homomorphism

$$\rho : G \to \mathrm{End}(V), \quad \rho(g)\rho(h) = \rho(g \cdot h), \quad \rho(e) = id_V$$

The *representation space* is denoted by the tuple (ρ, V) but usually we shorten this to just V.

If we want the representation to be linear, we can choose it as

$$\rho : G \to \mathrm{GL}(V)$$

instead. Choosing a linear basis for V in $\mathbb{C}^n$ gives this as

$$\rho : G \to \mathrm{GL}_n(\mathbb{C}).$$

Definition 2.2. A linear map $A : V \to W$ is an *intertwiner*, if under representations $\rho_V : G \to \mathrm{GL}(V)$ and $\rho_W : G \to \mathrm{GL}(W)$ it holds that

$$A \circ \rho_V(g) = \rho_W(g) \circ A \quad \forall g \in G.$$

Denote the space of all intertwiners between V and W of the group G as $\mathrm{Hom}_G(V, W)$, similarly denote its endomorphisms as $\mathrm{End}_G(V) \coloneqq \mathrm{Hom}_G(V, V)$.

Definition 2.3. If our vector space V is equipped with a (Hermitian?) inner product, or in other words we are given a representation $\pi : G \to \mathrm{U}(\mathcal{H}) \hookrightarrow \mathrm{GL}(\mathcal{H})$ where $\mathcal{H}$ is a complex Hilbert space, then π is unitary if

$$\pi(g)^\dagger \pi(g) = \pi(g)\pi(g)^\dagger = I.$$

This directly entails that $\pi(g^{-1}) = \pi(g)^\dagger$.

Definition 2.4. Given reps (ρ_V, V) and (ρ_W, W), the *tensor product representation* $(\rho_{V\otimes W}, V \otimes W)$ is defined by its action on the elements

$$\rho_{V\otimes W}(g)(v \otimes w) = \rho_V(g)v \otimes \rho_W(g)w$$

Definition 2.5. As above, the *direct sum representation* $(\rho_{V\oplus W}, V \oplus W)$

$$\rho_{V\oplus W}(g) = \begin{pmatrix} \rho_V(g) & \\ & \rho_W(g) \end{pmatrix}$$

Definition 2.6. A representation ρ is called *irreducible* if it cannot be written as a direct sum representation.

If G is finite, we can group these irreducibles (irreps) into isomorphism classes, and we denote these

$$\mathrm{Irr}(G) \cong \{[\rho_i]\}_{i\in I}.$$

Theorem 2.1. *There is a relation on the dimensions of the representations (cf. [Eti+11, Thm 3.1 (ii)])*

$$\sum_{i\in \mathrm{Irr}(G)} (\dim(\rho_i))^2 = |G|.$$

Lemma 2.1 (Schur's Lemma (cf. [Sch05] [IR19, Thm. 7.5])). Schur's Lemma *states that for irreps A and B of G, we can give its endomorphisms as a complex multiple of the identity (cf.). Further it holds that*

$$\mathrm{Hom}_G(A,B) = \begin{cases} \mathbb{C}\cdot \mathrm{id}_A, & \text{if } A \cong B; \\ 0, & \text{if } A \ncong B. \end{cases} \tag{2}$$

By isomorphism and basis choice, we can directly express this as a matrix representation

$$\mathbb{C}\cdot \mathrm{id}_A \cong \mathbb{C}\otimes \mathrm{id}_A \cong \mathrm{id}_A \otimes \mathbb{C} \cong \mathrm{id}_{\dim(A)} \otimes \mathbb{C}.$$

Definition 2.7 (Canonical decomposition (cf. [Ser77, §2.6, Thm. 8] [FH04, Prop. 1.8])). A representation V is semisimple (or isotypic) if it decomposes as

$$V \cong \bigoplus_{i\in I} V_i^{\oplus n_i} \cong \bigoplus_{i\in I} \mathbb{C}^{n_i} \otimes V_i \cong \bigoplus_{i\in I} V_i \otimes \mathbb{C}^{n_i}$$

by grouping isomorphic simples. Here we call n_i the multiplicity of its corresponding irrep and $\mathbb{C}^{n_i}$ the *multiplicity-space*. Canonically, we can define the multiplicity space as the vector space $\mathrm{Hom}_G(V_i, V)$ [FH04, Cor. 2.16].

Due to a theorem by Peter & Weyl [**<empty citation>**], the unitary representations of any compact G are semisimple. So there are two things goind on here (i believe). First, for any *finite* group G, its associated category $\mathrm{Rep}(G)$ is semisimple by a theorem after ???. For non-finite groups this does not hold in general. However if they are unitary, the representations also decompose as a direct sum. But is this not just a consequence of diagonalization? This also fits with the intuition for the braid factors in anyons.

I misunderstood this i think. This is some different theorem

Therefore, it must hold for any irreps R_a, R_b of G that their tensor product is completely reducible with the decomposition

$$R_a \otimes R_b \cong \bigoplus_{c\in I} R_c^{\oplus N_{ab}^c} \cong \bigoplus_{c\in I} R_c \otimes \mathbb{C}^{N_{an}^c}$$

with N_{ab}^c called *fusion-multiplicities*. If

$$N_{ab}^c \leq 1 \quad \forall a,b,c$$

the decomposition is said to be *multiplicity-free*.

On the operators between tensor products, i.e. $T \in \mathrm{End}_G(R_a \otimes R_b)$ this yields a formula

$$T \cong \bigoplus_c \underbrace{\mathrm{id}_{R_c}}_{\text{fixed}} \otimes \underbrace{\mathrm{Mat}_{N_{ab}^c \times N_{ab}^c}(\mathbb{C})}_{\text{degrees of freedom}}. \tag{3}$$

This holds because, if we apply a rep of g, it acts only on where? what?

Anyhow, this decomposition is essentially fixed by the representation, the degrees of freedom are captured in the vector space that relates the multiplicity spaces.

Note as well that the appearance of the *direct sum* here (cf. Equation (28)) is fully characterized by its injection and projection maps, for which we provide a graphical calculus.

$$|a,b;c,\mu\rangle : R_c \to R_a \otimes R_b \quad \text{and} \quad \langle c,\mu;a,b| : R_a \otimes R_b \to R_c$$



(4)

These maps are more commonly known as *Clebsch-Gordan tensors*. Note that the equations

$$\langle v,d;a,b|\,|a,b;c,\mu\rangle = \delta_{\mu\nu}\delta_{cd}\,\mathrm{id}_{R_c} \tag{5}$$

and

$$\sum_{c,\mu} |a,b;c,\mu\rangle\,\langle\mu,c;a,b| = \mathrm{id}_{R_a\otimes R_b}\,. \tag{6}$$

hold. As a consequence of this relation, the outer product of two Clebsch-Gordan spans the space $\mathrm{End}_G(R_a \otimes R_b)$.

Example 2.1. The simple objects of $G = \mathrm{U}(1)$ are given by the one-dimensional representations

$$\chi_n : U(1) \to \mathbb{C}^\times, \quad e^{i\theta} \mapsto e^{in\theta} \tag{7}$$

and each representation of $\mathrm{U}(1)$ decomposes as

$$V \cong \bigoplus_{n\in\mathbb{Z}} \chi_n^{\oplus m_n}.$$

Its fusion rule is given by

$$\chi_n \otimes \chi_m = \chi_{n+m}. \tag{8}$$

Example 2.2 (Representations theory of $\mathrm{SU}(2)$ (cf. [Hal15, Thm. C.1]))**.** Let $G = \mathrm{SU}(2)$ and let V_i and V_j be two spin irreps of charge i and j. These combine to

$$(\pi_{V_i\otimes V_j}, V_i \otimes V_j) = \bigoplus_{k\in|i-j|}^{i+j} (\pi_{V_k}, V_k) \tag{9}$$

in what is referred to as the *Clebsch-Gordan-series*. The dimension of a representation V_j here is given by

$$\dim(V_j) = 2j + 1 \tag{10}$$

Example 2.3. Lets look at an example of spin irreps in the $\mathrm{SU}(2)$ theory. The combined basis of two spin-$\frac{1}{2}$ spaces is given by the collection $\{|\uparrow\rangle, |\downarrow\rangle\}^{\otimes 2}$. By the fusion rule eq. (9), the spins combine as

$$\frac{1}{2} \otimes \frac{1}{2} = 0 \oplus 1$$

with its associated basis $\{|1,1\rangle, |1,0\rangle, |1,-1\rangle, |0,0\rangle\}$. We call this basis the *coupled basis*, indexed as $|j, j_z\rangle$ where j is the total spin and j_z is the magnetization? However we can also give a formal notation for this. Denote a state in the spin basis as $|j,m\rangle$ where j is the spin and m is its magnetization? In the case that we have singular spin $1/2$ states, the second labels is redundant since

$$|\uparrow\rangle = \left|\frac{1}{2},\frac{1}{2}\right\rangle \quad |\downarrow\rangle = \left|\frac{1}{2},-\frac{1}{2}\right\rangle.$$

Read up on this here.

Hmm i dont like this being an example. Should we even use the Lie theory here? I dont think we can get around it so put the

If we are given more than two representations, lets say we are given R_a, R_b and R_c we will noteice that the spaces $(R_a \otimes R_b) \otimes R_c$ and $R_a \otimes (R_b \otimes R_c)$ are only related by an isomorphism. Applying eq. (4) twice gives

$$(R_a \otimes R_b) \otimes R_c \cong \bigoplus_d R_d \otimes \left(\bigoplus_e \mathbb{C}^{N_{ab}^e} \otimes \mathbb{C}^{N_{ec}^d} \right),$$
$$R_a \otimes (R_b \otimes R_c) \cong \bigoplus_d R_d \otimes \left(\bigoplus_f \mathbb{C}^{N_{bc}^f} \otimes \mathbb{C}^{N_{af}^d} \right).$$

and we can give the relating map

$$F_d^{abc} : \bigoplus_e \mathbb{C}^{N_{ab}^e} \otimes \mathbb{C}^{N_{ec}^d} \to \bigoplus_f \mathbb{C}^{N_{bc}^f} \otimes \mathbb{C}^{N_{af}^d}. \tag{11}$$

Example 2.4. For $G = \mathrm{SU}(2)$ the fusion rules eq. (9) are multiplicity-free, so $F_j^{j_1 j_2 j_3}$ is a matrix with rows $j_{12} \in j_1 \otimes j_2$ (such that $j \in j_{12} \otimes j_3$) and columns $j_{23} \in j_2 \otimes j_3$ (such that $j \in j_1 \otimes j_{23}$). In the physics literature this is written in terms of Wigner's $6j$*-symbols*

introduce j_1 as a spin notation like in the singh et al papers.

$$\begin{Bmatrix} j_1 & j_2 & j_{12} \\ j_3 & j & j_{23} \end{Bmatrix},$$

via the *Racah recoupling formula* [MMS17]

cyten [26c] uses these racah formulas instead of tradtional 6j? Not sure about the difference.

$$\left[F_j^{j_1 j_2 j_3}\right]_{j_{12}, j_{23}} = (-1)^{j_1+j_2+j_3+j} \sqrt{(2j_{12}+1)(2j_{23}+1)} \begin{Bmatrix} j_1 & j_2 & j_{12} \\ j_3 & j & j_{23} \end{Bmatrix}. \tag{12}$$

On states, this can be give like

$$|(j_1 j_2) j_{12},\ j_3;\ j, m\rangle = \sum_{j_{23}} \left[F_j^{j_1 j_2 j_3}\right]_{j_{12}, j_{23}} |j_1,\ (j_2 j_3) j_{23};\ j, m\rangle\,.$$

Now we wish to do an actual example for 3 SU(2) spins.

For $j = \frac{1}{2}$ both $j_{12}, j_{23} \in \{0, 1\}$ are allowed, so F is a 2×2 matrix. The sign factor in eq. (12) is $(-1)^2 = 1$, and the $6j$-symbols give

matrix notation as for anyons!

$$F_{\frac{1}{2}}^{\frac{1}{2}\frac{1}{2}\frac{1}{2}} = \begin{pmatrix} -\frac{1}{2} & \frac{\sqrt{3}}{2} \\ \frac{\sqrt{3}}{2} & \frac{1}{2} \end{pmatrix} \qquad (\text{rows } j_{12} = 0, 1;\ \text{columns } j_{23} = 0, 1).$$

2.2 Application: The XXZ Hamiltonian

The motivation for constructing symmetric tensor networks is as follows: Assume we have a (unitary) representation of a group G denoted $U(g)$. Furthermore assume we have a Hamiltonian that commutes with $[U(g)^{\otimes L}, H] = 0$, equivalently, H admits a *global symmetry* $U(g)^{\otimes L} H (U(g)^{\otimes L})^\dagger$. Then there are different presentations of H with respect to a certain basis by *simultaneous diagonalization*[HJ12, Thm 1.3.12], as shown in Figure 2.1. Notably these presentation induce a *block-sparse* structure on the respective operators

We now show how to obtain this block-sparse structure and more by example of the XXZ hamiltonian. Much of this material is covered by [Alv12; CMY25; SV12; MG02] and others.

References seem out of place here.

Definition 2.8. The *XXZ Hamiltonian* on N sites is

$$H = J \cdot \sum_{i=1}^{N-1} (S_i^x S_{i+1}^x + S_i^y S_{i+1}^y + \Delta S_i^z S_{i+1}^z), \tag{13}$$

acting on the basis $|s_1 s_2 \dots s_n\rangle$ where $s_i \in \{\uparrow, \downarrow\}$.

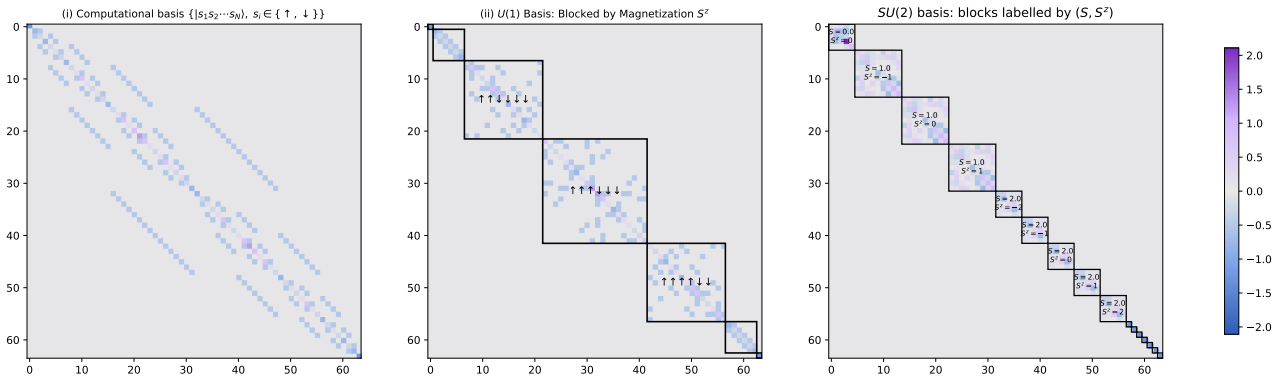


Figure 2.1: Three different presentations of the XXZ Hamiltonian for $N = 6$. (i): In the default computational basis of spin vectors. (ii) In a reordering of the computational basis into equal spin up (down) classes, i.e. no spin up, one spin up etc..., here U is simply a permutation matrix. (iii) After changing the basis into the (S^2, S_Z) basis.

Note that by definition, the XXZ-Hamiltonian is a local hamiltonian of the form

$$H = \sum_i h_{i,i+1}$$

This operator admits both a $\mathrm{U}(1)$ symmetry, and a $\mathrm{SU}(2)$ symmetry for $\Delta = 1$. We will handle the (simpler) abelian case first. The representations of $\mathrm{U}(1)$ here are induced by the operator

$$S^z_{\text{tot}} = \sum_{i=1}^{N} S^z_i \quad \text{with action} \quad U(\theta) : \theta \in \mathrm{U}(1) \mapsto e^{-i\theta 2 S^z_{\text{tot}}}$$

It can be shown that H is invariant under $U(\theta)$, and it suffices to show this invariance for the generators S^z_{tot}:

$$U(\theta) H U(\theta)^\dagger = H \quad \underset{\text{suffices by A.4}}{\Longleftrightarrow} \quad [H, S^z_{\text{tot}}] = 0. \tag{14}$$

Diagonalizing both operators yields a block structure as in Figure 2.1(ii). Note however that the diagonalization here essentially produced a permutation matrix, and we just reordered the basis to group by total spins. If we want to state this in terms of irreps, we have to realize that $\mathbb{C}^2 \cong \chi_{-1} \oplus \chi_1$. By eq. (8) we can state this space as the direct sum

So this is quite strange, i have to use $2S^z_{tot}$ here, otherwise we get non-integer weights which makes no sense.

$$\begin{aligned}(\mathbb{C}^2)^{\otimes 6} &\cong \chi_{-6} \oplus \chi_{-4}^{\oplus 6} \oplus \chi_{-2}^{\oplus 15} \oplus \chi_0^{\oplus 20} \oplus \chi_2^{\oplus 15} \oplus \chi_4^{\oplus 6} \oplus \chi_6 \\ &\cong (\mathbb{C}^1 \otimes \chi_{-6}) \oplus (\mathbb{C}^6 \otimes \chi_{-4}) \oplus (\mathbb{C}^{15} \otimes \chi_{-2}) \oplus (\mathbb{C}^{20} \otimes \chi_0) \oplus (\mathbb{C}^{15} \otimes \chi_2) \oplus (\mathbb{C}^6 \otimes \chi_4) \oplus (\mathbb{C}^1 \otimes \chi_6).\end{aligned}$$

Recall that for $\mathrm{U}(1)$, all representations are 1-dimensional, then by Schur's lemma (3), we can write H as its operator in $\mathrm{End}_{\mathrm{U}(1)}((\mathbb{C}^2)^{\otimes 6})$ as

$$H \underset{S \in \{-6,-4,-2,\,0,\,2,\,4,\,6\}}{\cong} \bigoplus \mathrm{Mat}_{m_S \times m_S}(\mathbb{R}) \otimes \mathrm{id}_{\chi_S} \underset{S \in \{-6,-4,-2,\,0,\,2,\,4,\,6\}}{\cong} \bigoplus \mathrm{Mat}_{m_S \times m_S}(\mathbb{R}) \quad \text{where } m_S \in \{1, 6, 15, 20, 15, 6, 1\}$$

and so we should expect 7 blocks in total, just as given in Figure 2.1. To store these blocks, we need to store an effective $1^2 + 6^2 + 15^2 + 20^2 + 15^2 + 6^2 + 1^1 = 934$ floating point numbers.

If we consider the XXZ model (2.8) with $\Delta = 1$ (so the XXX model), a nonabelian $\mathrm{SU}(2)$ symmetry emerges. Here, the generators of $\mathrm{SU}(2)$ are given by

$$S^\alpha_{tot} = \sum_{i=1}^{N} S^\alpha_i, \quad \alpha \in \{x, y, z\},$$

and its representation is $U(g_{\vec{n}}) = e^{i\theta \vec{n} \cdot S_{tot}}$. The Hamiltonian is invariant under

$$U(g) H U(g)^\dagger = H, \quad g \in \mathrm{SU}(2)$$

which is equivalent to $[H, S^\alpha_{tot}] = 0$ for all α. Since

$$[S^x_{tot}, S^y_{tot}] = iS^z_{tot}$$

(and all its permutations) we cannot diagonalize these operators simultaneously and we need to consider a new operator. The *Casimir operator*

$$S^2_{tot} = \sum_{a\in\{x,y,z\}} (S^a_{tot})^2$$

commutes with all the generators since

appendix?

$$[S^2_{tot}, S^b_{tot}] = \sum_a \left(S^a_{tot}[S^a_{tot}, S^b_{tot}] + [S^a_{tot}, S^b_{tot}]S^a_{tot}\right) = i\sum_{a,c} \varepsilon_{abc}\left(S^a_{tot}S^c_{tot} + S^c_{tot}S^a_{tot}\right) = 0,$$

We take the Casimir operator and the S^z_{tot} operator to diagonalize with. The eigenvalue $S(S+1)$ of S^2_{tot} labels the irrep V_S. S^z_{tot} then labels the states inside of the irrep, within V_S its eigenvalues $M \in \{-S, \dots, S\}$ distinguish the $2S+1$ basis vectors. However since the irreps also occur with a multiplicity, we need yet another label to differentiate these blocks. This is done just like for the CG-tensors with another label α. Then the states look like $|S, m, \alpha\rangle$.

Justify why we choose Sz and Stot2

Sx Sz sy are not constant on the irrep?

Or we can again do the representation theoretical approach. Let V_j denote the spin-j irrep of SU(2). We have that $\mathbb{C}^2 \cong V_{\frac{1}{2}}$ and by eq. (9) that

$$(\mathbb{C}^2)^{\otimes 6} \cong V_{\frac{1}{2}}^{\otimes 6} \cong 5V_0 \oplus 9V_1 \oplus 5V_2 \oplus V_3 \cong (\mathbb{C}^5 \otimes V_0) \oplus (\mathbb{C}^9 \otimes V_1) \oplus (\mathbb{C}^5 \otimes V_2) \oplus (\mathbb{C} \otimes V_3)$$

Now our operator is a morphism in $\mathrm{End}_G(V_{\frac{1}{2}}^{\otimes 6})$ and thus by Schur's lemma (2)

$$H \cong \bigoplus_{S\in\{0,1,2,3\}} \mathrm{Mat}_{m_S\times m_S}(\mathbb{C}) \otimes I_{2S+1} \cong \bigoplus_{S\in\{0,1,2,3\}} I_{2S+1} \otimes \mathrm{Mat}_{m_S\times m_S}(\mathbb{C}). \tag{15}$$

So while we only have to store the matrix elements as blocks, i.e. here we have to store only four of them, in our picture we get 16 blocks because of the multiplicities. If we were to store this Hamiltonian as its initial matrix, we would have to store $64 \times 64 = 4096$ floats. If we only store the 4 blocks, this is reduced to $5^2 + 9^2 + 5^2 + 1^2 = 132$ floats[1]. If we take the case that N is even larger, i.e. the case $N = 10$, the full operator requires 1 048 576 floats while the symmetric operator only needs 16 796. Generally we have about two orders of magnitude less.

Do exact asymptotics here!

While this big picture overview conveys the general idea, we should do some of the calculations by hand. In order to get an interpretation of the matrix coefficients in *eq.* (15), we restrict ourselves to the case of the XXZ-Hamiltonian for $N = 2$ spins:

$$H = J \cdot (S^x_1 S^x_2 + S^y_1 S^y_2 + S^z_1 S^z_2) = J \cdot S_1 \cdot S_2$$

This acts on the space $V_{\frac{1}{2}} \otimes V_{\frac{1}{2}}$ and thus on the basis $\{|\uparrow\rangle, |\downarrow\rangle\}^{\otimes 2}$ This has two distinct fusion outcomes, V_0 and V_1. The basis of the coupled states is given by $|1,1\rangle, |1,0\rangle, |1,-1\rangle$ (the triplet state) and $|0,0\rangle$ (the singlet state), and we want to reformulate $S_1 \cdot S_2$ in the coupled spin basis. We can do this purely algebraically by rearranging

$$S^2 = S_1^2 + S_2^2 + 2(S_1 \cdot S_2) \implies S_1 \cdot S_2 = \frac{1}{2}(S^2 - S_1^2 - S_2^2).$$

Since $S_i^2 = s(s+1)$ is fixed for $s = \frac{1}{2}$ here, we ought to write

$$H = \frac{J}{2}(S^2 - 3/2)$$

and this again splits into the sectors for $S = 0$ and $S = 1$

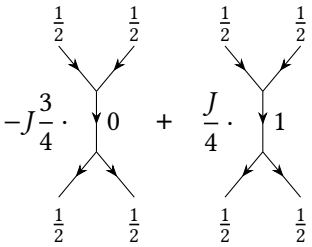
$$c_0 = \frac{J}{2}(0 - 3/2) = -\frac{3}{4}J \quad \text{and} \quad c_1 = \frac{J}{2}(2 - 3/2) = \frac{1}{4}J$$

[1] As we will see when discussing the actual algorithms behind these symmetric tensors, there is still more structure to be stored. However, these methods will still be more efficient than storing the full numerical Hamiltonian. We do exact calculations in chapter 4.

which are exactly the two degrees of freedom we get by the direct sum. So we expect our operator, now in $\bigoplus_{S\in\{0,1\}} \mathrm{End}_G(V_S)$ to look like

$$\left(\begin{array}{c|ccc} -\frac{3}{4}J & 0 & 0 & 0 \\ \hline 0 & \frac{1}{4}J & 0 & 0 \\ 0 & 0 & \frac{1}{4}J & 0 \\ 0 & 0 & 0 & \frac{1}{4}J \end{array}\right).$$

in the dense representation. However we can also still represent this as an element of $\mathrm{End}_G(V_{1/2}^{\otimes 2})$ *factored* through its fusion result. This notation might for now come out of nowhere, but remember how we characterized the graphical calculus for the Clebsch-Gordan tensors in (4). The graphical calculus here reads: Change base into the coupled basis, apply the two degrees of freedom that we just calculated, and change the base back into the uncoupled basis.



(16)

Note that the definition above is indeed that of a *projector*. Define this as $\mathcal{P}^c_{i,i+1}$ and note that $P_{i,i+1}P_{i,i+1} = P_{i,i+1}$.

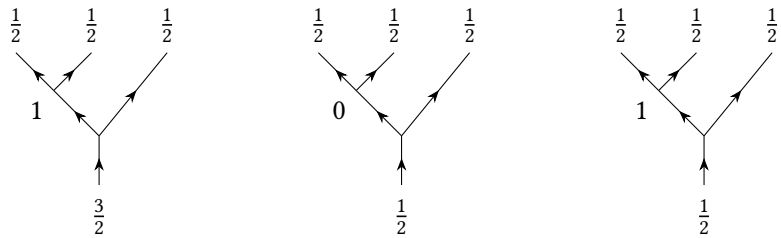
Now we consider the same Hamiltonian with $N = 3$, our state space is $(V_{1/2}\otimes V_{1/2})\otimes V_{1/2} = V_{3/2}\oplus V_{1/2}\oplus V_{1/2}$ with a multiplicity 2 on the $1/2$ state and the Operator is

$$H = \frac{1}{2}(S_1\cdot S_2 + S_2\cdot S_3)$$

and consider that we chose the left bracketing when combining the spins. We have the basis

$$\left|\frac{3}{2},1\right\rangle,\quad \left|\frac{1}{2},0\right\rangle,\quad \left|\frac{1}{2},1\right\rangle.$$

We can represent this in terms of the graphical calculus in eq. (4) as



Recall that the XXX-Hamiltonian is a *local* hamiltonian, and we can write it as

$$H = \sum_{i=1}^{n} h_{i,i+1}.$$

In this case, the part of each sitewise operator $h_{i,i+1} = S_i\cdot S_{i+1}$ that carries the *interaction* of the Hamiltonian is the same.

The derivation for $S_1\cdot S_2$ is the same as above, we simply need to include the the action on the new $\frac{3}{2}$ sector. For total spin $\frac{1}{2}$ we have the intermediate couplings $\{0,1\}$, which incurs $\mathrm{diag}(1/4,-3/4)$, and for the total spin $\frac{3}{2}$ we only have the coupling with 1 and we incur a factor of $\frac{1}{4}$.

Now, for the second part $\mathrm{id}\otimes(S_2\cdot S_3)$, where $S_2\cdot S_3\in\mathrm{End}_G(V_{1/2}\otimes V_{1/2})$. This means we cannot directly apply it to the left-aligned basis and we need to perform the recoupling move F from eq. (12) first. This leaves us with the expression

I swapped the columns here, but i should follow the convention from the 6j definition.

$$S_2 \cdot S_3 = F_{1/2}^{1/2,1/2,1/2}\,\mathrm{diag}(1/4, -3/4)(F_{1/2}^{1/2,1/2,1/2})^T = \begin{pmatrix} -\frac{1}{2} & \frac{\sqrt{3}}{4} \\ \frac{\sqrt{3}}{4} & 0 \end{pmatrix}.$$

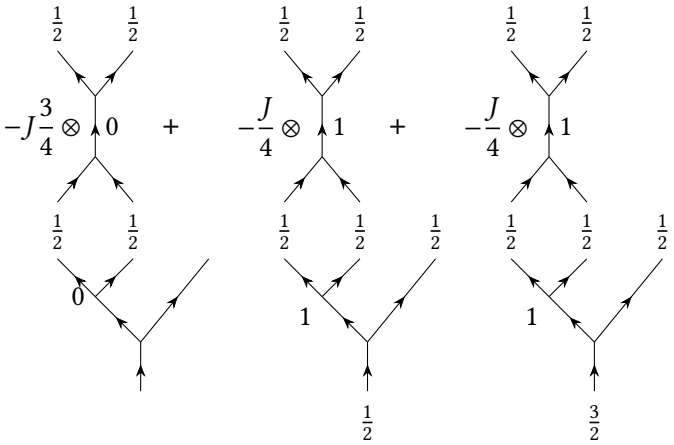
on the spin-$\frac{1}{2}$ sector and $\frac{1}{4}$ on the spin-$\frac{3}{2}$ sector. We show both the sparse (left) and the dense representation of these operators.

$$S_1 \cdot S_2 = \underbrace{\mathrm{id}_{V_{1/2}}}_{\dim 2} \otimes \begin{pmatrix} 1/4 & 0 \\ 0 & -3/4 \end{pmatrix} \oplus \underbrace{\mathrm{id}_{V_{3/2}}}_{\dim 4} \otimes (1/4)$$

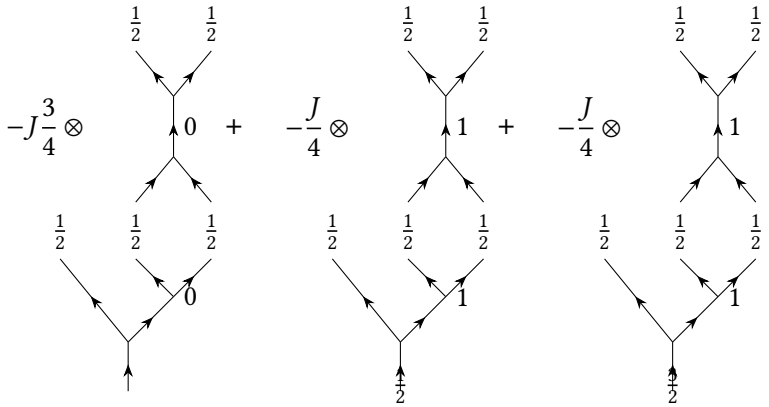
$$S_2 \cdot S_3 = \underbrace{\mathrm{id}_{V_{1/2}}}_{\dim 2} \otimes \begin{pmatrix} -\frac{1}{2} & \frac{\sqrt{3}}{4} \\ \frac{\sqrt{3}}{4} & 0 \end{pmatrix} \oplus \underbrace{\mathrm{id}_{V_{3/2}}}_{\dim 4} \otimes (1/4)$$

$$\begin{pmatrix} \frac{1}{4} & -\frac{\sqrt{3}}{4} & 0 & 0 & 0 & 0 & 0 & 0 \\ -\frac{\sqrt{3}}{4} & \frac{3}{4} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & \frac{1}{4} & -\frac{\sqrt{3}}{4} & 0 & 0 & 0 & 0 \\ 0 & 0 & -\frac{\sqrt{3}}{4} & \frac{3}{4} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & -\frac{1}{2} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{1}{2} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & -\frac{1}{2} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{1}{2} \end{pmatrix}$$

Visually we explain this by defining the operators as before and including the ground state trees:



(17)



(18)

where we have to apply a recoupling before we can apply $S_2 \cdot S_3$ and we have to apply another one to move back.

Note that we have implicitely given the action of a *reshape* operation. If we may implement this, in presentation ???

There is a third symmetry with $\mathbb{Z}_2$ under the operator

$$R = \exp(i\pi \sum_j S_j^x)$$

which can be interpreted as a *spin-flip*. Both $\mathbb{Z}_2$ and $\mathrm{U}(1)$ commute with H, so both are in the symmetry group. We can combine both of these symmetries by their semidirect product, because the do no commute $RU(\theta)R^{-1} = U(-\theta)$ [CMY25]. So we get a new symmetry group

$$\mathrm{U}(1) \rtimes \mathbb{Z}_2 = O(2) \subseteq \mathrm{SU}(2).$$

unfortunately i missed this possiblity, but i think this is not as nice to use as $\mathrm{U}(1)$. And we still want our abelian example. But im not sure how to include this as of yet.

2.3 Symmetric Tensors

So far we have applied the symmetric formalism to Hamiltonians, which are endomorphisms $\mathrm{End}_G(\,)$ between some finite vector spaces, and we have an interpretation of local Hamiltonians acting between sites. This section will show how to express arbitrary symmetric tensors as intertwiners, and thus also constructing symmetric MPSs, MPOs and MERAs.

Let G be a group, and all representations thereof are finite-dimensional.[2] Fix representations $\rho_i \colon G \to \mathrm{GL}V_i$ for $i = 1, \dots, n$ and $\sigma_j \colon G \to \mathrm{GL}W_j$ for $j = 1, \dots, m$, and set

$$V = V_1 \otimes \cdots \otimes V_n, \qquad \rho_V(g) = \rho_1(g) \otimes \cdots \otimes \rho_n(g),$$
$$W = W_1 \otimes \cdots \otimes W_m, \qquad \rho_W(g) = \sigma_1(g) \otimes \cdots \otimes \sigma_m(g).$$

The dual representation on W_j^* is $\sigma_j^*(g)f = f \circ \sigma_j(g)^{-1}$. A tensor with n outgoing and m incoming legs is an element

$$T \in V_1 \otimes \cdots \otimes V_n \otimes W_m^* \otimes \cdots \otimes W_1^* \;\cong\; V \otimes W^*,$$

on which G acts by

$$\Pi(g) = \rho_1(g) \otimes \cdots \otimes \rho_n(g) \otimes \sigma_m^*(g) \otimes \cdots \otimes \sigma_1^*(g). \tag{19}$$

Recall the definition of an intertwiner definition 2.2 as an element of $\mathrm{Hom}_G(-,-)$. For standard tensors, we had the equivalence between an object of $V \otimes W^*$ and the maps in $\mathrm{Hom}(V, W)$. Here we wish to show the same for symmetric tensors.

Definition 2.9. A tensor $T \in V \otimes W^*$ is *symmetric* (or G-invariant) if $\Pi(g)T = T$ for all $g \in G$ [SPV10].

Let $\Phi \colon V \otimes W^* \to \mathrm{Hom}(W, V)$ be the linear isomorphism $\Phi(v \otimes f) = \big(w \mapsto f(w)\, v\big)$ as we have defined in chapter 1.

Dirac notation, correct label.

Lemma 2.2. *For all $g \in G$ and $T \in V \otimes W^*$,*

$$\Phi\big(\Pi(g)T\big) = \rho_V(g) \circ \Phi(T) \circ \rho_W(g)^{-1}.$$

Then T is symmetric iff. $\Phi(T) \in \mathrm{Hom}_G(W, V)$, i.e.

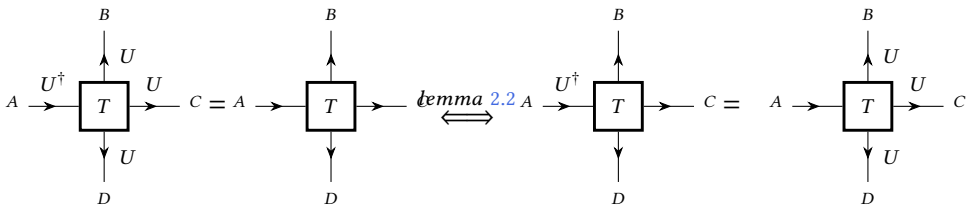
$$\rho_V(g) \circ \Phi(T) = \Phi(T) \circ \rho_W(g) \qquad \forall g \in G.$$

Proof. Both sides are linear in T, so take $T = v \otimes f$. Then $\Pi(g)T = \rho_V(g)v \otimes \big(f \circ \rho_W(g)^{-1}\big)$, and

$$\Phi(\Pi(g)T)(w) = f\big(\rho_W(g)^{-1}w\big)\, \rho_V(g)v = \rho_V(g)\, \Phi(T)\big(\rho_W(g)^{-1}w\big),$$

which is the first claim. Since Φ is injective, $\Pi(g)T = T$ holds iff $\rho_V(g)\Phi(T)\rho_W(g)^{-1} = \Phi(T)$, and multiplying on the right by the invertible $\rho_W(g)$ gives the second claim. □

Hence $V \otimes W^* \cong \mathrm{Hom}_G(W, V)$, and we identify T with $\Phi(T)$.



Globally invariant
$T \in \mathrm{Hom}(A, B \otimes C \otimes D)$

is an intertwiner
$T \in \mathrm{Hom}_G(A, B \otimes C \otimes D)$

[2] Compactness of G is not needed here; but we require it later for Schur's lemma. Or where?

Corollary 2.1 (Networks of symmetric tensors are symmetric, cf. [SPV10, Fig. 7(a)]). *Since a tensor network is built by composition and application of the (co)ev maps, both of which are intertwiners, the full network is also an intertwiner. Important there is that this entire discussion is only about FdVect or FdHilb for now, but i think the same applies to any braided category.*

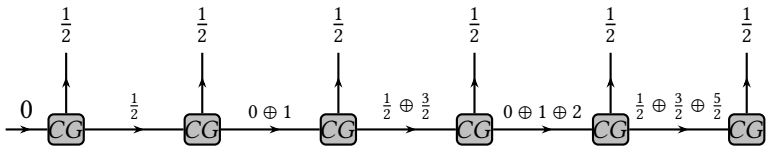
As an application of this more general tensor formalism, we can construct a symmetric MPS. Recall that a MPS is a contraction of trivalent tensors. Here, such a trivalent tensor is just an operator $\mathrm{Hom}_G(D_i \otimes P_i, D_{i+1})$.

Theorem 2.2 (Wigner-Eckart, cf. [Sel16, §2], [SV12, Eq. 23]). *For each trivalent symmetric tensor, there is a*

$$\mathrm{Hom}_G(K \otimes V, W) \cong \bigoplus_{a,b,c} \underbrace{\mathrm{Hom}(\mathbb{C}^{n_a} \otimes \mathbb{C}, \mathbb{C}^{m_b})}_{\text{reduced matrix elements}} \otimes \underbrace{\mathrm{Hom}_G(K \otimes V_a, V_b)}_{\text{Clebsch-Gordan tensor}}.$$

This is an instance of a more general theorem concerning tensors of arbitrary shape theorem 4.1 which we will prove in Chapter 4.

If we want to represent our ground state above as a MPS, we can first realize that the physical dimensions are always $\frac{1}{2}$, so this operator reduces to $\mathrm{Hom}_G(D_i \otimes V_{1/2}, D_{i+1})$. Then we can visualize it as such



and we can immediately observe an increasing bond dimension to the right.

We can do some analysis on the individual tensors here. If we choose the third tensor, of the form $\mathrm{Hom}_G(0 \oplus 1, \frac{1}{2} \otimes (\frac{1}{2} \oplus \frac{3}{2}))$, we can do some analysis over the CG-rule for SU(2).

There are 3 possible combination in the basis of this space, these are the spaces

$$(0, \frac{1}{2}, \frac{1}{2}); \quad (1, \frac{1}{2}, \frac{1}{2}); \quad (1, \frac{1}{2}, \frac{3}{2})$$

and all spaces have degeneracy 1, and so do their associated degeneracy tensors. This means that we have 3 degrees of freedom on the first tensor, and these are precisely the degrees captured by the degeneracy tensor associated to each space.

We also want to retain the ability to build MPSs by the SVD (cf. appendix A.1). Note that this requires the implicit notion of a reshape, however we already know how to express these tensors as block sparse operators (3). Take

svd appendix

$$T \cong \bigoplus_c T_c \otimes \mathrm{id}_{V_c}, \qquad T_c = U_c \Sigma_C W_c^*$$

and we can rewrite due to the identity $(A \otimes B) \cdot (C \otimes D) = (AC \otimes BD)$ as

$$T = \bigoplus_c (U_c \otimes \mathrm{id}_{V_c})(\Sigma_c \otimes \mathrm{id}_{V_c})(W_c^* \otimes \mathrm{id}_{V_c}).$$

Note that the information that is stored by the irreducible parts id_{V_c} does not change, it is already fixed by the symmetry.

In order to truncate these operators (i.e. achieving a smaller bond dimensions as in **??**)

$$T_c = \tilde{U}_c \tilde{\Sigma}_c \tilde{W_c}^*$$

we can apply the truncation blockwise

$$T = \bigoplus_c (\tilde{U}_c \otimes \mathrm{id}_{V_c})(\tilde{\Sigma}_c \otimes \mathrm{id}_{V_c})(\tilde{W_c}^* \otimes \mathrm{id}_{V_c}).$$

Effectively we can reduce an operators bond dimension by reducing its multiplicity and thus reducing the dimensions of the degeneracy matrix

$$V = \bigoplus_c V_c^{n_c} \mapsto \bigoplus_c V_c^{\chi_c}.$$

Now, this means that the bond dimension at the index of decomposition is reduced to χ. Generally, te effective bond dimensions are given by $\sum_c \chi_c \dim(V_c)$.

TODO MPO operator

Recall the XXX-Hamiltonians MPO operator with bond dimension 5 from Equation (1). This means that the bonds are $D_i \cong \mathbb{C}^5 \cong V_{1/2}^{\otimes 2} \oplus V_0 \cong V_0 \oplus V_1 \oplus V_0$. We can also represent the symmetric MPO by a matrix

$$\begin{pmatrix} \mathrm{id} & S & 0 \\ 0 & 0 & S \\ 0 & 0 & \mathrm{id} \end{pmatrix}$$

where $S = (S_x, S_y, S_z)$, with physical legs $V_{1/2}$ and the virtual bond V_1.

All the reasoning for deriving this symmetric MPO in the papers i saw [Fis25a], [SO20] was done after we have defined the fusiontrees formally. Maybe this entire section should go after categories? Talk about this in the meeting.

Note that in this section, we have not explained how to apply reshapes and contractions to these tensors. We will explain all of these operations as part of a more general formalism on *categories* in Chapter 4.

3

Anyons I : Introduction

This chapter introduces the theory of *anyons* not by their algebra (cf. chapter 4) but by their physical instantion.

In the contemporary literature, anyons are introduced as *defects* in a 2 dimensional space (manifold? compact?) with *exotic exchange statistics* on its wave functions. These exchange statistics differ from bosons or fermions in that we allow particles to pick up an arbitrary complex phase $e^{i\theta}$ on exchanges in the abelian case, and a unitary transformation in the non-abelian case. Here we introduce as one would have found by doing elementary analysis on the *parameter space* of point-like particles. We argue why these systems are important and why these are called *topological*. Ending this chapter we sketch the motivations for translating this approach into an algebraic description using *categories* and their corresponding string calculi.

Is this string calc or only diagrams?

3.1 Configuration Spaces & The Braid Group

The original article by Leinaas and Myrheim [LM77] identifies a generalized theory of *identical* particles that are point-like. Denote the ambient space that such a particle inhabits by Σ and then the space of n identical particles possible coordinates by $\Sigma^n \coloneqq \bigtimes^n \Sigma$. Since the particles are assumed to be indistinguishable, any permutation, so an action of Sym_n (cf. appendix A.1) necessarily leads to the same physical state, so we denote the full space of our system by the quotient

$$\Sigma^n/\mathrm{Sym}_n.$$

However we also want no two particles occupying the same space[1], so the diagonal of overlapping points $\Delta = \{x \in \Sigma^n \mid x_i = x_j \text{ for some } i \neq j\}$ needs to be excluded as well [Wu84]. This leads to the expression

rephrase!

$$\mathrm{Conf}_n(\Sigma) = \left\{(x_1, \cdots, x_n) \in \Sigma^{\times^n} \mid \forall_{i=j} x_i \neq x_j\right\}/\mathrm{Sym}_n \tag{20}$$

which is defined to be the *configuration space* of n unordered particles. However as we will see, anyons come in different "types", and we consider only particles of the same type to be identical. This entails that we also need a concept o a configuration space that is not ordered. We follow [SS22] with the following definition:

original publications.

Definition 3.1. The *ordered Configuration Space* of n particles of possibly differnt types (or labeled particles with the same type) on a surface of dimension D is

$$\underset{\{1,\ldots,n\}}{\mathrm{Conf}}\,(\Sigma) \coloneqq (\Sigma)^n - \Delta^n_\Sigma$$

and the *unordered Configuration Space* of n unlabeled identical particles is defined as

$$\mathrm{Conf}_n(\Sigma) \coloneqq \underset{\{1,\ldots,n\}}{\mathrm{Conf}}\,(\Sigma)/\mathrm{Sym}_n.$$

Note that while [LM77] conjectured computations for strange exchange statistics in both 2 and 3 dimensions, the concept of an anyon as a 1 dimensional representation of the braid group is due to [Wil90].

we do nt know what this is yet.

[1] If this were the case, then the resulting exchange statistics would be bosonic [Wu84, Footnote 12].

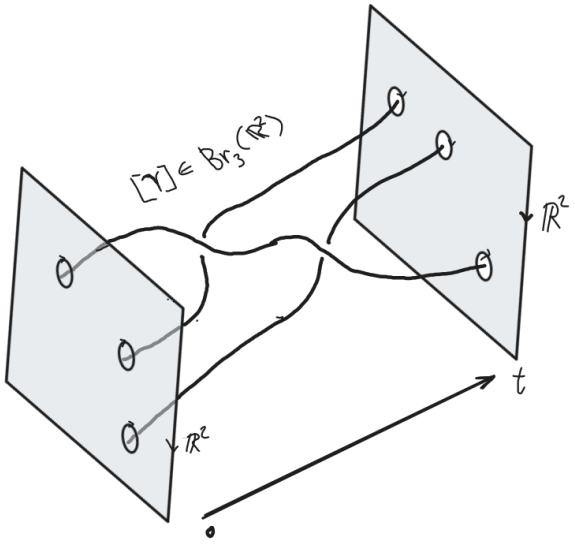


Figure 3.1: The configuration space of 3 particles in the plane $\mathbb{R}^2$ at time 0 and time t. There are two braids between the spaces.

Now we want to classify all the possible motions particles can take in this space. Here we let these motions account for time evolutions. The *motion group* of the configuration space is given by the spaces *homotopy classes* (cf. Appendix A.2), which are given by the fundamental group(oid)[2] of the space. Informally, this group collects all the loops at some configuration x_0 (a point in the configuration space) up to homotopy. We want to differentiate two cases here: If $D = 2$ it is known these spaces correspond to the *(pure) braid groups* on n strands denoted

$$\pi_1(\mathrm{Conf}_n(\Sigma)) = \mathrm{Br}_n(\Sigma) \quad \text{and} \quad \pi_1(\underset{\{1,\dots,n\}}{\mathrm{Conf}}(\Sigma)) = \mathrm{PBr}_n(\Sigma).$$

If we choose Σ as $\mathbb{R}^2$ we can give an presentation in terms of generators:

Definition 3.2 (Braid Group, cf. [Art25] [Wil18, Def. II])**.** The braid group (on the plane $\mathbb{R}^2$) is defined in terms of its generators:

$$\mathrm{Br}_n(\mathbb{R}^2) \simeq \left\langle \sigma_1, \cdots, \sigma_{n-1} \;\middle|\; \begin{array}{c} \underset{i+1<j}{\forall} \left(\sigma_i\sigma_j = \sigma_j\sigma_i\right), \\ \underset{1\le i<n-1}{\forall} \underbrace{\left(\sigma_i\,\sigma_{i+1}\,\sigma_i = \sigma_{i+1}\,\sigma_i\,\sigma_{i+1}\right)}_{\text{Yang–Baxter relation}} \end{array} \right\rangle, \tag{21}$$

We can also give these generators graphically, as well as view a composite in fig. 3.1.

$$\sigma_i \coloneqq \left[\; \cdots \;\middle|\; \underset{i-1}{} \;\middle|\; \underset{i}{\times} \;\middle|\; \underset{i+1}{} \;\middle|\; \underset{i+2}{} \;\middle|\; \cdots \;\right] \tag{22}$$

However now if $D >= 3$ we know that both of these coincide with the symmetric group Sym_n instead. In fact there is an exact sequence

$$1 \longrightarrow \mathrm{PBr}_n(\Sigma) \hookrightarrow \mathrm{Br}_n(\Sigma) \longrightarrow \mathrm{Sym}_n \longrightarrow 1$$

and this is quite best understood forgetting what an anyon is, the configuration space intrinsically has no knowledge of what it is either, it only knows its points. In the case of the ordered configuration space these

[2] One should rather use the groupoid here, since in a real scenario these particles likely will not hit the same return point. But the only part that i know of that defines it this way is the source [Neo16].

points are each endowed with a label, and these may yet correspond to identical particles in the physical layer. So the pure braid group requires a motion to end up at its exact starting configuration, so it has no permutation of labels. The braid groups generator σ_i hover can be directly mapped to the symmetric groups s_i, thus we say there is a group homomorphism $\mathrm{Br}_n \to \mathrm{Sym}_n$. Since the pure braid group does not induce *any* permutation but the trivial one, it must be in the kernel of this homomorphism.

Compare this to the presentation of the symmetric group (cf. appendix A.1) to see that Sym_n is a *subgroup* of $\mathrm{Br}_n(\mathbb{R}^2)$.

We can also do an example on there 2-sphere, since there is an extra relation of the $\mathbb{R}^2$ braid group. Generally, this braid group arises as a *quotient* of the Artin group, but this does not hold in general.

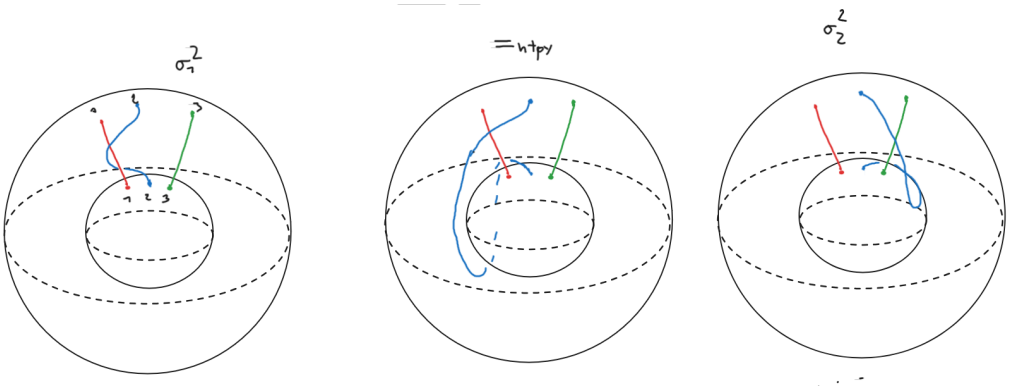


Figure 3.2: 3 Anyons on the sphere S^2 with the equivalence of the braids σ_1^2 and σ_2^2 [Sim23]. Implicitly, we are given the configuration spaces $\mathrm{Conf}_3(S^2)$.

Analyzing this case further yields that

$$\pi_1(\mathrm{Conf}_3(S^2)) \cong Br(S^2) \cong \mathbb{Z}_4 \boxtimes \mathbb{Z}_3$$

and thus finite. In fact, we know about the braid group on spheres that the order of $Br_n(S^2)$ is given by

$$\left|Br_n(S^2)\right| = \begin{cases} |\mathbb{Z}_2| = 2 & \text{if } n = 2, \text{ [FB62, p. 255]} \\ |???| = 12 & \text{if } n = 3, \text{ [nLa26, Prop 2.2]} \\ \infty & \text{if } n \geq 4. \end{cases} \tag{23}$$

Remark 3.1. This is not to assert that anyons (rather anyonic excitations) do not exist 3 or more dimensions, this simply states that there are no *point-like defects* in 3 or more dimensions that behave as anyons. Sheet and string-like defects can exhibit anyonic exchange statistics and higher fusion categories [KTZ20; BMM19; BWC06]. Generally, defects that are of 2 dimensions less than the space of their material can be considered having anyonic exchange statistics.

To reiterate, the configuration space here takes the role of a *classical* parameter space. Any quantum process we give on this state should ideally *parameterized* by the configuration space itself. To make this a formal definition, we associate a Hilbert space to each point in the configuration space by using a *Hilbert bundle* (cf. appendix A.2).

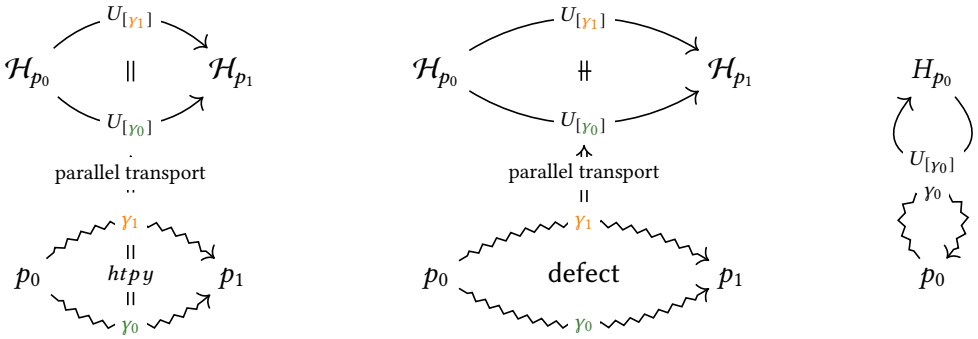
$$\mathcal{H} \xrightarrow{\pi} \underset{\{1,\dots,n\}}{\mathrm{Conf}}(\Sigma),$$

i.e. to each point $x \in \underset{\{1,\dots,n\}}{\mathrm{Conf}}(\Sigma)$ there is an associated Hilbert space $\mathcal{H}_x$ called the *fiber* of x. Furthermore, a path through the configuration space $[\gamma] : x_0 \to x_1 \in \underset{\{1,\dots,n\}}{\mathrm{Conf}}(\Sigma)$ is lifted to some unitary representation

$$\rho([\gamma]) \in U(\mathcal{H}_{x_0} \to \mathcal{H}_{x_1})$$

by *parallel transport*. Since we care about the topology of the system, we would like our paths to be *homotopy protected*. So any topological deformation (so not cuttings or crossings) can be resolved back to itself.

However if we follow two paths in the configuration space that are *not* homotopic, we want *distinctly robust transformations* on the fibers of the space. We can illustrate this as follows (adapted after [SS23, Fig. H])



and this is only the case when the corresponding connection is *flat*. Another case is when the connection is trivial, lets say we are working over a time-axis $\mathbb{R}$, then the bundle $\mathcal{H} \to \mathbb{R} \cong \mathcal{H} \times \mathbb{R}$.

The closed paths in the respective groups form a *monodromy* in the configuration space and a *holonomy* in the Hilbert Bundle, and it is these generators that yield us unitary gates that are universal for *topological quantum computation* (cf. Section 5.2)

However, in the usual literature anyons are also identified over their *fusion tensor category* (cf. Chapter 4). Although this correspondence is well known and the categorical interpretation can be considered standard, a formal treatment and proof were only given recently. These are pointed out in [Val21a], and formally rigorous derivations of this fact are given in [SS23]. However we will not give a derivation of this result and instead assume it. Perhaps a more natural introduction to this correspondence is given by an excursion over *mapping class groups (MCGs)* as done in [Bur16].

Check if htis is a full proof.

withing another theory, not general?

Now we need to introduce anther assumption, that anyons are spatially localized phenomena. This means that in every n-anyon system we can look at a k-anyon subsystem (such that they are adjacent) and apply evolutions only within this subsystem. Then the other anyons outside of the system are not affected by this. Note that this correlates with the inclusion $\mathrm{Br}_k \hookrightarrow \mathrm{Br}_n$ since while we could always take the restriction

$$\rho_{|\mathrm{Br}_k} : \mathrm{Br}_k \to \mathrm{U}(\mathcal{H}_n).$$

we are now allowed, by the postulate of locality that we can also take this restriction with a proper subsystem $H_k \subseteq H_n$ as

$$\rho_{|\mathrm{Br}_k} : \mathrm{Br}_k \to \mathrm{U}(\mathcal{H}_k).$$

Remarkably, these postulates gave us a type of *symmetry* for these anyonic systems. Assume that we are working with identical particles. Let O be observables on the Hilbert space and let $\rho : \mathrm{Br}_n \to \mathrm{U}(\mathcal{H})$ a unitary linear representation of $\mathcal{H}$ under the action of Br_n. THen there is a symmetry

$$[O, \rho(\sigma_i)] = 0 \quad \forall 0 \leq i < n.$$

For a system of $n = k = 2$ particles, and their representation ρ with the only generator σ_1. Since $\rho(\sigma_1)$ is unitary, we can diagonalize it and thus we get $\dim(\mathcal{H})$ eigenvectors $e^{i\theta_X} \in \mathrm{U}(1)$. We can elicit a first appearance of non-abelian behaviour under the Artin-relation **??** $[\rho(\sigma_1), \rho(\sigma)_2]$ not necessarily being zero here.

compare to the symmetry case.

Now choose $n \geq 3$, $k = 2$ still.

There is one issue here. We assumed that anyons were indistinguishable with no degrees of freedom. However, languages like the twist and papers such as [Val] mention that they have a self twisting. [Bur16] uses the *framed-braid-group* for this. How to proceed?

4

Categorical Networks and the Algebraic Theory of Anyons

In the existing literature it is well known that all of the concepts we have seen up to know are modeled by a formalism called *tensor categories*:

1. fdVect_k and fdHilb tensors are the most trivial and motivating case and trivial build tensor categories.
2. $\mathrm{Rep}(G)$ (for G finite) as in chapter 2 are one of the primary motivations for construction fusion tensor categories. [ENO17].
3. All anyonic topological orders (i.e species of quasiparticles as in section 3.1) can be classified by *modular tensor categories (MTCs)* [Kit06][RSW09].

Literature 4.1 (Algebraic Theory of Anyons)**.** The algebraic theory of anyons appears in many different forms. In this work we use the term *modular tensor category*, but other terms are *braided fusion category*, *braided 6j fusion system*, *modular tensor systems* and *modular functors*. Monographs introducing the theory of anyons include [Sim23, Parts I-II], [RW17, Chapter 2]. Other reviews and summaries (although this list is non-exhaustive) are given in [Bur16], [Val21a; Val], [Kit06, Appendix E], [Bon07, Chapter 2], [BSS08].

check this

Literature 4.2 (Category Theory)**.** Category theory was introduced in the 1940s by the work of Eilenberg and MacLane [**???**], originally a construct in algebraic topology. Although at first considered as a part of "abstract nonsense"[1], category theory has since found applications all across algebra, logic, computer science and mathematical physics. Introductions to category theory can be found in the standard references [Mac78] and [**leinster???**], modern treatments include [Rie] and [Vak25].

Literature 4.3 (Tensor Categories)**.** The parts most relevant to us here concern monoidal, braided monoidal and tensor categories. Originally investigated by Grothendieck, tensor categories are part of a larger program on *tannakian categories*, and are in whole motivated by representation theoretic problems.

Monoidal Categories were introduced by MacLane and Benabou, tensor categories are due to Deligne [Del90]

earlier reference

The correspondence between the penrose notation (i.e. planar diagrams) and the calculus provdided by tensor categories is explained in great detail in Joyal and Street [JS91], although earlier accounts such as [RT91] already make use of such a string calculus. For larger reviews see [Sel11] and [BL11].

We have also seen a method for evaluating networks built in $\mathrm{Rep}(G)$ in chapter 2 by decomposing a network into its fusion tree and it's degeneracy spaces. As an immediate consequence, this yields an algorithm for evaluating generalized *anyonic* and *categorical* tensor networks, as in [Unf24, § 4][DH25] [Pfe12; Pfe+10] [Aye17] [Sch+20, § ?]. This decomposition and its associated algorithms are also given formally in Theorem 4.1.

In light of this we provide a bottom-up construction of the theory of fusion categories in section 4.1, each concept will carry along an example for representations, a physical interpretation in terms of anyonic worldlines as well as the correspondence to the ATS systems of Penrose [Pen71]. We formalize the aforementioned decomposition in section 4.2 and list the algorithms necessary for achieving a tensor network out of these theories.

or tqfts?

[1] In a non-derogatory way. This quote is due to Steenrod?

4.1 Tensor & Fusion Categories

The general idea of a category is that it admits a sort of *interface*. As noticed, there are many similarities in data and structure between the entities we already dealt with, i.e. the tensor product of vector space, the tensor product of representations, of Hilbert spaces etc.... We can view all these concepts as a category with additional constraints added. But first the basic definition of a category:

Definition 4.1. A *category* C consists of

Mention consisting of data?

- a set of *objects* $\mathrm{ob}(C)$,
- for every pair $A, B \in \mathrm{ob}(C)$, a *hom-set* $\mathrm{Hom}_C(A, B)$ of *morphisms* $f : A \to B$,
- for every $A \in \mathrm{ob}(C)$, an *identity morphism* $\mathrm{id}_A : A \to A$,
- for every triple $A, B, C \in \mathrm{ob}(C)$, a *composition* map

$$\circ : \mathrm{Hom}_C(B, C) \times \mathrm{Hom}_C(A, B) \longrightarrow \mathrm{Hom}_C(A, C) \qquad A \xrightarrow{f} B \xrightarrow{g} C \text{ with } A \xrightarrow{g \circ f} C$$[2]

such that composition is *associative*, i.e. $(f \circ g) \circ h = f \circ (g \circ h)$ whenever defined, and *unital*, i.e. $f \circ \mathrm{id}_A = f = \mathrm{id}_B \circ f$ for all $f : A \to B$.

Example 4.1. Some examples here:

1. The category fdVect_k is the category of finite dimensional vector spaces and its morphisms are the linear operators between vector spaces.
2. The category fdHilb is the category whose objects are Hilbert spaces and whose morphisms are the bounded linear operators between them. Here we can alternatively take the objects to be the tuple $(\mathcal{H}, \langle -, - \rangle)$.
3. The category Set admits sets as objects and functions between sets as the morphisms.
4. The category $\mathrm{Rep}(G)$ has all the representations of G as its objects and the intertwiners of G as its morphisms which we denote by $\mathrm{Hom}_G(-, -)$ here.
5. The product category $C \times \mathcal{D}$ holds as objects tuples $(a \in C, b \in \mathcal{D})$ and as morphisms tuples of maps $(f \in \mathrm{Hom}_C(a_1, a_2), g \in \mathrm{Hom}_{\mathcal{D}}(b_1, b_2))$.

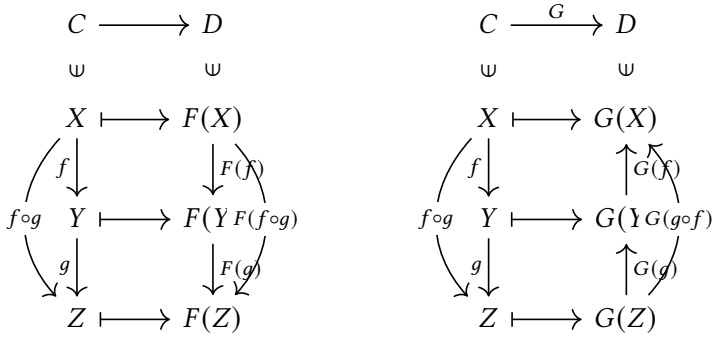
We can take categories and associate them with each other by use of a *functor*. This should be familiar for users of any functional programming language. If preferred this can be seen as the association of a homomorphism for categories alike for groups.

Definition 4.2. A *covariant functor* F between the categories C and $\mathcal{D}$, denoted as $F : C \to \mathcal{D}$ assigns to each object $A \in C$ an object $F(A) \in \mathcal{D}$ as well as to every morphism in $f : A \to B \in \mathrm{Hom}_C$ a morphism $F(f) : F(A) \to F(B) \in \mathrm{Hom}_{\mathcal{D}}$.

A *contravariant functor* $G : C \to \mathcal{D}$ assigns the objects the same, but reverses morphisms. $g : X \to Y \mapsto G(g) : G(Y) \to G(X)$.

[2] These diagrams are mostly but in part created with the great software https://q.uiver.app.

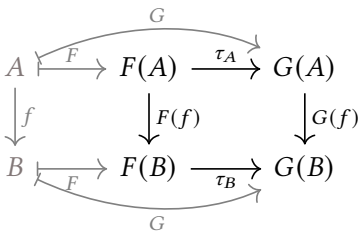
Both of these must satisfy the relation $F(\mathrm{id}_X) = \mathrm{id}_{F(X)}$ as well as the respective composition laws as in the diagram below:



covariant covariant

We can also relate functor with each other through maps between them. An important relation here will be *naturality*:

Definition 4.3. A transformation τ between functors $F, G : \mathcal{C} \rightrightarrows \mathcal{D}$ is *natural* if the diagram commutes for all $A : \mathrm{ob}(\mathcal{C})$ and $f : \mathrm{Hom}_{\mathcal{C}}(A, B)$:



Example 4.2. Here are two familiar examples on vector- and Hilbert-spaces:

1. Consider the category of Hilbert spaces fdHilb and assign it a *forgetful* functor

$$\begin{aligned} \omega : \mathrm{fdHilb} &\longrightarrow \mathrm{fdVect}_{\mathrm{k}}, \\ (\mathcal{H}, \langle \cdot, \cdot \rangle_{\mathcal{H}}) &\longmapsto \mathcal{H}, \\ \mathcal{B}(\mathcal{H}, \mathcal{K}) \ni T &\longmapsto T \in \mathrm{Hom}(\mathcal{H}, \mathcal{K}). \end{aligned}$$

   that takes each Hilbert space to its underlying vector space, and each bounded linear operator to its linear operator between vector spaces. So the inner product structure is forgotten.

2. Take the category $\mathrm{fdVect}_{\mathrm{k}}$. We can regard taking dual spaces $(-)^* : V \to \mathrm{Hom}(V, \mathbb{C})$ as a functor, as

$$f^* : W^* \to V^*$$

   fits the notion of contravariance (**??**) and furthermore, $(g \circ f)^* = f^* \circ g^*$.

3. A *bifunctor* $F : \mathcal{C} \times \mathcal{D} \to \mathcal{E}$ is a functor from the product category (5) $\mathcal{C} \times \mathcal{D}$ to E.

We will now explore how to use category theory to model quantum processes. We want this to be able to do a list of things. First, given two systems A and$B$, we want the combined system to be an object as well. Given processes, i.e. operators on each individual system $f : A \to A'$,$g : B \to B'$, there should be a combined process. Additionally, structures such as entanglement and no-cloning should be readily apparent in this theory.

Do we need to give an intro to these properties?

Contemplate the tensor product $\otimes_{\mathbb{C}}$ of vector spaces V,W. We can define this using a *universal property*. We can make this tensor product into one of these interfaces using the following definition:

Definition 4.4 (Monoidal Category (cf. [Mac78], [Sel11], [KZ22, Def. 3.4.17])). A *monoidal category* C is a category that admits a unit object $I \in ob(C)$ and a bifunctor $\otimes : C \times C \to C$, as well as 3 constraining isomorphisms

$$\rho_A : A \otimes I \cong A \qquad \lambda_B : I \otimes B \cong B \qquad \alpha_{ABC} : (A \otimes B) \otimes C \cong A \otimes (B \otimes C)$$

such that the following *triangle* and *pentagon*[3] diagrams commute:

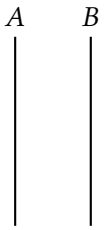
$$\begin{array}{ccc} (A \otimes I) \otimes B & \xrightarrow{\alpha_{A,I,B}} & A \otimes (I \otimes B) \\ & {\scriptstyle \rho_A \otimes \mathrm{id}_B} \searrow \quad \swarrow {\scriptstyle \mathrm{id}_A \otimes \lambda_B} & \\ & A \otimes B & \end{array} \qquad \begin{array}{ccc} ((A \otimes B) \otimes C) \otimes D & \xrightarrow{\alpha_{A,B,C} \otimes id_D} & (A \otimes (B \otimes C)) \otimes D \\ {\scriptstyle \alpha_{A \otimes B,C,D}} \downarrow & & \downarrow {\scriptstyle \alpha_{A,B \otimes C,D}} \\ (A \otimes B) \otimes (C \otimes D) & & A \otimes ((B \otimes C) \otimes D) \\ & {\scriptstyle \alpha_{A,B,C \otimes D}} \searrow \quad \swarrow {\scriptstyle id_A \otimes \alpha_{B,C,D}} & \\ & A \otimes (B \otimes (C \otimes D)) & \end{array}$$

For simplicity we denote a monoidal category as the sextuple $(C, \otimes, I, \alpha, \lambda, \rho)$.

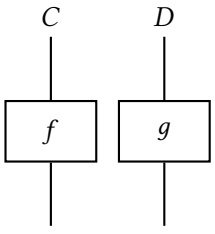
Note that because we defined $- \otimes - : C \times C \to C$ to be a *(bi)functor*, we immedeately gain the follwing relations by functoriality:

$$(f \otimes g) \circ (f' \otimes g') = (f \circ f') \otimes (g \circ g') \tag{24}$$
$$\mathrm{id}_{A \otimes B} = \mathrm{id}_A \otimes \mathrm{id}_B$$

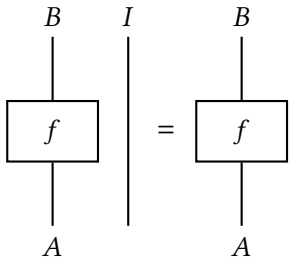
Graphical Notation 4.1. We denote the elemenets of this Category by simple lines, and the morphisms as boxes between these lines. Since the tensor product corresponds to paralell execution, we describe tensored objects as parallel lines and tensored operators as parallel squares.



Parallel Objects: $A \otimes B$



Parallel Morphisms: $f \otimes g : A \otimes B \to C \otimes D$



The commuting condition in definition 4.4 is an instance of the *MacLane-coherence-theorem* [Mac78] (for monoidal categories). This essentially assert that no matter how many objects we look at, their tensoring are isomorphic to each other without specifying another isomorphism on higher spaces, but instead reusing the α associator on 3 strands is enough.

forgot theorem number

Example 4.3. Some examples of this tensor product here:

1. In fdVect_k, we denote the tensor product as $\otimes_k$, and this is the default kronecker-product on k-vector spaces. The monoidal unit in this case is the field k.

   Does k need to be algebraically closed?

2. The category $\mathrm{Rep}(G)$ is monoidal and has a tensor product by definition 2.4:

   $$(V \otimes_k W, \rho_{V \otimes_k W})$$

   and on morphisms $f \in \mathrm{Hom}_G(V_1, V_2), g \in \mathrm{Hom}_G(W_1, W_2)$ this gives

   $$(f \otimes g)(v \otimes w) = f(v) \otimes g(w) \in \mathrm{Hom}_G(V_1 \otimes_k W_1, V_2 \otimes_k W_2)$$

3. Take for example the Fib anyon theory. We can take products of anyons $a \otimes b$ and we can also identify the vacuum type I over $a \otimes I = a$ and so on. However we also know that these anyons come in pairs, so they ought to be in a fusion channel. For now we do not have a concept of antiparticles or of fusion channels. We can only give the pairings of the anyons by bracketings of the monoidal tensors.

   rephrase!

[3] Medieval mathematicians used to include the diagonal composite morphisms in this diagram, however, after a few stake burning incidents, this practice was abandoned.

Definition 4.5. Let $(C, \otimes_C, 1_C)$ and $(D, \otimes_D, 1_D)$ be monoidal categories. A *monoidal functor* is a functor $F : C \to D$ that preserves the monoidal structure

Definition 4.6. Let $(\mathcal{V}, \otimes_{\mathcal{V}}, I, \alpha, \lambda, \rho)$ be a monoidal category. A category C *enriched over* $\mathcal{V}$ (or a $\mathcal{V}$-*category*) consists of

1. for every $A, B \in \mathrm{ob}(C)$, a *hom-object* $\mathrm{Hom}_C(A, B) \in \mathcal{V}$ (as opposed to Set in normal category),
2. for every triple $A, B, C \in \mathrm{ob}(C)$, a *composition morphism* in $\mathcal{V}$

$$\circ_{A,B,C} : \mathrm{Hom}_C(B, C) \otimes_{\mathcal{V}} \mathrm{Hom}_C(A, B) \longrightarrow \mathrm{Hom}_C(A, C),$$

3. for every $A \in \mathrm{ob}(C)$, a *unit morphism* in $\mathcal{V}$

$$j_A : I \longrightarrow \mathrm{Hom}_C(A, A),$$

This immedeately gives us another defintion:

$$\mathrm{Hom}_C(-, X) : C \to \mathcal{V} \quad \text{is a contravariant functor.}$$

Definition 4.7 (Braided Monoidal Category, (cf. [JS91])). A *braided monoidal category* is a monoidal category $(C, \otimes, I, \alpha, \lambda, \rho)$ equipped with an additional isomorphism, the *braiding*

$$\beta_{A,B} : A \otimes B \;\cong\; B \otimes A,$$

on A and B satisfying

$$\beta_{A,B\otimes C} = (\mathrm{id}_A \otimes \beta_{A,C}) \circ (\beta_{A,B} \otimes \mathrm{id}_C) \quad \text{and} \quad \beta_{A\otimes B,C} = (\beta_{A,C} \otimes \mathrm{id}_B) \circ (\mathrm{id}_A \otimes \beta_{B,C}). \tag{25}$$

As a consequence, the *hexagon* relations commute for all $A, B, C \in C$:

$$\begin{array}{ccccc}
(A \otimes B) \otimes C & \xrightarrow{\alpha} & A \otimes (B \otimes C) & \xrightarrow{\beta_{A,B\otimes C}} & (B \otimes C) \otimes A \\
{\scriptstyle \beta_{A,B}\otimes \mathrm{id}_C}\downarrow & & & & \downarrow{\scriptstyle \alpha} \\
(B \otimes A) \otimes C & \xrightarrow{\alpha} & B \otimes (A \otimes C) & \xrightarrow{\mathrm{id}_B \otimes \beta_{A,C}} & B \otimes (C \otimes A)
\end{array}
\qquad
\begin{array}{ccccc}
A \otimes (B \otimes C) & \xrightarrow{\alpha^{-1}} & (A \otimes B) \otimes C & \xrightarrow{\beta} & C \otimes (A \otimes B) \\
{\scriptstyle \beta}\downarrow & & & & \downarrow{\scriptstyle \alpha^{-1}} \\
A \otimes (C \otimes B) & \xrightarrow[\alpha^{-1}]{} & (A \otimes C) \otimes B & \xrightarrow[\beta]{} & (C \otimes A) \otimes B
\end{array}$$

Definition 4.8. A braided monoidal category is *symmetric* if $\beta_{B,A} \circ \beta_{A,B} = \mathrm{id}_{A\otimes B}$ for all A, B.

Lemma 4.1 ([Eti+15, Prop. 8.1.10]). *The* Yang-Baxter equation *(a representation of eq. (21)) holds for the braiding:*

$$(\beta_{A,B} \otimes \mathrm{id}_C)(\mathrm{id}_B \otimes \beta_{A,C})(\beta_{B,C} \otimes \mathrm{id}_A) = (\mathrm{id}_A \otimes \beta_{B,C})(\mathrm{id}_B \otimes \beta_{A,C})(\mathrm{id}_A \otimes \beta_{A,B})$$

In fact, we can take any braiding to represent a generator σ_i of Br_n by using

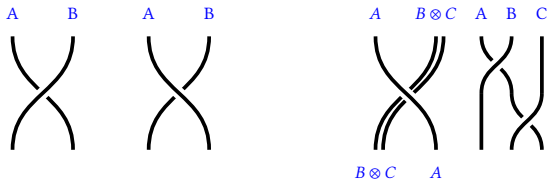
$$\sigma_i \mapsto \mathrm{id}_{\bigotimes_{j=1}^{i-1} A_j} \otimes \beta_{A_i,A_{i+1}} \otimes \mathrm{id}_{\bigotimes_{j=i+2}^{n} A_j}.$$

Do we need R for this or can we do this with the braiding op as such?

If the category is symmetric (4.8), then these braid representations also form representations for the symmetric group (cf. 45).

Graphical Notation 4.2. Now the graphical calculus changes, as we need to regard whether the braidings defined by β are over or under. We denote the simple braidings as

Check for the correct definition of the ambient space here.



Example 4.4. 1. In fdVect_k this mapping is trivial.

2. For representations, there exist transformations on the basis states, for example the theory Rep(SU(2)) picks up the phase $(-1)^{j_a+j_b+j_c}$. For all categories built from representations of groups, these are symmetric.

3. For anyons, these correspond to the phase picked up when exchaning the anyons. For abelian anyons, these braidings are symmetric indeed, however for nonabelians these create non-symmetric categories.

Notation!

If we want to use contractions between matchings legs, and overall want the ability to take duals just like in FD$\mathbb{C}$Vec, we need both a concept of duality and the evaluation maps. This is also the categorification of the *creation* and *annihilation* maps from Chapter 5. Let this motivate the following definition:

Definition 4.9. A monoidal category $(C, \otimes, 1)$ is *rigid* if, for every $A \in C$, there exists a *dual object* A^* with the following maps:

$$\mathrm{ev}_A : A^* \otimes A \to 1 \quad \text{and} \quad \mathrm{coev}_A : 1 \to A \otimes A^*$$

subject to the diagrammatic constraints

$$A \xrightarrow{\lambda_A} 1 \otimes A \xrightarrow{\mathrm{coev}_A \otimes \mathrm{id}_A} (A \otimes A^*) \otimes A \xrightarrow{\alpha_{A,A^*,A}} A \otimes (A^* \otimes A) \xrightarrow{\mathrm{id}_A \otimes \mathrm{ev}_A} A \otimes 1 \xrightarrow{\rho_A} A$$
$$\mathrm{id}_A$$

$$A^* \xrightarrow{\rho_{A^*}} A^* \otimes 1 \xrightarrow{\mathrm{id}_{A^*} \otimes \mathrm{coev}_A} A^* \otimes (A \otimes A^*) \xrightarrow{\alpha_{A^*,A,A^*}} (A^* \otimes A) \otimes A^* \xrightarrow{\mathrm{ev}_A \otimes \mathrm{id}_{A^*}} 1 \otimes A^* \xrightarrow{\lambda_{A^*}} A^*$$
$$\mathrm{id}_{A^*}$$

Example 4.5. These maps appear both for traditional vector space but also have an anyonic interpretation:

1. In fdVect_k, the dual object for every vector space V is given by its *dual-space* $V^* \coloneqq \mathrm{Hom}(V, k)$. The evaluation and coevaluation maps here correspond to

$$\mathrm{ev}_V : V \otimes V^* \to k, \quad \langle f| \otimes |v\rangle \mapsto \langle f|v\rangle \in k$$

$$\mathrm{coev}_V : k \to V^* \otimes V,$$

2. In anyon theories we generally want the ability to take antiparticles for each species of anyon, and furthermore, we want to be able to *create* and *annihilate* these anyons. So the dual object corresponds to the antiparticle, and the ev and coev maps either produce a particle-antiparticle pair from the vacuum or annihilate them (to a state containing the vacuum).

Graphical Notation 4.3. Now we have to introduce this notion of duality on the level of the graphical calculus. We denote an object in the category by an oriented arrow, and denote its dual object by the reverse.

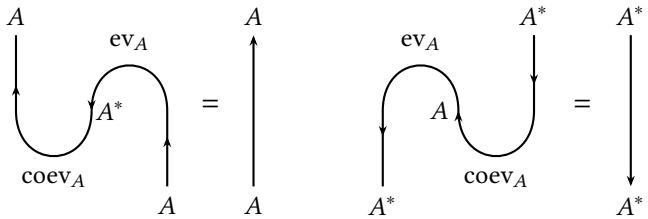
Do these need to point up or down?



Similarly the ev_A and coev_A maps induce a graphical menaing:

coev_A :  and ev_A :  .

We can give the two constraint (aptly named the *snake equations*) as a "tightening" of a string:



Observe that in the graphical calculus, we denoted this as a bent arrow from $A \to A^*$. In fact, we gain the following identities:

$$\mathrm{Hom}_C(X \otimes A, Y) \cong \mathrm{Hom}_C(X, A^* \otimes Y) : f \mapsto$$

We can also build dualities for morphisms here: Let $f : A \to B$ be arbitrary and define

$$f^* : B^* \to A^* = (\mathrm{ev}_B \otimes \mathrm{id}_{A^*})(\mathrm{id}_{B^*} \otimes f \otimes \mathrm{id}_{A^*})(\mathrm{id}_{B^*} \otimes \mathrm{coev}_A)$$

This gives us a few additional theorems:

1. $\mathrm{id}_A^* = \mathrm{id}_{A^*}$
2. $(g \circ f)^* = f^* \circ g^*$
3. $(f \otimes g)^* = g^* \otimes f^*$
4. $f \mapsto f^*$ is linear

Also notice by this that $(-)^*$ is a contravariant functor.

Further the natural observation that $A \cong A^{**}$ holds should be encoded in these categories.

Definition 4.10. A *pivotal structure* on a rigid monoidal category is given by a natural isomorphism $\mathrm{id} \to (-)^{**}$, i.e. a collection of isomorphisms that identify $\{A \to A^{**}\}$.

This allows us to define a notion of a traced category.

Definition 4.11. We can define a notion of a *right trace* on any endomorphism $V \to V$ in such categories:

$$\mathrm{tr}_R \quad \coloneqq \quad 1 \xrightarrow{\mathrm{coev}_A} A \otimes A^* \xrightarrow{(\delta_A \circ (-)) \otimes \mathrm{id}_{A^*}} A^{**} \otimes A^* \xrightarrow{\mathrm{ev}_{A^*}} 1$$

Definition 4.12. A pivotal rigid braided monoidal category is called *spherical* if

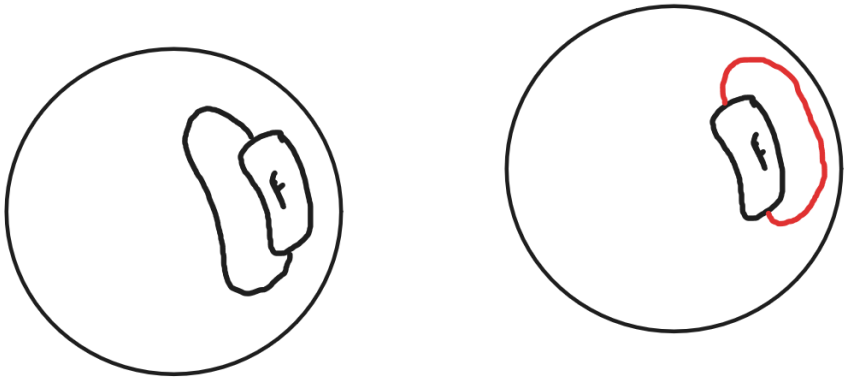
$$\mathrm{tr}_R = \mathrm{tr}_L$$

and we can further define the *quantum dimension* of an object A as

$$\dim(A) = \mathrm{tr}(\mathrm{id}_A) \tag{26}$$

Graphical Notation 4.4. The usage of *spherical* here derives from the fact that we can arbitrarily reorder our string calculus on the surface of the 2-sphere: Take for example the trace where the connecting strand is swapped around the back of the sphere.

Does this lead naturally into the ribbon being the only obstacle to planarity?



Lemma 4.2. *State without proof, in any spherical category C we can make the following statements about the trace. Let $A \in C$ and $f, g \in \mathrm{End}(A)$ arbitrary, let*

$$\mathrm{tr} : \mathrm{End}_C(A) \to \mathrm{End}_C(\mathrm{I})$$

1. $\mathrm{tr}(f^*) = \mathrm{tr}(f)$
2. $\mathrm{tr}(f \otimes g) = \mathrm{tr}(f) \cdot \mathrm{tr}(g)$
3. $\mathrm{tr}(h \circ k) = \mathrm{tr}(k \circ h)$
4.

Likely more, but which ones do we actually need?

So there is two reasons that we need this morphism here. One is to resolve the third Reidemeister move (which we havent introduced yet!) and another is that we can suppose anyons to have an intrinsic rotation, however, this would contradict that we considered them equivalent particles with no degrees of freedom. I think [Bur16] uses a deformed braid group for this, but I would need to double check this.

Definition 4.13. A *ribbon* category is a braided monoidal rigid category $(C, \otimes, 1, \alpha, \gamma, \rho, \beta, \mathrm{ev}, \mathrm{coev})$ with maps $\theta_A : A \to A$ that realizes a *self-braiding* of the object. This is subject to

$$\theta_{A\otimes B} = (\theta_A \otimes \theta_B) \circ \beta_{B,A} \circ \beta_{A,B}$$

as well as

$$\theta_{A^*} = (\theta_A)^*$$

Example 4.6. Since this is an endomorphism, physically this acts by a phase.

Definition 4.14. Given a linear category C with objects $\{A_j\}_{j=1}^N \subseteq \mathrm{ob}(C)$, an object $\bigoplus_{i=1}^n A_i \in C$ is called the *direct sum* if it is equipped with the injection and projection map

$$\iota_i : A_j \to \bigoplus A_i \quad \text{and} \quad \pi_i : \bigoplus A_j \to A_i$$

such that

$$\pi_i \circ \iota_i = \mathrm{id}_{A_i} \, \forall i \quad \text{and} \quad \sum_i \iota_i \pi_i = \mathrm{id}_{\bigoplus A_i}$$

Definition 4.15 (Semisimple Category, cf. [Bai17])**.** A *semisimple* category[4] is a category in which every object $S \in \mathrm{ob}(C)$ decomposes as a finite direct sum

$$S \cong \bigoplus_i S_i, \qquad S_i \text{ simple} \tag{27}$$

where a *simple* object is an object, where for any other object A that admits an injective map, the object is either the 0 morphism or an isomorphism to S . If we group these simples by isomorphism, we can restate (27) equivalently as

this is vector space intuition, not sure if this works. Maybe we need to introduce monomorphism? Need to review this definition for general categories.

$$S \cong \bigoplus_i S_i^{\oplus n_i}$$

where $n_i = \dim \mathrm{Hom}_C(S_i, S)$. Note that by our assumption of $\mathrm{fdVect}_{\mathrm{k}}$-enrichment, all the n_is are finite. Further denote the set of simple objects by $\mathrm{Irr}(C) \subseteq \mathrm{ob}(C)$.

Example 4.7.

1. In $\mathrm{Rep}(G)$, we know that either the representations are simple (Irreps) or they decompose as some direct sum representation.

Is this just the first part of schurs lemma?

Erichment first.

2. In Anyonic theories, we primarily consider simple objects $a, b, c \ldots$ and tensor products between them. Then we assign each combination of anyons a fusion outcome $a \otimes b = c \oplus d \oplus \cdots$.

Lemma 4.3 (Schur's lemma [Sch05])**.** *If A and B are simple objects in a linear category over an algebraically closed field k (this is always $\mathbb{C}$ in our case), then it holds that*

$$\mathrm{Hom}(A, B) = \begin{cases} k \cdot \mathrm{id}_S, & \text{if } A \cong B; \\ 0, & \text{if } A \ncong B. \end{cases}$$

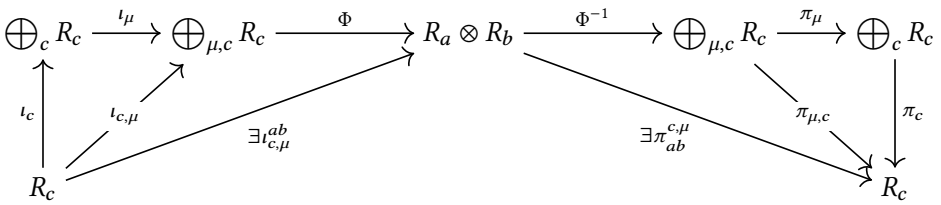
Proof [IR19, Theorem 7.5].

[4] There are multiple definitions of semisimplicity and semisimple categories, and here we are using the definition of *object simplicity*. A review of these definitions is given in [Bai17].

Now assume that we take a semisimple category that is also monoidal. By semisimplicity, any of the objects of the form $R_a \otimes R_b$ are isomoprhic to $\bigoplus R_c$ where $R_a, R_b, \{R_c\}$ are all simple. A simple consequence of Lemma 4.3 produces

$$\dim \mathrm{Hom}_C(R_a \otimes R_b, R_c) = \dim \mathrm{Hom}_C(\bigoplus_c N_d^{ab} R_d, R_c) = \dim \bigoplus_d \mathrm{Hom}_C(R_d, R_c)^{\oplus N_d^{ab}} \overset{lemma\ 4.3}{=} \dim k^{N_c^{ab}}$$

Call this isomorphism $\Phi : \bigoplus_{c,\mu} R_c \to R_a \otimes R_b$. Then we can extend projection and injection maps in dependence of the tensor product by the following diagram:



(28)

Furthermore, these admit similar constraints as the direct sum:

$$\pi_{ab}^{c,\mu} \iota_{d,\nu}^{ab} = \pi_{c,\mu} \Phi^{-1} \Phi \iota_{d,\nu} = \delta_{cd} \delta_{\mu\nu} \mathrm{id}_{R_c} \tag{29}$$

and

$$\sum_{c,\mu} \iota_{c,\mu}^{ab} \pi_{ab}^{c,\mu} = \Phi \left(\sum_{c,\mu} \iota_{c,\mu} \pi_{c,\mu} \right) \Phi^{-1} = \Phi\, \mathrm{id}_{\bigoplus R_c}\, \Phi^{-1} = \mathrm{id}_{R_a \otimes R_b} \tag{30}$$

We take these maps as above and claim they form a basis of the spaces $\mathrm{Hom}_C(a \otimes b, c)$ and $\mathrm{Hom}_C(c, a \otimes b)$.

Proof. First we show linear independence. Let $\{x_\mu\}_1^{N_c^{ab}}$ be a set of scalars and suppose that $\sum_\mu x_\mu \iota_{c,\mu}^{ab} = 0$. Observe that

$$\pi_{c,\mu}^{ab} \left(\sum_\mu x_\mu \iota_{c,\mu}^{ab} \right) = \sum x_\mu \delta_{\mu,\nu}\, \mathrm{id}_{R_c} = x_\nu\, \mathrm{id}_{R_c}$$

and since $\mathrm{id}_{R_c} \neq 0$ these form a linearly independent set.

Now let $f \in \mathrm{Hom}_C(R_c, R_a \otimes R_b)$ be an arbitrary morphism. By (30), we can write

$$f = \mathrm{id}_{R_a \otimes R_b} \circ f = \sum_{d,\nu} \iota_{d,\nu}^{ab} (\pi_{ab}^{d,\nu} \circ f \in \mathrm{Hom}_C(R_c, R_d)) \overset{Lemma\ 4.3}{=} \sum_{d,\nu} \iota_{d,\nu}^{ab} \delta_{R_c \cong R_d} \lambda_\nu\, \mathrm{id}_{R_c} = \sum_\nu \lambda_\nu \iota_{c,\nu}^{ab}$$

and f is an arbitrary linear combination and thus

$$\mathrm{span}_{\mathbb{C}} \left\{ \iota_{d,\nu}^{ab} \right\}_{\nu=1}^{N_{ab}^c} \cong \mathrm{Hom}_C(R_c, R_a \otimes R_b)$$

and the statement follows dually for $\mathrm{Hom}_C(R_a \otimes R_b, R_c)$. □

If we look at Hom spaces of a higher order such as $\mathrm{Hom}_C(R_d, (R_a \otimes R_b) \otimes R_c)$, we have to choose a bracketing of the tensor products since we can only associate $(R_a \otimes R_b) \otimes R_c \cong R_a \otimes (R_b \otimes R_c)$. Under the (covariant) functor $\mathrm{Hom}_C(-, R_d)$ this introduces a pushforward

$$(\alpha_{abc}^d)^* : \mathrm{Hom}_C((R_a \otimes R_b) \otimes R_c, R_d) \cong \mathrm{Hom}_C(R_a \otimes (R_b \otimes R_c), R_d)$$

and a pullback under $\mathrm{Hom}_C(R_d, -)$

$$(\alpha_d^{abc})^* : \mathrm{Hom}_C(R_d, (R_a \otimes R_b \otimes R_c)) \cong \mathrm{Hom}_C(R_d, R_a \otimes (R_b \otimes R_c)).$$

We can also define a basis for these extended spaces. In fact, lets generalize this to the space $\mathrm{Hom}_C(R_b, \bigotimes_{i=1}^n R_{a_i})$ where we assume this is fully left-algined bracketing.

We define the higher bases recursively: Let $n > 2$ and intermediate labels $e_1, e_2, \ldots, e_{n-2} \in \mathrm{Irr}(C)$ and multiplicity indices $\mu_1, \ldots, \mu_{n-1}$. We fix some notation to let $\vec{\mu}_{k-1}$ to denote the tuple of all labels up to μ_{k-1}. Then we can define

$$\iota_{d,\hat{e}_{n-2},\hat{\mu}_{n-1}}^{a_1 \cdots a_n} \coloneqq \left(\iota_{e_{n-2},\vec{e}_{n-3},\vec{\mu}_{n-2}}^{a_1 \cdots a_{n-1}} \otimes \mathrm{id}_{R_{a_n}} \right) \circ \iota_{d,\mu_{n-1}}^{e_{n-2} a_n}. \tag{31}$$

and

$$\pi^{a_1\cdots a_n}_{e_{n-2}} \tag{32}$$

We can assume this holds by induction for $(n-1)$:

$$\pi^{a_1\cdots a_{n-1}}_{e;\vec{e},\vec{\mu}} \circ \iota^{a_1\cdots a_{n-1}}_{e';\vec{e}',\vec{\mu}'} = \delta_{ee'}\,\delta_{\vec{e}\vec{e}'}\,\delta_{\vec{\mu}\vec{\mu}'}\ \mathrm{id}_{R_e}\,. \tag{33}$$

$$\sum_{e,\vec{e},\vec{\mu}} \iota^{a_1\cdots a_{n-1}}_{e;\vec{e},\vec{\mu}} \circ \pi^{a_1\cdots a_{n-1}}_{e;\vec{e},\vec{\mu}} = \mathrm{id}_{\bigotimes_{i<n} R_{a_i}}\,. \tag{34}$$

The inductive step then easily evaluates:

$$\underbrace{\pi^{a_1\cdots a_n}_{c;\vec{e},\vec{\mu}} \circ \iota^{a_1\cdots a_n}_{d;\vec{f},\vec{\nu}}}_{\in \mathrm{Hom}_C(R_d,R_c)} \overset{(24)}{=} \pi^{e_{n-2}a_n}_{c,\mu_{n-1}} \circ \left[\left(\pi^{a_1\cdots a_{n-1}}_{e_{n-2};\vec{e}_{n-3},\vec{\mu}_{n-2}} \circ \iota^{a_1\cdots a_{n-1}}_{f_{n-2};\vec{f}_{n-3},\vec{\nu}_{n-2}}\right) \otimes \mathrm{id}_{R_{a_n}}\right] \circ \iota^{f_{n-2}a_n}_{d,\nu_{n-1}} \tag{35}$$

$$\overset{(33)}{=} \pi^{e_{n-2}a_n}_{c,\mu_{n-1}} \circ \left[\delta_{e_{n-2}f_{n-2}}\,\delta_{\vec{e}_{n-3}\vec{f}_{n-3}}\,\delta_{\vec{\mu}_{n-2}\vec{\nu}_{n-2}}\ \mathrm{id}_{R_{e_{n-2}}} \otimes \mathrm{id}_{R_{a_n}}\right] \circ \iota^{f_{n-2}a_n}_{d,\nu_{n-1}} \tag{36}$$

$$\overset{(24)}{=} \delta_{\vec{e}_{n-2}\vec{f}_{n-2}}\,\delta_{\vec{\mu}_{n-2}\vec{\nu}_{n-2}} \cdot \pi^{e_{n-2}a_n}_{c,\mu_{n-1}} \circ \left[\mathrm{id}_{R_{e_{n-2}}\otimes R_{a_n}}\right] \circ \iota^{e_{n-2}a_n}_{d,\nu_{n-1}} \tag{37}$$

$$= \delta_{\vec{e}_{n-2}\vec{f}_{n-2}}\,\delta_{\vec{\mu}_{n-2}\vec{\nu}_{n-2}} \cdot \pi^{e_{n-2}a_n}_{c,\mu_{n-1}} \circ \iota^{e_{n-2}a_n}_{d,\nu_{n-1}} \tag{38}$$

$$\overset{(29)}{=} \delta_{cd}\delta_{\vec{e}_{n-2}\vec{f}_{n-2}}\,\delta_{\vec{\mu}_{n-1}\vec{\nu}_{n-1}}\,\mathrm{id}_{R_c}\,. \tag{39}$$

We can proceed for completeness likewise.

$$\begin{aligned}
&\sum_{d,\vec{e},\vec{\mu}} \iota^{a_1\cdots a_n}_{d;\vec{e},\vec{\mu}} \circ \pi^{a_1\cdots a_n}_{d;\vec{e},\vec{\mu}} \\
&\overset{(31)}{=} \sum_{d,\vec{e},\vec{\mu}} \left(\iota^{a_1\cdots a_{n-1}}_{e_{n-2};\vec{e}_{n-3},\vec{\mu}_{n-2}} \otimes \mathrm{id}_{R_{a_n}}\right) \circ \iota^{e_{n-2}a_n}_{d,\mu_{n-1}} \circ \pi^{e_{n-2}a_n}_{d,\mu_{n-1}} \circ \left(\pi^{a_1\cdots a_{n-1}}_{e_{n-2};\vec{e}_{n-3},\vec{\mu}_{n-2}} \otimes \mathrm{id}_{R_{a_n}}\right) \\
&= \sum_{d,\vec{e}_{n-3},\vec{\mu}_{n-2}} \left(\iota^{a_1\cdots a_{n-1}}_{e_{n-2};\vec{e}_{n-3},\vec{\mu}_{n-2}} \otimes \mathrm{id}_{R_{a_n}}\right) \circ \left[\sum_{d,\mu_{n-1}} \iota^{e_{n-2}a_n}_{d,\mu_{n-1}} \circ \pi^{e_{n-2}a_n}_{d,\mu_{n-1}}\right] \circ \left(\pi^{a_1\cdots a_{n-1}}_{e_{n-2};\vec{e}_{n-3},\vec{\mu}_{n-2}} \otimes \mathrm{id}_{R_{a_n}}\right) \\
&\overset{(30)}{=} \sum_{d,\vec{e}_{n-3},\vec{\mu}_{n-2}} \left(\iota^{a_1\cdots a_{n-1}}_{e_{n-2};\vec{e}_{n-3},\vec{\mu}_{n-2}} \otimes \mathrm{id}_{R_{a_n}}\right) \circ \mathrm{id}_{R_{e_{n-2}}\otimes R_{a_n}} \circ \left(\pi^{a_1\cdots a_{n-1}}_{e_{n-2};\vec{e}_{n-3},\vec{\mu}_{n-2}} \otimes \mathrm{id}_{R_{a_n}}\right) \\
&= \sum_{\vec{e}_{n-2},\vec{\mu}_{n-2}} \left(\iota^{a_1\cdots a_{n-1}}_{e_{n-2};\vec{e}_{n-3},\vec{\mu}_{n-2}} \otimes \mathrm{id}_{R_{a_n}}\right) \circ \left(\pi^{a_1\cdots a_{n-1}}_{e_{n-2};\vec{e}_{n-3},\vec{\mu}_{n-2}} \otimes \mathrm{id}_{R_{a_n}}\right) \\
&\overset{(24)}{=} \sum_{\vec{e}_{n-2},\vec{\mu}_{n-2}} \left(\iota^{a_1\cdots a_{n-1}}_{e_{n-2};\vec{e}_{n-3},\vec{\mu}_{n-2}} \circ \pi^{a_1\cdots a_{n-1}}_{e_{n-2};\vec{e}_{n-3},\vec{\mu}_{n-2}}\right) \otimes \left(\mathrm{id}_{R_{a_n}} \circ \mathrm{id}_{R_{a_n}}\right) \\
&\overset{(34)}{=} \mathrm{id}_{\bigotimes_{i<n} R_{a_i}} \otimes \mathrm{id}_{R_{a_n}} \\
&= \mathrm{id}_{\bigotimes_{i=1}^{n} R_{a_i}}\,.
\end{aligned}$$

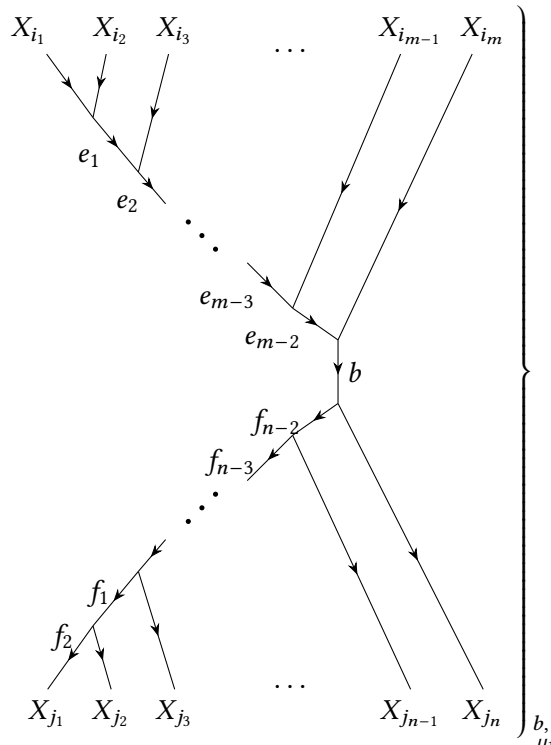
Let $e := e_{n-2}$, $\iota' = \iota^{a_1\cdots a_{n-1}}_{e;\vec{e}_{n-3},\vec{\mu}_{n-2}}$, and π' likewise. Then

$$\sum_{d,\vec{e},\vec{\mu}} \iota^{a_1\cdots a_n}_{d;\vec{e},\vec{\mu}} \circ \pi^{a_1\cdots a_n}_{d;\vec{e},\vec{\mu}} = \sum_{e,\vec{e}_{n-3},\vec{\mu}_{n-2}} (\iota' \otimes \mathrm{id}_{R_{a_n}}) \circ \left[\sum_{d,\mu_{n-1}} \iota^{ea_n}_{d,\mu_{n-1}} \circ \pi^{ea_n}_{d,\mu_{n-1}}\right] \circ (\pi' \otimes \mathrm{id}_{R_{a_n}}).$$

In fact, we can also form a "canoncial" basis for the spaces $\mathrm{Hom}_C(X^{\otimes n}, X^{\otimes n})$ as given in [Val21b]

this needs a lot of conventions?

$$\mathrm{Hom}\left(\bigotimes_{k=1}^{m} X_{i_k}, \bigotimes_{l=1}^{n} X_{j_l}\right) = \mathrm{span}_{\mathbb{C}}\left\{ \right.$$



$$\left. \right\}_{\substack{b, e_1, \ldots, e_{m-2}, f_1, \ldots, f_{n-2} \\ \mu_1, \ldots, \mu_{m-2}, \nu_1, \ldots, \nu_{n-2}}} \tag{40}$$

Now, what this construction allows us to do is as follows: Consider the decompositions

$$\mathrm{Hom}_C((A \otimes B) \otimes C, D) \cong \bigoplus_{e \in \mathrm{Irr}(C)} \mathrm{Hom}_C(A \otimes B, e) \otimes_k \mathrm{Hom}_C(e \otimes C, D)$$
$$\mathrm{Hom}_C(A \otimes (B \otimes C)) \cong \bigoplus_{f \in \mathrm{Irr}(C)} \mathrm{Hom}_C(A \otimes f, D) \otimes_k \mathrm{Hom}_C(B \otimes C, f)$$

Which is induced by inverse of the isomorphism refer to basis

$$\mu \otimes_k \nu \mapsto (\mu \otimes \mathrm{id}_c) \circ \nu.$$

Using this we can coordinatize the pushforward as a matrix

$$F_D^{ABC} : \bigoplus_{e \in \mathrm{Irr}(C)} \mathrm{Hom}_C(A \otimes B, e) \otimes_k \mathrm{Hom}_C(e \otimes C, D) \to \bigoplus_{f \in \mathrm{Irr}(C)} \mathrm{Hom}_C(A \otimes f, D) \otimes_k \mathrm{Hom}_C(B \otimes C, f)$$

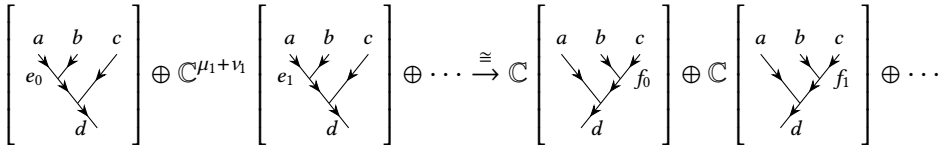
where the entries of this matrix are called the $6j$-symbols after Wigner. However to actually get an element we need to specify 10 parameters[5], they are determined on the basis vectors as such:

$$\iota_{e,\mu}^{ab} \otimes_k \iota_{f,\nu}^{ec} = \sum_{f,\alpha,\beta} [F_d^{abc}]_{(e\mu\nu)}^{(f\alpha\beta)} \iota_{d,\beta}^{af} \otimes_k \iota_{f,\alpha}^{bc} \tag{41}$$

and one should realize that this is really just a change of base matrix.

Graphically, we can represent this as

$$F_d^{abc} : \mathbb{C}^{\mu_0+\nu_0} \left[\right. \quad \left. \right] \oplus \mathbb{C}^{\mu_1+\nu_1} \left[\right. \quad \left. \right] \oplus \cdots \xrightarrow{\cong} \mathbb{C} \left[\right. \quad \left. \right] \oplus \mathbb{C} \left[\right. \quad \left. \right] \oplus \cdots$$



We can do the same construction for the braiding:

$$R_c^{ab} : \mathrm{Hom}(R_a \otimes R_b, R_c) \to \mathrm{Hom}(R_b \otimes R_a, R_c)$$
$$\iota_{c,\mu}^{ba} = \sum_{\nu} [R_c^{ab}]_{\mu}^{\nu} \iota_{c,\nu}^{ab}$$

[5] Wigner (on SU(2)) studied theories that were *multiplicity-free*, so 6 symbols were sufficient.

Example 4.8. Lets look at some examples of semisimple monoidal categories.

1. We know that representations admit semisimple theories and are endowed with a tensor product. These can be viewed as enrichments as the morphisms in Rep() are just linear maps. In the example of SU(2), the basis maps $\iota_{c,\nu}^{ab}$ are equal to the *Clebsch-Gordan* coefficients, in fact these are multiplicity free and thus we can omit the index ν.

CG-coeffs can be indexed here (they are not symbolic only!)

2. In anyon theories however, these bases are fully symbolic. They denote the fusion and splitting spaces of an anyon.

Now we would like the category that has all of these statements combined:

Definition 4.16. A *fusion-category* is a semisimple, fdVect_k-enriched rigid braided monoidal category, where each simple object has only *finitely many* isomorphism classes. Furthermore, we required that I (the monoidal unit) is simple, equivalently $\mathrm{End}_C(I) \cong k$.

Remark 4.1. While each representation category $\mathrm{Rep}(G)$ forms a fusion category for finite G, this is not the case for infinite G such as SU(2) or U(1). These violate the condition that there are finitely many simples. We can remedy this by instead substituting a deformation of the underlying Lie-algebra as $U_q(\mathfrak{sl}_2)$. This has various reasons, one of them is that we only wish to consider gapped theories for our topological orders

From the discussion above we can synthesize an alternative but equivalent definition of a fusion category. This is the definition that is provided in the physics literaturs such as in [Kit06] or [Bon07; BSS08].

Definition 4.17 ([Mos25])**.** A *fusion category* can be restated in its vector space form as a finite set of labels (sectors?) L and vector spaces $V_c^{ab} \coloneqq \mathrm{Hom}_C(a \otimes b, c)$ for all $a, b, c \in L$, as well as the isomorphisms

$$F_d^{abc} : \bigoplus_e V_e^{ab} \otimes V_d^{ec} \to \bigoplus_e V_d^{ae} \otimes V_e^{bc}$$

such that the pentagon holds:

$$\begin{array}{ccccc}
 & & \bigoplus_{e,l} V_e^{ab} \otimes V_f^{cd} \otimes V_g^{el} & & \\
 & \nearrow^{id_{V_e^{ab}} \otimes F_g^{ecd}} & & \searrow^{F_g^{abl} \otimes id_{V_l^{cd}}} & \\
\bigoplus_{e,f} V_e^{ab} \otimes V_f^{ec} \otimes V_g^{fd} & & & & \bigoplus_{k,l} V_g^{ak} \otimes V_k^{bl} \otimes V_l^{cd} \\
 & \searrow_{F_f^{abc} \otimes id_{V_g^{fd}}} & & \nearrow_{F_k^{bcd} \otimes id_{V_g^{ak}}} & \\
 & & \bigoplus_{f,h} V_h^{af} \otimes V_f^{bc} \otimes V_g^{hd} \xrightarrow[F_g^{afd} \otimes id_{V_f^{bc}}]{} \bigoplus_{k,h} V_g^{ak} \otimes V_h^{bc} \otimes V_k^{hd} & &
\end{array} \tag{42}$$

The unit objects are given by

So far we have arrived at the structure of a fusion category. While we do not have quantum systems based upon the objects of these categories, we can model some processes through the hom-spaces, and we have $\mathbb{C}$-enrichment for them as well. To model quantum systems, we need them to be Hilbert spaces as well. Clasically this is done by defining a *unitary structure* on the fusion category.

So why can we not jst do a Hilber-Enrichment?

Definition 4.18. Let C be a fusion category over $\mathbb{C}$. A *unitary structure* on C is a family of maps

We could give this as a functor, but this has a weird target.

$$(-)^\dagger : \mathrm{Hom}_C(A, B) \to \mathrm{Hom}_C(B, A)$$

which is the identity on objects and satisfies

1. *(involution)* $(\lambda f + \mu g)^\dagger = \bar{\lambda} f^\dagger + \bar{\mu} g^\dagger$ and $f^{\dagger\dagger} = f$.

2. *(Contravariance)* $(f \circ g)^\dagger = g^\dagger \circ f^\dagger$ and $\mathrm{id}_A^\dagger = \mathrm{id}_A$.
3. *(Positivity)* $f^\dagger \circ f = 0 \implies f = 0$.
4. *(Monoidality)* $(f \otimes g)^\dagger = f^\dagger \otimes g^\dagger$, and the associator and unitors are unitary, e.g. $\alpha^\dagger_{A,B,C} = \alpha^{-1}_{A,B,C}$.
5. *(Duality)* the pivotal isomorphism $j_A : A \to A^{**}$ is unitary, equivalently $(f^\dagger)^* = (f^*)^\dagger$ for all f.

A fusion category together with this structure is called a *unitary fusion category*.

Proposition 4.1. *In a unitary fusion category:*

1. $\mathrm{Hom}_C(A, B)$ *is a finite-dimensional Hilbert space for all* A, B.
2. *The bases can be chosen with* $\pi^{c,\mu}_{ab} = (\iota^{ab}_{c,\mu})^\dagger$, *so each* ι *is an isometry,* Φ *is unitary, and* $\iota\iota^\dagger$ *is an orthogonal projection. The higher fusion-tree bases are orthonormal.*
3. *The F-symbols* F^{abc}_d *are unitary matrices.*
4. *We can dfine an inner product on* C *by*
$$\langle f, f \rangle = \mathrm{tr}(f^\dagger \circ f)$$
5. *Reciprocity* $\mathrm{Hom}(R_c, R_a \otimes R_b) \cong \mathrm{Hom}(R_a^* \otimes R_c, R_b)$ *is unitary up to the factor ???.*
6. *If* C *is braided with braiding* β, *then* $\beta^\dagger_{A,B} = \beta^{-1}_{A,B}$, *so the R-symbols are unitary. In the ribbon case the twist satisfies* $\theta^\dagger_A = \theta^{-1}_A$.

Now there is only one last structure missing before we can model anyons, and this is perhaps the least intuitive of all of these structures. While foundational for the physical interpretations of these theories, it is of little interest to our tensor formalisms and we will not go into depth on this.

Definition 4.19. In a braided fusion category C with $A, B \in C$ define the *S-matrix*
$$S_{AB} \coloneqq \mathrm{tr}(\beta_{B,A} \circ \beta A, B).$$
If S is invertible, we call the category *modular*.

4.2 Decompositions for Categorical Networks

This chapter generalizes the decomposition in [SV12, Equation 27] to both Fusion Categories and the Representation categories of compact groups. Note that in [DH25], such a generalization is given for representations of finite groups, which does not include $SU(2)$. However I do not yet know if we should do so, since I not yet sure how to argue the F moves here. If we instead take the deformation $U_q(sl_2)$ we get finiteness (and fusion cats?), unfortunately $U_q(sl_2)$ is not a group

Definition 4.20. Recall that in a semisimple category C we can express every object S as a finite direct sum over isomorphism classes
$$S \cong \bigoplus_i S_i^{\oplus n_i}$$
and we also work purely over $\mathbb{C}$ here. We now wish to mirror *isotypic decomposition*definition 2.7 within categories, i.e. we wish to categorify this construction. Let V be a vector space of dimension n, i.e. choose a basis $v_1, \ldots, v_n$ and define
$$V \otimes C \coloneqq \bigoplus_{i=1}^n C$$
and we call V the *multiplicity space* of the object $C \in C$.

Furthremore, we can assert that $- \otimes - : \mathrm{fdVect}_k \times C \to C$ is a functor.

Lemma 4.4. *Let C be a k-linear category and let D, E be finite-dimensional k-vector spaces. For any objects $R, S \in C$, there is a natural isomorphism*

$$\mathrm{Hom}_C(D \otimes R, E \otimes S) \cong \mathrm{Hom}_k(D, E) \otimes \mathrm{Hom}_C(R, S).$$

Proof. Let $m = \dim_k D$ and $n = \dim_k E$. Proof by isomorphism chains:

Redo this over a universal property, this is not natural.

$$\begin{aligned}
\mathrm{Hom}_C(D \otimes R, E \otimes S) &\cong \mathrm{Hom}_C(R^{\oplus m}, S^{\oplus n}) && \text{by ??} \\
&\cong \bigoplus_{i=1}^{m} \bigoplus_{j=1}^{n} \mathrm{Hom}_C(R, S) && \text{by additivity of } \mathrm{Hom}_C \\
&\cong k^{mn} \otimes \mathrm{Hom}_C(R, S) && \text{by ??} \\
&\cong \mathrm{Hom}_k(D, E) \otimes \mathrm{Hom}_C(R, S) && \text{since } \mathrm{Hom}_k(D, E) \cong k^{mn}
\end{aligned}$$

(The last as s) □

Theorem 4.1 (Decomposition of Semisimple Categories)**.** *Let C be a semisimple, monoidal, k-linear category whose tensor product is k-linear in each variable, with simple objects $\{R_a\}_{a \in L}$ and let V_i, W_i be a collection of (not necessarily simple) objects in C. Every morphism*

$$T \in \mathrm{Hom}_C\left(\bigotimes_{i=1}^{n} V_i, \bigotimes_{i=1}^{m} W_i\right)$$

admit a decomposition as

$$T \cong \bigoplus_{\hat{a}, \hat{b}} \mathrm{Hom}_k(D_{\hat{a}}, E_{\hat{b}}) \otimes_k \mathrm{Hom}_C(R_{\hat{a}}, R_{\hat{b}}),$$

where $\hat{a} = (a_1, \ldots, a_n)$ and P is a k-vector space of dimension ???

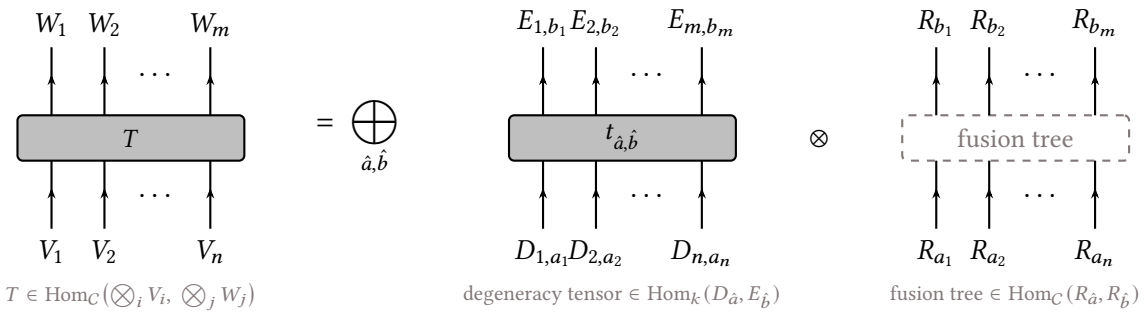
Proof.

$$\begin{aligned}
&\mathrm{Hom}_C\left(\bigotimes_i V_i, \bigotimes_j W_j\right) \\
&\quad\cong \mathrm{Hom}_C\left(\bigotimes_i \left(\bigoplus_{a_i} D_{i,a_i} \otimes R_{a_i}\right), \bigotimes_j \left(\bigoplus_{b_j} E_{j,b_j} \otimes R_{b_j}\right)\right) && (??) \\
&\quad\cong \mathrm{Hom}_C\left(\bigoplus_{\hat{a}} D_{\hat{a}} \otimes R_{\hat{a}}, \bigoplus_{\hat{b}} E_{\hat{b}} \otimes R_{\hat{b}}\right) && (\text{linearity of } \otimes) \\
&\quad\cong \bigoplus_{\hat{a}, \hat{b}} \mathrm{Hom}_C\left(D_{\hat{a}} \otimes R_{\hat{a}}, E_{\hat{b}} \otimes R_{\hat{b}}\right) && (\text{additivity of } \mathrm{Hom}_C) \\
&\quad\cong \bigoplus_{\hat{a}, \hat{b}} \mathrm{Hom}\left(D_{\hat{a}}, E_{\hat{b}}\right) \otimes \mathrm{Hom}_C\left(R_{\hat{a}}, R_{\hat{b}}\right) && (\text{Lemma 4.4})
\end{aligned}$$

In a basis expansion this then looks like

$$\cong \bigoplus_{\hat{a}, \hat{b}} \sum_{e, f, \mu, \nu} \underbrace{t^{\hat{a}, \hat{b}}_{\vec{e}, \vec{\mu}, \vec{f}, \vec{\nu}}}_{\mathrm{Hom}\left(D_{\hat{a}}, E_{\hat{b}}\right)} \otimes_k \left(\pi^{\hat{a}}_{c, \vec{e}, \vec{\mu}} \circ \iota^{\hat{b}}_{c, \vec{f}, \vec{\nu}}\right)$$

□



Corollary 4.1 (Operators are block matrices [KZ22])**.** *thm header. An operator*

$$Hom(V,W) \cong Hom_C\Big(\bigoplus_a R_a^{\oplus d_a}, \bigoplus_b R_b^{\oplus e_b}\Big) \cong \bigoplus_{a,b} Hom(D_a, E_b) \otimes Hom_C(R_a, R_b)$$

and by Schurs lemma this is just

$$\bigoplus_{a \in Irr(V) \cap Irr(W)} Hom(D_a, E_a) \cong \bigoplus_{a \in Irr(V) \cap Irr(W)} Mat_{d_a \times e_a}(k)$$

Corollary 4.2. *Let $Q_{\mu,\eta}$ be a basis of $Hom_C\big(\bigotimes_{\hat{a}} R_{\hat{a}}, \bigotimes_{\hat{b}} R_{\hat{b}}\big)$ where μ denotes the multiplicities and η the internal fusions. In a* multiplicity-free *Category we will omit μ as a label.*

Example 4.9. Easier. However, now assign $V_{\frac{1}{2}} \otimes V_{\frac{1}{2}}$ to each leg in the example above. Each leg then becomes $V_0 \oplus V_1 \cong \mathbb{C} \otimes V_0 \oplus \mathbb{C} \otimes V_1$. Then we can start with the decomposition.

$$\begin{aligned}
&\mathrm{Hom}_C\left(\bigoplus_{a,b\in(0,1)} (\mathbb{C}\otimes\mathbb{C}) \otimes V_a \otimes V_b \bigoplus_{b\in(0,1)} \mathbb{C}\otimes V_b\right) \\
&= \bigoplus_{a,b,c\in(0,1)} \mathrm{Hom}_C((\mathbb{C}\otimes\mathbb{C}) \otimes (V_a \otimes V_b), \mathbb{C}\otimes V_c) \\
&= \bigoplus_{a,b,c\in(0,1)} \mathrm{Hom}(\mathbb{C}\otimes\mathbb{C},\mathbb{C}) \otimes \mathrm{Hom}_{\mathrm{Rep}(SU(2))}(V_a\otimes V_b, V_c)
\end{aligned}$$

Example 4.10 (XXZ-Hamiltonian)**.** Consider a 4 leg tensor in $\mathrm{End}(V_{1/2} \otimes V_{1/2})$, this decomposes as

$$A$$

Example 4.11. Lets look at a tensor with in legs $(\tau \otimes \tau)$ and τ and out legs τ.

$$\begin{aligned}
&Hom_C(((k\otimes 1)\oplus(k\otimes\tau))\otimes(k\otimes\tau), k\otimes\tau) = Hom_C() \\
&\mathrm{Hom}_C\big(((k\otimes 1)\oplus(k\otimes\tau))\otimes(k\otimes\tau),\ k\otimes\tau\big) \\
&\cong \mathrm{Hom}_C\big((k\otimes 1\otimes k\otimes\tau)\oplus(k\otimes\tau\otimes k\otimes\tau),\ k\otimes\tau\big) \\
&\cong \mathrm{Hom}_C(k\otimes 1\otimes k\otimes\tau,\ k\otimes\tau)\oplus\mathrm{Hom}_C(k\otimes\tau\otimes(k\otimes\tau),\ k\otimes\tau) \\
&\cong \mathrm{Hom}(k\otimes k,k)\otimes\mathrm{Hom}_C(1\otimes\tau,\tau)\oplus\mathrm{Hom}(k\otimes k,k)\otimes\mathrm{Hom}_C(\tau\otimes\tau,\tau).
\end{aligned}$$

Example 4.12 (Anyonic matrix operator)**.** Finally, assume we have a $4 \longrightarrow 4$ anyon operator with one leg on each side, representing the space $\mathrm{Hom}_{\mathrm{Fib}}((\tau\otimes\tau\otimes\tau\otimes\tau),(\tau\otimes\tau\otimes\tau\otimes\tau))$. Here we consider it as an operator. The full decomposition is $2 + 3\tau$. Hence we obtain

$$\begin{aligned}
&\mathrm{Hom}_{\mathrm{Fib}}\big((\mathbb{C}^2\otimes 1)\oplus(\mathbb{C}^3\otimes\tau),(\mathbb{C}^2\otimes 1)\oplus(\mathbb{C}^3\otimes\tau)\big)\,. \\
&= \bigoplus_{a,b\in(1,\tau)} \mathrm{Hom}_C(\mathbb{C}^{m_a}\otimes a,\ \mathbb{C}^{m_b}\otimes b) = \bigoplus_{a,b\in(1,\tau)} \mathrm{Hom}(\mathbb{C}^{m_a},\mathbb{C}^{m_b})\otimes Hom_C(a,b)
\end{aligned}$$

where

$$m_1 = 2, \qquad m_\tau = 3.$$

For this case, Schurs lemma applies. So we have

$$= Hom(k^2,k^2)\oplus Hom(k^3,k^3)$$

which is just a block matrix with indexing $(1_1, 1_2, \tau_1, \tau_2, \tau_3)$.

4.2.1 Implementing Fusion Trees

Before we begin to build categorical tensors, we will show how to implement a fusion-tree. Primarily, since a tensors is just an arbitrary morphisms in $\mathrm{Hom}_C(a \otimes b \otimes c \cdots, x \otimes y \otimes \cdots)$ we will store its domain and its codomain as two lists of charges a, b, c : `charge`. Specifics here would allow `int` for $\mathbb{Z}_2$ and `float` for SU(2).

For the fusion trees, the most basic building blocks are the trivalent verteces $\iota^{ab}_{c,\mu}$ and $\pi^{c,\mu}_{ab}$. A larger tree is then stored as an array of these vertices, so we have a correspondence

$$[\iota^{ab}_{c,\mu}, \iota^{cd}_{e,\nu}, \iota^{ef}_{g,\omega}, \ldots, \pi^{z,\delta}_{xy}] \equiv \pi^{z,\delta}_{xy} \circ \cdots \circ (\iota^{ef}_{g,\omega} \otimes \cdots \otimes \mathrm{id}_w) \circ (\iota^{cd}_{e,\nu} \otimes \mathrm{id}_f \otimes \cdots \otimes \mathrm{id}_w) \circ (\iota^{ab}_{c,\mu} \otimes \mathrm{id}_d \otimes \mathrm{id}_f \otimes \cdots \otimes \mathrm{id}_w)$$

Here w is the last input leg, and z is the output of the last fusion. We also represent morphisms in $\mathrm{Hom}_C(a, b)$ (so single lines) as trivalent vertices by its unitors, so as $\mathrm{Hom}_C(a \otimes I, b) \cong \mathrm{Hom}_C(a, b \otimes I)$ and this extra vertex is not captured by the shape unless I is the only object in the codomain. Also note that the ordering of these nodes does not matter in general.

Keep this in mind for the implementation.

example

However we only save this entire array once, and we leave in the labels as variables. An actual tensor will have charges associated to the labels in its domain, so for example a will carry and associated SU(2) charge of $\frac{1}{2} \oplus \frac{3}{2}$. Furthermore, we are not given the charges for the intermediate indeces that appear in the fusion tree, and these need to be deteremined from the domain and codomain, thus we implement the procedure `tree.charge_sectors()`. A charge sector here is simply a mapping between each label and one of its possible charges. All the charge sectors are then a list of these mappings. Consider the trees (multiplicity free)

$$(\pi^{ab}_c \otimes \mathrm{id}_d) \circ (\pi^{cd}_e)$$

where the domain is $[a, b, c] \cong [\tau, \tau, \tau]$ and the codomain is $[e] \cong [\tau]$. It is now only left to determine the factor c, and we can show there are two valid paths, one with $c = \tau$ and one with $c = I$. Thus the determined charge sectors are

$$\big[[\tau, \tau, \tau, I, \tau], [\tau, \tau, \tau, \tau, \tau]\big].$$

If we had instead chosen $[a, b, c] \cong [\tau \oplus I, \tau \oplus I, \tau \oplus I]$ and $[e] \cong [I \oplus \tau]$, we get charge sectors

$$\big[[I, I, I, I, I], [I, I, \tau, I, \tau], [I, \tau, I, \tau, \tau], [\tau, I, I, \tau, \tau], [I, I, \tau, I, \tau], [\tau, \tau, I, \tau, \tau], [\tau, \tau, I, I, I], [\tau, \tau, \tau, I, \tau], [\tau, \tau, \tau, \tau, \tau]\big]$$

so it suffices to only save the *geometry* of the tree, and all of its charge sectors.

Now we wish to apply all the reshape operations to these trees. We know that F-moves create a superposition of trees when applied, but we only care about the geometry for now. Also recall that F-moves act locally on a tree, for the geometry this entails that a local rearrangement is enough. Recall that an fmove acts on the local space $\mathrm{Hom}(a \otimes b \otimes c, d)$ and thus in the array representation on spaces such as $[\iota^{ab}_c, \iota^{e}_{cd}]$. There are also F-moves defined on non-simple trees that are of the form $[\iota^{a,\mu}_{bc}, \pi^{cd}_{e,\nu}]$ also admit an F-move, in this case to the target $[]$. This convention is sometimes also chosen as the default definition of the F-move, and is denotes as $[F^{ab}_{cd}]^{f}_{e}$, especially in the context of spin chains [Bon07]

nicer!

Applying permutations to the trees is also only possible whenever therere is a clear target to apply this to, so the two charges need to be a adjacent (i.e. they are a part of the same trivalent vertex).

check the sources im referring to.

Now, if we actually want to do this on the level of tensors, we need to consider their coefficients in the matrix parts.

For each pair $(\hat{a}, \hat{b})$, the space $\mathrm{Hom}_C(R_{\hat{a}}, R_{\hat{b}})$ is spanned by tree arrays: a fusion tree $R_{\hat{a}} \longrightarrow R_c$ followed by a splitting tree $R_c \longrightarrow R_{\hat{b}}$,

$$\mathbb{T}_s = [\iota \ldots \iota,\ \pi \ldots \pi],$$

where s collects the coupled charge c, all internal charges and all vertex multiplicities. By Theorem 4.1 every tensor is then

$$T = \sum_s t_s \otimes \mathbb{T}_s, \qquad t_s \in \mathrm{Hom}_k(D_{\hat{a}}, E_{\hat{b}}),$$

so it is stored as one degeneracy tensor t_s per charge sector. These are the matrix parts: one axis per open leg, of length $\dim D_{i,a_i}$ or $\dim E_{j,b_j}$.

Since $\otimes$ is k-linear, $(D \otimes R) \otimes (D' \otimes R') \cong (D \otimes_k D') \otimes (R \otimes R')$. The associator and braiding are natural, so on these components they split as

$$\alpha_{D\otimes R,\, D'\otimes R',\, D''\otimes R''} = \alpha^{\mathrm{Vect}} \otimes \alpha_{R,R',R''}, \qquad \sigma_{D\otimes R,\, D'\otimes R'} = \tau_{D,D'} \otimes \sigma_{R,R'},$$

with τ the flip of vector spaces. Hence a move acts on the tree factor only, and never touches the matrix part except for permuting its axes under a braiding.

Lets give the braiding tensor explicitely. It is an element in $\mathrm{Hom}_C(a \otimes b, b \otimes a)$

$$\sum_{c,\mu} [R_c^{ab}]_\mu^\nu \otimes (\pi_{c,\nu}^{ba} \circ \iota_{ab}^{c,\mu})$$

Assume we are working over Fib. This is multiplicity free, and we can write the braiding as

$$c_{\tau,\tau} = e^{-4\pi i/5}(\pi_1^{\tau\tau} \circ \iota_{\tau\tau}^1) + e^{3\pi i/5}(\pi_1^{\tau\tau} \circ \iota_{\tau\tau}^1)$$

Should we consider the domain and the codomain as fused, we can also write it as

$$\bigoplus_{c \in 1,\tau} R_c^{\tau\tau} \otimes \mathrm{id}_c$$

5

Anyons II: Spin Chains & Tensor Networks

In this chapter, we give the applications of the categorical formalism to anyonic tensor networks. We also quickly review the simpler *spin-chain formalism* in Section 5.1.

One of the primary motivations for the study of anyons are their universality for quantum computation. In **??** we provide an interpretation of how both normal braiding circuits and a variation in terms of *weaves* can be simulated by the tensor network theory.

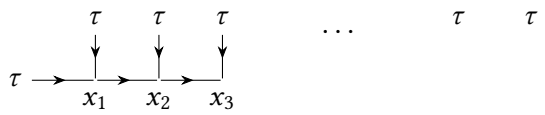
Literature 5.1. [PS15; Pfe+10; Pfe12; Pfe14; Sin+14]

5.1 1D Lattice Models & The Golden Chain

What is a 2 + 1D critical point? This is relevant for the distinction between 1D and 2D models. Is a 2D model the same as a fusion surface model?

TODO: Write an outlook section here on work like [LDV24] [Eck25] etc. This seems like a direct generalization of our approach.

In [Fei+07], anyonic chains, specifically in the Fib model, are introduced as a 1D lattice of n τ anyons, defined as the fusion space thereof.



Here, the state is fully characterized by its intermediate indeces $x_1 \dots x_{n-1}$, so we can choose a basis $|x_1 x_2 \cdots x_{n-1}\rangle$.

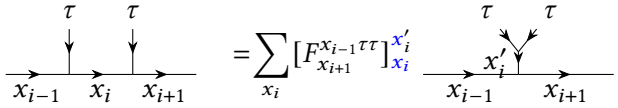
Similar to the Heisenberg-chain[1], the golden chain introduces a Hamiltonian that assigns -1 to the interaction $\tau \otimes \tau = 1$ and 0 to the interaction $\tau \otimes \tau = \tau$.

We can decompose the whole space $\mathrm{Hom}(\tau^{\otimes n+1}, \tau)$ into the local states

$$\mathrm{Hom}(\tau^{\otimes n+1}, \tau) = \bigoplus_{\substack{x_1, x_2, \dots, x_{n-1} \\ \text{all admissible paths}}} \mathrm{Hom}(\tau \otimes \tau, x_1) \otimes \mathrm{Hom}(x_1 \otimes \tau, x_2) \otimes \cdots \otimes Hom(x_{n-1} \otimes \tau, \tau) \tag{43}$$

with the admissability condition that there are not adjacents 1 in the sequence of x_is.

This is evaluated as a 3 local hamiltonian, as we can decompose the spin chain into local parts where two τ anyons are adjacent (check how they call it in the paper.).



$\in \mathrm{Hom}(x_{i-1} \otimes \tau, x_i)$
$\otimes \mathrm{Hom}(x_i \otimes \tau, x_{i+1})$

$\in \mathrm{Hom}(x_{i-1} \otimes x'_i, x_{i+1})$
$\otimes \mathrm{Hom}(\tau \otimes \tau, x'_i)$

Now we can apply the interaction based on the intermediate state[2] x'_i and apply another F-move back to

[1] Note that $\frac{1}{2} \otimes \frac{1}{2} = 0 \oplus 1$ in the Heisenberg SU(2) model.

[2] Technically we need a projector here read up again.

the chain basis:

$$H_{[i,i+1]} : |x_{i-1}x_i x_{i+1}\rangle \to |x_{i-1}x_i'' x_{i+1}\rangle \coloneqq \sum_{x_i' \in \{1,\tau\}} [F_{x_{i+1}}^{x_{i-1}\tau\tau}]_{x_i}^{x_i'} E_{x_i'} [F_{x_{i+1}}^{x_{i-1}\tau\tau}]_{x_{i'}}^{x_{i''}}$$

In the golden chain, we set $E_1 = -1$ and $E_\tau = 0$. Then (5.1) reduces to

$$H_{[i,i+1]} = -[F_{x_{i+1}}^{x_{i-1}\tau\tau}]_{x_i}^{1} [F_{x_{i+1}}^{x_{i-1}\tau\tau}]_{1}^{x_i''}$$

and we can write this three local hamiltonian as a matrix

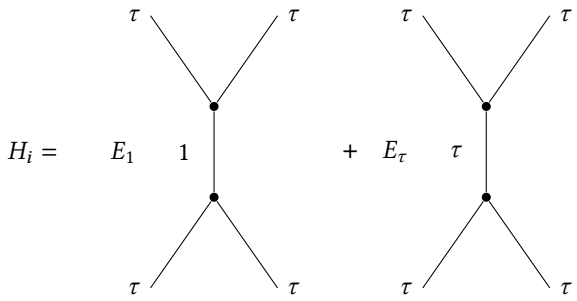
$$H_i = -J \begin{pmatrix} 1 & & & & \\ & 0 & & & \\ & & 0 & & \\ & & & \varphi^{-2} & \varphi^{-\frac{3}{2}} \\ & & & \varphi^{-\frac{3}{2}} & \varphi^{-1} \end{pmatrix}$$

This method is reviewed and extended in [Tre+08], with longer ranger interaction. It will become apparent that the construction given in these papers is a manualization of the tensor network constructions we will build here. Generalizations to arbitrary models are given in [Gil+13] as well as [BG17]. Generalized spin chains and dualities using module categories are given in [Eck25].

The tensor network formalism is handled in [Pfe+10], [Pfe12] and [Aye17], and in terms of categories this is done in [Unf24] and [DH25].

If we want to write this as a tensor contraction, we first define a hamiltonian in a similar fashion to chapter 1.

This takes the form of an operator in $Hom(\tau \otimes \tau, \tau \otimes \tau)$ as



(Note this is allowed due to $\mathbb{C}$-linearity of Fib.)

In this case we can view the golden chain approach chosen above as a manualization of the tensor network process.

We can also write H_i in its operator form matrix as

$$H_i \in \bigoplus_{a \in 1,\tau} Hom(\mathbb{C}, \mathbb{C})$$

as

$$H_i = -J \begin{array}{c} \\ 1 \\ \tau \end{array} \begin{array}{c} \begin{array}{cc} 1 & \tau \end{array} \\ \begin{pmatrix} E_1 & 0 \\ 0 & E_\tau \end{pmatrix} \end{array}$$

5.2 Topological Quantum Computation

Recall the basis theory of quantum computing. Here the basic computational unit is the *qubit*, spanned by unit vectors $|0\rangle = (1,0)^T$ and $|1\rangle = (0,1)^T$, these may be realized by a spin as in Chapter 2. We can also combine these by their tensor product $(\mathbb{C}^2)^{\otimes n}$ to form a n-qubit *register*. The computations themselves are linear function from $\mathbb{C}^{2^n} \to \mathbb{C}^{2^n}$ that are also unitary.

Some important single=qubit maps are the X, Y and Z gates. The *Hadamard-gate* H produces bell states, and the T-gate yields a phase on ???.

$$\mathrm{X} = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \qquad \mathrm{Y} = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix} \qquad \mathrm{Z} = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} \qquad \mathrm{H} = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix} \qquad \mathrm{T} = \begin{pmatrix} 1 & 0 \\ 0 & e^{i\pi/4} \end{pmatrix} \tag{44}$$

We can apply these gates sequentially by composition and in parallel by the tensor product. There are also gates on two qubits that are not realized as tensor products, these are called *multi-qubit* gates. A special class of those is called *controlled gates*, which act on the composite $|0x\rangle$ by identity and on the ket $|1x\rangle$ by a single qubit gate in the second register. One of the important gates is the *CNOT* gate, or controlled X gate which maps $|ab\rangle$ to $|a, a \oplus b\rangle$. Every quantum gate $U \in \mathrm{U}(2^n)$ can then be represented as a unitary in $\mathrm{U}(2^{n+1})$ of the form

These are not SU?

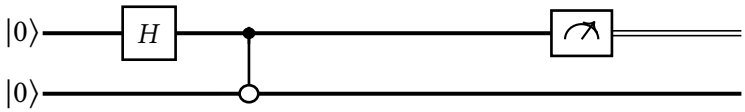
$$\mathrm{C}\,U = \begin{pmatrix} \mathrm{I}_{2\times 2} & 0 \\ 0 & U \end{pmatrix} \qquad \mathrm{CNOT} = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{pmatrix}$$

The last operation on quantum circuits are *measurements*. A measurement operator is a mapping from $\mathbb{C}^2 \to \{0, 1\}$. Measurements occur with respect to a certain basis, here as a represetative choose $\{|v_1\rangle, |v_2\rangle\}$. If we take an arbitrary superposition of a state $|\Phi\rangle = \alpha\,|v_1\rangle + \beta\,|v_2\rangle$ we get the result 0 with probability $|\alpha|^2$ and the result 1 with probability $|\beta|^2$. Note that $\alpha, \beta \in \mathbb{C}$. Finally, it is possible for states to differ by a global scalar $e^{i\theta} \in \mathrm{U}(1)$. These global phases do not affect the probabilities, therefore we choose to measure distances by the *trace norm*

$$||U - V|| = \sqrt{1 - \frac{|\,Tr(U^\dagger V)|}{2}}$$

which ignores the phase.

A collection of these gates is called a *quantum circuit*. While quantum circuits are a subset of tensor networks (over $\mathrm{fdVect}_\mathrm{k}$) [MS08], they admit their own string calculus, originally introduced by Deutsch [Deu89].



This string calculus is a hardware agnostic language for quantum circuits. Realizing quantum computation on a physical architecture is another problem and has its own architectures, which can be compiled from the circuit diagrams. Furthermore, physical computers may not be able to accommodate arbitrary gates in $\mathrm{SU}(2)$ but instead support only a finite and perhaps limited set of gates. If we can take an arbitrary but finite combination of these gates, and we can approximate any unitary $U \in \mathrm{SU}(2)$ up to precision ϵ, we call such a gate set *universal*[3].

Projective Space!

Theorem 5.1 (Solovay-Kitaev [Kit97]). *Given a finite $G \subseteq \mathrm{SU}(2)$, such that $\langle G\rangle$ is dense in $\mathrm{SU}(2)$, then we can approximate every $U \in \mathrm{SU}(2)$ by a sequence of elements S from G such that $||U - S|| \leq \epsilon$. The length of this sequence is within $O\left(\log^{3.97}(1/\epsilon)\right)$. Furthermore, this yields an efficient algorithm producing such an approximation in $O\left(\log^{2.71}(1/\epsilon)\right)$ [DN05].*

There are more efficient bounds now i think. Also this may not require inverses. [Kup25]

There is also a complexity perspective on quantum computation. Deutsch [Deu85] introduced the *quantum Turing machine* as a model. Bernstein and Vazirani [BV97] adapted this model as a formal complexity class called BQP.

Motivation by Feynman and Manin

Now we move towards defining topological quantum computation. But first some history:

In a first instance, Witten [Wit89] showed that the knot invariants in $\mathrm{SU}(2)_k$ TQFTs at the roots of unity correspond to special values of the *Jones polynomial* [Jon85]. While we will not define this problem, it is know that this is a BQP-complete problem [JVW90] at roots of unity. A complete analysis was given later by Aharonov, Jones, and Landau [AJL06], the resulting algorithm is now known as the *AJL*-algorithm.

[3] We cannot achieve exact universality by a finite set as $\mathrm{SU}(2)$ is uncountable.

Topological quantum computation in its protean form was first conjectured by Kitaev [Kit03, Appendix E][4]. Mainly what was given is, that the braidings of anyons can contribute unitary transforms, and that we can measure by fusing together anyons. As this work is based on a physical realization of the toric code [Kit97], the computations by exchanges are fault tolerant . In other words they are topologically protected.

This is not true, Clifford magic gates are not

Similarly to the originally conjectured quantum computing by Feynman to simulated quantum processes, Freedman [Fre98] conjectured a *topological quantum field computer* to simulate TQFTs a year later, based on both the work of Kitaev and Witten. Here it was even considered to be a stronger class of computation than standard quantum computing. By combined work, Freedman, Kitaev, and Wang [FKW02] showed that while TQC is both fault tolerant and at most as powerful as standard quantum computation by way of the Jones polynomial, it does not provide stronger algorithms than BQP. The reverse direction, that it is at least as strong as BQP is shown by Freedman, Larsen, and Wang [FLW00]. The universality given here corresponds only to certain theories such as $\mathrm{SU}(2)_k$ for $k = 3$ and $k \geq 5$.

How do we build such a computer: By the discussion above, we need 3 things:

- A ground state that has high enough dimension to represent a quantum bit, i.e. is $\cong \mathbb{C}^2$.
- A option to time evolve the system by application of unitaries.
- An option to read out the information of the evolved state (*measurement*).

Computing with nonabelian anyons requires a universal gate set built from representations of the *braid-group*:

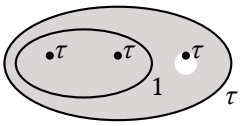
Definition 5.1 (Braid Group (algebraic))**.**

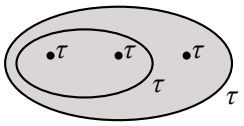
$$Br(\mathbb{R}^2) = \left\langle \begin{array}{c} \sigma_i\sigma_j = \sigma_j\sigma_i, \quad |i-j| \geq 2 \\ \sigma_i\sigma_{i+1}\sigma_i = \sigma_{i+1}\sigma_i\sigma_{i+1}, \quad i \in [1, n-2] \end{array} \right\rangle$$

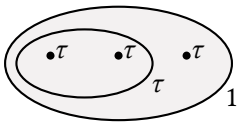
and also the *pure-braid-group*:

Introduce the pure braid group?

By the construction of anyon chains in **??**, we know that the space of 3 anyons fusing to τ is in fact 2-dimensional. This motivates a choice of qubits as the following:

$|0\rangle =$ 

$|1\rangle =$ 

$|NC\rangle =$ 

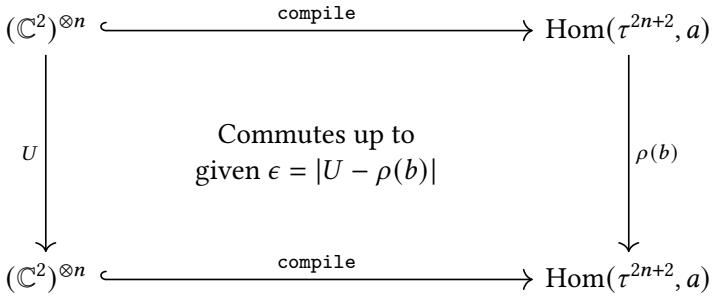
and we say that $\mathbb{C}^2$ is *embedded* in $V^{\tau\tau\tau}$. We could choose a different presentations, where each anyon is drawn from the vacuum as

TODO!

Then we can build multi-qubit circuits by simply stacking these diagrams (inside their own fusion channel). This is what is called a *sparse-encoding*. A dense encoding would correspond to just choosing the smallest space of appropriate dimension that includes one of the $(\mathbb{C}^2)^{\otimes d}$.

The solovay-kitaev theorem guarantees a polylog time and space algorithm for approximating any gate in $\mathrm{SU}(d)$ to a desired accuracy. This, and similar procedures are realized by the functor `compile`. A *topological quantum compiler* essentially only translates the qubits to their Hom space equivalences and translates the qudit-gates to their braid group representations. There are of course special compilation methods, the one we cover here takes first a decomposition of normal gates into CNOTs and then applies injection weaving, so we only need to lower single qubit gates to braids.

[4]The citation reads that this paper was accepted for publication in 2003, however a preprint was released in 1997.



Because this square commutes and `compile` runs in polynomial time, we can in fact assert that the *QFC* can simulate *BQP*.

However, compiling qudits for braids is a *hard* (computationally expensive) and slow process, as the dimension of the embedding is usually a constant factor > 1 larger than the dimension of the desired qudit space.

This process (and these do depend on the topology of the ground state) and the indivuual gates are given in [HG24], [GHM24].

The work of Simon et al. [Sim+06] provides a alternative using the pure braid group. Remember that although we can represent any weave generator as

$$T_{ij} = \sigma_i \sigma_{i+1} \cdots \sigma_{j-1} \sigma_j^2 \sigma_{j-1} \cdots \sigma_{i+1} \sigma_i$$

a weave is effectively realized by moving only one of the particles (strands) around $n-1$ fixed strands. As it turns out, the weave representations are also dense in SU(()2). (cite). This can also solve our problem of effectively compiing controlled gates. Remember the presentation we chose for our qubits: The $|0\rangle$ qubit has an intermediate channel of 1 and the $|1\rangle$ qubit has an intermediate channel of τ. We also know that braiding a 1-type anyon around any other anyon shoulde have no effect. By our construction of fusion we know that the tau anyons in the 1 channel can only produce 1 when fused (i.e. $\tau \otimes \tau = 1$). Should we have a superposition of qubits $\alpha |0\rangle + \beta |1\rangle$ this means we produce a composite anyon of type 0 with probability α^2 and a composite anyon of type τ with probability β^2. Now if we find a weave realizing our controlled gate on the lower qubit, we will have succesfully synthesized a controlled gate. For this, we only need to *inject* the composite anyon in the lower qubit *without* affecting the state of our system. After fusing in our top qubit, we effectively end up with a braiding problem in PBr_n, however since we only want the τ part to have an effect we can reduce this to Br_n. Practically, this means that we need to find a traditional braid $X \in Br_3$ such that the representation of

or doing so invertibly

$$\sigma_3$$

We can use *weaves* to build *injection gates* which in turn realize control gates. Then, by a universality argument of [FLW00] we can approximate any other multiqubit gate, as single qubit gates can be computed in little time after an algorithms[5][6] of [KBS13] based on number theoretic methods (which are already mentioned in [FW07]).

Rewrite this.

Specifically, there is a decomposition of an arbitrary gate in $(\mathbb{C}^2)^{\otimes n}$ into a universal set of `CNOT` and $\mathbb{C}^2$ unitaries. However we must also ensure that the decomposition in into CNOT does not produce nonadjacent control-target pairs, as we can not freely cross the braids over another qubit. This can however also be a problem for standard qubit architectures, which is why we can decompose every unitary circuit into its *linear nearest neighbour* (LNN) form [KMS07; SWD11]. Such decompositions are already implemented in languages such as `Qiskit`[7][Jav+24].

Now all that remains to do is to reexpress a traditional working gate set in the Fib basis of sets.

[5] We provide an implementation of this process in `Haskell` [Rie25]

[6] Although we cannot reach the claimed bound of ϵ but instead reach $\sqrt{\epsilon}$

[7] `Qiskit` documentation: https://quantum.cloud.ibm.com/docs/en/api/qiskit

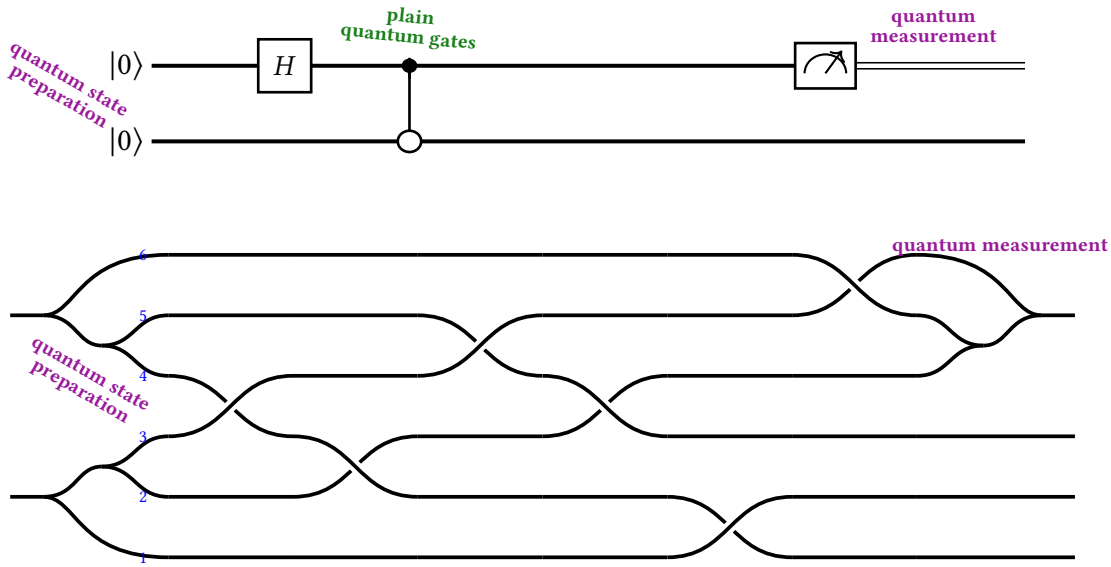


Figure 5.1: A standard 2-qubit quantum circuit and a 2-qubit Fibonacci anyon circuit.

A

Appendix

A.1 Some Group Theory

We recall some of the elementary definitions and results on groups as used in this work, mainly in Chapter 2 and Chapter 3.

Definition A.1. A *group* $(G, \cdot)$ consists of a (possibly infinite) set of elements and a binary op $\cdot : G \times G \to G$ such that

1. $(a \cdot b) \cdot c = a \cdot (b \cdot c)$
2. There exists an $\mathrm{id}_G \in G$ s.t. $\mathrm{id}_G \cdot a = a \cdot \mathrm{id}_G = a$
3. Each $a \in G$ is invertible with $a \cdot a^{-1} = a^{-1}a = \mathrm{id}_G$

Definition A.2. A group $(G, \cdot)$ is called *abelian*, if for all $a, b \in G$ it holds that $a \cdot b = b \cdot a$, its contrapose is called a *non-abelian* group.

Recall that we say a group is *generated* by a set of elements if we can write any other element as the concatenation of the generating set. Here by example:

Definition A.3. The *symmetric group* of n points S_n is given by

$$S_n = \left\langle s_1, \ldots, s_{n-1} \;\middle|\; \begin{array}{c} s_i^2 = id \\ s_i s_{i+1} s_i = s_{i+1} s_i s_{i+1} \\ s_i s_j = s_j s_i, |i-j| \geq 2 \end{array} \right\rangle. \tag{45}$$

Definition A.4 (Representation theory of the symmetric group)**.** For $n \geq 2$ the symmetric group S_n is equipped with two one-dimensional representations, the *trivial representation* where every generator is equal to the identity, and the *sign representation* which equips each generator with its relative permutation parity.

Definition A.5. The *General Linear Group* GLn of degree n is the set of $n \times n$ matrices that are invertible, and its operation $\cdot$ is that of matrix multiplication.

A.2 Some Algebraic Topology

Let spaces X, Y be topological in the following, and let all functions be continuous unless stated otherwise.

Definition A.6. A *homotopy* between two continuous functions $f, g : X \to Y$ is a continuous function

$$H : X \times [0,1] \to Y \text{ such that } H(x,0) = f(x) \text{ and } H(x,1) = g(x) \tag{46}$$

Definition A.7. A *loop* with basepoint x_0 on X is a function

$$\gamma : [0,1] \to X \quad \text{such that} \gamma(0) = \gamma(1) = x_0 \tag{47}$$

Definition A.8 (Bundle)**.** A *vector (Hilbert) bundle* over a space X is a collection of vector (Hilbert) spaces $\{V_x\}_{x\in X}$ so to say a parameterization of vector spaces dependent on a base space. This base space could be any of a manifold, a topological space, a scheme and others. Here we mostly care about differentiable manifolds and topoplogical spaces such as configuration spaces. An element V_x is called the *fiber* at $x \in X$.

Now that we have the spaces parameterized over the base space, we want to encode this dependence, hence the definition of a *connection*.

Definition A.9 (Connection)**.** A *connection* on a bundle $V \longrightarrow X$ assigns to each path $\gamma : [0,1] \to X$ a homomorphism on the paths

$$\hat{\gamma} : V_{\gamma(0)} \to V_{\gamma(1)}.$$

This assignment is also called *parallel transport*. If

A.3 Tensor Decompositions & Algorithms

Definition A.10. The *singular value decomposition* of a complex operator A of shape (m, n) is a decomposition of the form

$$A = U\Sigma V^*$$

where

$$U \in \mathrm{U}(m \times m), \quad \Sigma \in \mathrm{Mat}_{m\times n}(\mathbb{R}) \quad \text{and} \quad V \in \mathrm{U}(n \times n)$$

and Σ is further diagonal, in fact is it made up of the *singular values* of A.

This also enables us to perform a low rank appoximation of an operator by truncating Σ to only keep its r largest singular values.

A.4 Proofs for Chapter 2

This is the proof for statement Equation (14) on page 16.

Lemma A.1. *The XXZ Hamiltonian admits a symmety under $U(\theta) = e^{i\theta S^z_{tot}}$ and diagonalizing with S^z_{tot} suffices.*

Proof. Actually should we just cite a proof or is this of interest? [Has24, § 4.2]. Recall that when two operators commute, we can *simultaneously diagonalize* them [HJ12, Theorem 1.3.12].

$$[H, S^z_{\text{tot}}] = \sum_i^N [\sigma^x_i \sigma^x_{i+1} + \sigma^y_i \sigma^y_{i+1} + \Delta\sigma^z_i \sigma^z_i, \sum_{j=1}^N \sigma^z_j]$$

and since operators on different sites commute, only $j = i$ and $j = i + 1$ contribute. So we ought to write

$$\sum_i^N [\sigma^x_i \sigma^x_{i+1} + \sigma^y_i \sigma^y_{i+1} + \Delta\sigma^z_i \sigma^z_i, \sigma^z_i + \sigma^z_{i+1}]$$

and now using the identity $[AB, C] = A[B, C] + [A, C]B$:

$$[\sigma^x_i \sigma^x_{i+1}, \sigma^z_i + \sigma^z_{i+1}] = -2i(\sigma^y_i \sigma^x_{i+1} + \sigma^x_i \sigma^y_{i+1})$$

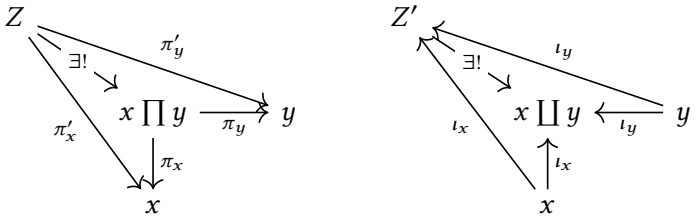
$$[\sigma^y_i \sigma^y_{i+1}, \sigma^z_i + \sigma^z_{i+1}] = 2i(\sigma^x_i \sigma^y_{i+1} + \sigma^y_i \sigma^x_{i+1})$$

which cancel out. Furtermore, the z spins commute as well. Thus the whole sum is 0 and we are done. □

A.5 Supplementary categorical constructions for Chapter 4

Note on universal properties: When dealing with categories, we are fundamentally dealing with abstractions that behave alike to a known construction. These behaviours are made precise by the use of a *universal property*. We illustrate this with the definition of a (co)product:

Definition A.11. Let C be a category and $x, y \in ob(C)$. Their *product* is an object $x \prod y$ in C with morphisms $\pi_x : x \prod y \to x$ and $\pi_y : x \prod y \to y$. Their *coproduct* is defined as the object $x \coprod y$ together with the maps $\iota_x : x \to x \coprod y$ and $\iota_y : y \to x \coprod y$. Both of these definitions are universal with their properties.



This should all go in the anyon chapter? or later on into categories. Lets show how to build the splitting of a fusion tree

$$\mathrm{Hom}(a \otimes b \otimes d, e) = \bigoplus_c \mathrm{Hom}(a \otimes b, c) \otimes \mathrm{Hom}(c \otimes d, e)$$

First we ought to give an mapping

$$\Theta_c : \mathrm{Hom}(a \otimes b, c) \otimes_k \mathrm{Hom}(c \otimes d, e) \to \mathrm{Hom}(a \otimes b \otimes d, e)$$

by

$$\mu \otimes_k \nu \mapsto (\mu \otimes \mathrm{id}_d) \circ \nu$$

A.5.1 TQFT

Hmm, where does this go?

A TQFT is defined by a finite set of labels L, and a rule that assigns each compact cpa-space a vector space and each map between them (a cobordism) a linea map between spaces. A TQFT is thus a functor between the category $(Bord, \sqcup, \emptyset) \to$ Vect.

We can interpret this as assigning each ??? bordism its space of states under the functor and each cobordism its space of operators.

A.6 Overview of existing implementations

Table A.1: Overview of tensor network libraries and their support for symmetries and features relevant toward symmetry. If a specific symmetry is given, the software is restricted to that symmetry. A previous list of existing implementations in 2018 is also given in [Hub18]. Note that implementations such as the original MATLAB implementation by Singh, Pfeifer, and Vidal [SPV10] are not publicly available or named.

	Metadata			Features							
Library/Backend	**Year**	**Language**	**License**	**Abelian** §2	**Non-Abelian** §2	**Categorical** §4	**Product Symmetries** §??	**Fermionic Number**	**GPU**	**Autodiff**	**Software & Publication**
MPToolkit (L)[1]	2002	C++	GPL-3.0	✓	✓	✗	✓	?	✗	✗	[26b; MG02]
QSpace (L)	2012	Matlab	Apache-2.0	✓	✓	✗	?	?	?	?	[Wei24; Wei12a]
CheMPS2 (L)	2013	C++	GPL-2.0	✓	✓	Ferm	✗	✓	?	?	[Wou+14]
Uni10 (L)	2015	Python, C++	?	?	?	?	?	?	?	?	[KHC15]
SyTen (L)	2017	C++	✗[7]	?	?	?	?	?	?	?	[Hub+; Hub17]
SymLibrary (R)	2019	Matlab	MIT	✓	✓	$SU(2)_k$	✓	✓	?	?	[McM19; MSB20]
TensorNetwork (L)	2019	Python	?	✓	✗	✗	✗	✗	?	?	[Rob+19]
TeNPy (L)	2019	Python	Apache-2.0	✓	✗	✗	?	?	?	?	[Hau+24]
ITensor (L)	2022	Julia	Apache-2.0	✓	✓	?	?	?	?	?	[FWS22]
Block2 (L)	2023	C++	GPL-3.0	✓	SU(2)	✗	✗	?	?	?	[26a; Zha+23]
ChemTensor (L)	2023	C	Apache-2.0	✓	SU(2)	✗	✗	?	✗	✗	[Men26]
grassmanntn (R)	2023	Python	Apache-2.0			sVect[3]	?	✓	?	?	[Yos26]
QuantumTea (L)	2024	Python	?	✓	?	✗	?	?	?	?	[Bac+26]
Cyten[2] (L)(B)	2024	Python & C++	Apache-2.0	✓	✓	✓	✓	✓	✗	?	[26c; Unf24]
YASTN	2025	Python	?	✓	✗	✗	✓	?	✓	✓	[Ram+25]
TensorKit.jl[5] (L)	2025	Julia	MIT	✓	✓	✓	✓	?	?	✓	[DH25]
Cytnx[4]	2024	C++	?	✓	✗	✗	✓	?	✓	?	[Wu+25]
TeMFpy[8] (B)	2025	Python	MIT	✗	✗	✗	✗	✓	✗	✗	[HS25]
abeliantensors	2026	Python	MIT	$U(1), \mathbb{Z}_n$	✗	✗	✗	✗	✗	✗	[Hau26]
symmray[6] (B)	2025	Python	?	✓	✗	sVect	✓	✓	?	✓	[Gao+25; Gra26b]
Telum.jl	2026	Julia	?	✓	✓	$Ferm$	?	✓	?	?	[Lee26]
Su2tn (R)	2025	Python	None	✗	SU(2)	✗	✗	7	?	?	[Fis25b; Fis25a]

[1] (L) indicates that this is a standalone Library, (B) indicates that this is a backend for another library and (R) indicates that this is a research script and likely not maintained.
[2] Cyten originated as a branch of TenPy and was later split into its own repository [Tenpy pull 309].
[3] This supports a variant called *Grassman Tensor Networks*.
[4] Ideas from Cytnx were eventually adapted into Cyten.
[5] Part of the `QuantumKit.jl` ecosystem.
[6] symmray functions as a backed for the tensor network software quimb [Gra18; Gra26a]
[7] Syten is a closed source program available among collaborations.
[8] TeMFpy is built on top of Tenpy.

Bibliography

[26a] *Block-Hczhai/Block2-Preview*. block-hczhai, Sept. 17, 2026. URL: https://github.com/block-hczhai/block2-preview (visited on 09/17/2026) (cit. on p. 53).

[26b] *Mptoolkit/Mptoolkit*. mptoolkit, Sept. 7, 2026. URL: https://github.com/mptoolkit/mptoolkit (visited on 09/17/2026) (cit. on p. 53).

[26c] *Tenpy/Cyten*. Tensor Network Python Package, Sept. 16, 2026. URL: https://github.com/tenpy/cyten (visited on 09/17/2026) (cit. on pp. 15, 53).

[Abr12] Samson Abramsky. *No-Cloning In Categorical Quantum Mechanics*. Mar. 19, 2012. DOI: 10.48550/arXiv.0910.2401. arXiv: 0910.2401 [quant-ph]. (Visited on 07/26/2026). Pre-published (cit. on p. 5).

[AJL06] Dorit Aharonov, Vaughan Jones, and Zeph Landau. *A Polynomial Quantum Algorithm for Approximating the Jones Polynomial*. Apr. 10, 2006. DOI: 10.48550/arXiv.quant-ph/0511096. arXiv: quant-ph/0511096. (Visited on 10/15/2026). Pre-published (cit. on p. 46).

[AK23] Fatimah Rita Ahmadi and Aleks Kissinger. "The ZX-Calculus as a Language for Topological Quantum Computation". In: *Journal of Physics A: Mathematical and Theoretical* 56.41 (Oct. 13, 2023), p. 415301. DOI: 10.1088/1751-8121/acef7e. arXiv: 2211.03855 [quant-ph]. (Visited on 11/21/2024) (cit. on p. 5).

[Alv12] G. Alvarez. "Implementation of the SU(2) Hamiltonian Symmetry for the DMRG Algorithm". In: *Computer Physics Communications* 183.10 (Oct. 2012), pp. 2226–2232. DOI: 10.1016/j.cpc.2012.04.025. arXiv: 1003.1919 [cond-mat.str-el]. (Visited on 10/09/2026) (cit. on p. 15).

[Art25] Emil Artin. "Theorie der Zöpfe". In: *Abhandlungen aus dem Mathematischen Seminar der Universität Hamburg* 4.1 (Dec. 1, 1925), pp. 47–72. DOI: 10.1007/BF02950718. (Visited on 09/11/2026) (cit. on p. 24).

[Aye17] Babatunde Moses Ayeni. "Studies of Braided Non-Abelian Anyons Using Anyonic Tensor Networks". PhD thesis. Macquarie University, Dec. 3, 2017. DOI: 10.25949/19435697.v1. arXiv: 1708.06476 [quant-ph]. (Visited on 06/28/2026) (cit. on pp. 27, 45).

[Bac+26] Flavio Baccari et al. *Quantum TEA: Qtealeaves*. Version 1.12.5. Zenodo, May 2026. DOI: 10.5281/zenodo.20109357 (cit. on p. 53).

[Bai17] James Bailie. *Semisimple Categories*. 2017. URL: https://jameshbailie.github.io/files/papers/2017-06-26-Semisimple-categories.pdf (visited on 07/08/2026) (cit. on p. 34).

[BG17] Matthew Buican and Andrey Gromov. "Anyonic Chains, Topological Defects, and Conformal Field Theory". In: *Communications in Mathematical Physics* 356.3 (Dec. 1, 2017), pp. 1017–1056. DOI: 10.1007/s00220-017-2995-6. arXiv: 1701.02800v2 [hep-th]. (Visited on 06/28/2026) (cit. on pp. 4, 45).

[Bia20] Jacob Biamonte. *Lectures on Quantum Tensor Networks*. Jan. 4, 2020. DOI: 10.48550/arXiv.1912.10049. arXiv: 1912.10049 [quant-ph]. (Visited on 05/11/2026). Pre-published (cit. on p. 2).

[BKP17] Parsa Bonderson, Christina Knapp, and Kaushal Patel. "Anyonic Entanglement and Topological Entanglement Entropy". In: *Annals of Physics* 385 (Oct. 1, 2017), pp. 399–468. DOI: 10.1016/j.aop.2017.07.018. (Visited on 07/05/2026) (cit. on p. 4).

[BL11] John C. Baez and Aaron D. Lauda. "A Prehistory of n -Categorical Physics". In: *Deep Beauty*. Ed. by Hans Halvorson. 1st ed. Cambridge University Press, Apr. 18, 2011, pp. 13–128. ISBN: 978-1-107-00570-9 978-0-511-97697-1. DOI: 10.1017/CBO9780511976971.003. (Visited on 09/28/2026) (cit. on p. 27).

[BMM19] Alex Bullivant, João Faria Martins, and Paul Martin. "Representations of the Loop Braid Group and Aharonov-Bohm like Effects in Discrete (3+1)-Dimensional Higher Gauge Theory". In: *Advances in Theoretical and Mathematical Physics* 23.7 (2019), pp. 1685–1769. DOI: 10.4310/ATMP.2019.v23.n7.a1. arXiv: 1807.09551 [math-ph]. (Visited on 10/15/2026) (cit. on p. 25).

[Bon07] Parsa Hassan Bonderson. "Non-Abelian Anyons and Interferometry". California Institute of Technology, June 4, 2007. DOI: 10.7907/5NDZ-W890. (Visited on 07/26/2026) (cit. on pp. 27, 38, 42).

[BSS08] Parsa Bonderson, Kirill Shtengel, and J. K. Slingerland. "Interferometry of Non-Abelian Anyons". In: *Annals of Physics* 323.11 (Nov. 1, 2008), pp. 2709–2755. DOI: 10.1016/j.aop.2008.01.012. (Visited on 07/26/2026) (cit. on pp. 4, 27, 38).

[Bur16] Simon Burton. *A Short Guide to Anyons and Modular Functors*. Oct. 17, 2016. DOI: 10.48550/arXiv.1610.05384. arXiv: 1610.05384 [quant-ph]. (Visited on 07/15/2026). Pre-published (cit. on pp. 5, 26, 27, 34).

[BV97] Ethan Bernstein and Umesh Vazirani. "Quantum Complexity Theory". In: *SIAM Journal on Computing* 26.5 (1997), pp. 1411–1473. DOI: 10.1137/S0097539796300921. eprint: https://doi.org/10.1137/S0097539796300921 (cit. on p. 46).

[BWC06] John C. Baez, Derek K. Wise, and Alissa S. Crans. *Exotic Statistics for Strings in 4d BF Theory*. May 9, 2006. DOI: 10.48550/arXiv.gr-qc/0603085. arXiv: gr-qc/0603085. (Visited on 10/15/2026). Pre-published (cit. on p. 25).

[CB08] Gregory M. Crosswhite and Dave Bacon. "Finite Automata for Caching in Matrix Product Algorithms". In: *Physical Review A* 78.1 (July 29, 2008), p. 012356. DOI: 10.1103/PhysRevA.78.012356. (Visited on 05/18/2026) (cit. on p. 11).

[CMY25] Weiguang Cao, Yuan Miao, and Masahito Yamazaki. "Global Symmetries of Quantum Lattice Models under Non-Invertible Dualities". In: *SciPost Physics Core* 8.4 (Oct. 14, 2025), p. 070. DOI: 10.21468/SciPostPhysCore.8.4.070. (Visited on 10/09/2026) (cit. on pp. 15, 20).

[CWV26] Bram Vancraeynest-De Cuiper, Weronika Wiesiolek, and Frank Verstraete. "Les Houches Lecture Notes on Tensor Networks". In: *SciPost Physics Lecture Notes* (June 23, 2026), p. 128. DOI: 10.21468/SciPostPhysLectNotes.128. arXiv: 2512.24390 [cond-mat.str-el]. (Visited on 06/27/2026) (cit. on p. 2).

[Del02] P. Deligne. "Categories Tensorielles". In: *Moscow Mathematical Journal* 2.2 (2002), pp. 227–248. DOI: 10.17323/1609-4514-2002-2-2-227-248. (Visited on 09/23/2026) (cit. on p. 2).

[Del90] Pierre Deligne. "Catégories Tannakiennes". In: *The Grothendieck Festschrift, Volume II*. Ed. by Pierre Cartier et al. Vol. 87. Progress in Mathematics. Boston, MA: Birkhäuser, 1990, pp. 111–195 (cit. on p. 27).

[Deu85] David Deutsch. "Quantum Theory, the Church–Turing Principle and the Universal Quantum Computer". In: *Proceedings of the Royal Society of London. A. Mathematical and Physical Sciences* 400.1818 (July 8, 1985), pp. 97–117. DOI: 10.1098/rspa.1985.0070. (Visited on 10/19/2026) (cit. on p. 46).

[Deu89] David Elieser Deutsch. "Quantum Computational Networks". In: *Proceedings of the Royal Society of London. A. Mathematical and Physical Sciences* 425.1868 (Sept. 8, 1989), pp. 73–90. DOI: 10.1098/rspa.1989.0099. (Visited on 10/12/2026) (cit. on p. 46).

[DH25] Lukas Devos and Jutho Haegeman. *TensorKit.Jl: A Julia Package for Large-Scale Tensor Computations, with a Hint of Category Theory*. Version 1. 2025. DOI: 10.48550/arxiv.2508.10076. (Visited on 06/01/2026). Pre-published (cit. on pp. 2, 4, 27, 39, 45, 53).

[DN05] Christopher M. Dawson and Michael A. Nielsen. *The Solovay-Kitaev Algorithm*. Aug. 23, 2005. DOI: 10.48550/arXiv.quant-ph/0505030. arXiv: quant-ph/0505030. (Visited on 07/15/2026). Pre-published (cit. on pp. 5, 46).

[DT24] Clement Delcamp and Apoorv Tiwari. "Higher Categorical Symmetries and Gauging in Two-Dimensional Spin Systems". In: *SciPost Physics* 16.4 (Apr. 23, 2024), p. 110. DOI: 10.21468/SciPostPhys.16.4.110. arXiv: 2301.01259 [hep-th]. (Visited on 06/08/2026) (cit. on p. 5).

[Eck25] Luisa Eck. "Quantum Lattice Models from Fusion Categories". PhD thesis. University of Oxford, 2025. DOI: 10.5287/ora-zozm87o5w. (Visited on 06/28/2026) (cit. on pp. 44, 45).

[ENO17] Pavel Etingof, Dmitri Nikshych, and Viktor Ostrik. *On Fusion Categories*. Apr. 28, 2017. DOI: 10.48550/arXiv.math/0203060. arXiv: math/0203060. (Visited on 09/23/2026). Pre-published (cit. on p. 27).

[Eti+11] Pavel Etingof et al. *Introduction to Representation Theory*. Feb. 1, 2011. DOI: 10.48550/arXiv.0901.0827. arXiv: 0901.0827 [math.RT]. (Visited on 10/23/2026). Pre-published (cit. on pp. 12, 13).

[Eti+15] Pavel Etingof et al. *Tensor Categories*. Vol. 205. Mathematical Surveys and Monographs. Providence, Rhode Island: American Mathematical Society, July 22, 2015. ISBN: 978-1-4704-2024-6 978-1-4704-2349-0. DOI: 10.1090/surv/205. (Visited on 10/24/2026) (cit. on p. 31).

[FB62] Edward Fadell and James Van Buskirk. "The Braid Groups of E2 and S2". In: *Duke Mathematical Journal* 29.2 (June 1962), pp. 243–257. DOI: 10.1215/S0012-7094-62-02925-3. (Visited on 07/28/2026) (cit. on p. 25).

[Fei+07] Adrian Feiguin et al. "Interacting Anyons in Topological Quantum Liquids: The Golden Chain". In: *Physical Review Letters* 98.16 (Apr. 20, 2007), p. 160409. DOI: 10.1103/PhysRevLett.98.160409. (Visited on 06/28/2026) (cit. on pp. 2, 4, 44).

[FH04] William Fulton and Joe Harris. *Representation Theory*. Vol. 129. Graduate Texts in Mathematics. New York, NY: Springer, 2004. ISBN: 978-3-540-00539-1 978-1-4612-0979-9. DOI: 10.1007/978-1-4612-0979-9. (Visited on 10/24/2026) (cit. on p. 13).

[Fis25a] Victor Fischer. "Tensor Network Methods with Non-Abelian Symmetries". MA thesis. Munich: Technische Universität München, 2025 (cit. on pp. 22, 53).

[Fis25b] Victor Fischer. *Victorfischer2000/Su2tn*. Feb. 4, 2025. URL: https://github.com/victorfischer2000/su2tn (visited on 08/17/2026) (cit. on p. 53).

[FKW02] Michael H. Freedman, Alexei Kitaev, and Zhenghan Wang. "Simulation of Topological Field Theories by Quantum Computers". In: *Communications in Mathematical Physics* 227.3 (June 1, 2002), pp. 587–603. DOI: 10.1007/s002200200635. arXiv: quant-ph/0001071. (Visited on 10/14/2026) (cit. on pp. 2, 47).

[FLW00] Michael Freedman, Michael Larsen, and Zhenghan Wang. *A Modular Functor Which Is Universal for Quantum Computation*. Feb. 1, 2000. DOI: 10.48550/arXiv.quant-ph/0001108. arXiv: quant-ph/0001108. (Visited on 08/13/2026). Pre-published (cit. on pp. 47, 48).

[FNW92] M. Fannes, B. Nachtergaele, and R. F. Werner. "Finitely Correlated States on Quantum Spin Chains". In: *Communications in Mathematical Physics* 144.3 (Mar. 1, 1992), pp. 443–490. DOI: 10.1007/BF02099178. (Visited on 07/15/2026) (cit. on p. 9).

[Fre98] Michael H. Freedman. "P/NP, and the Quantum Field Computer". In: *Proceedings of the National Academy of Sciences* 95.1 (Jan. 6, 1998), pp. 98–101. DOI: 10.1073/pnas.95.1.98. (Visited on 07/25/2026) (cit. on pp. 5, 47).

[FW07] Michael H. Freedman and Zhenghan Wang. "Large Fourier Transforms Never Exactly Realized by Braiding Conformal Blocks". In: *Physical Review A* 75.3 (Mar. 15, 2007), p. 032322. DOI: 10.1103/PhysRevA.75.032322. arXiv: cond-mat/0609411. (Visited on 08/03/2026) (cit. on p. 48).

[FWS22] Matthew Fishman, Steven R. White, and E. Miles Stoudenmire. "The ITensor Software Library for Tensor Network Calculations". In: *SciPost Phys. Codebases* (2022), p. 4. DOI: 10.21468/SciPostPhysCodeb.4 (cit. on p. 53).

[Gao+25] Yang Gao et al. “Fermionic Tensor Network Contraction for Arbitrary Geometries”. In: *Physical Review Research* 7.2 (May 27, 2025), p. 023193. DOI: 10.1103/PhysRevResearch.7.023193. (Visited on 05/04/2026) (cit. on p. 53).

[GHM24] Lachezar S. Georgiev, Ludmil Hadjiivanov, and Grigori Matein. *Diagonal Coset Approach to Topological Quantum Computation with Fibonacci Anyons*. Apr. 3, 2024. DOI: 10.48550/arXiv.2404.01779. arXiv: 2404.01779 [quant-ph]. (Visited on 08/13/2026). Pre-published (cit. on p. 48).

[Gil+13] C. Gils et al. “Anyonic Quantum Spin Chains: Spin-1 Generalizations and Topological Stability”. In: *Physical Review B* 87.23 (June 17, 2013), p. 235120. DOI: 10.1103/PhysRevB.87.235120. (Visited on 06/28/2026) (cit. on pp. 4, 45).

[Gra18] Johnnie Gray. “Quimb: A Python Library for Quantum Information and Many-Body Calculations”. In: *Journal of Open Source Software* 3.29 (2018), p. 819. DOI: 10.21105/joss.00819 (cit. on p. 53).

[Gra26a] Johnnie Gray. *Jcmgray/Quimb*. Sept. 17, 2026. URL: https://github.com/jcmgray/quimb (visited on 09/17/2026) (cit. on p. 53).

[Gra26b] Johnnie Gray. *Jcmgray/Symmray*. June 21, 2026. URL: https://github.com/jcmgray/symmray (visited on 06/30/2026) (cit. on p. 53).

[Hal15] Brian C. Hall. *Lie Groups, Lie Algebras, and Representations: An Elementary Introduction*. Vol. 222. Graduate Texts in Mathematics. Cham: Springer International Publishing, 2015. ISBN: 978-3-319-13466-6 978-3-319-13467-3. DOI: 10.1007/978-3-319-13467-3. (Visited on 10/24/2026) (cit. on p. 14).

[Has24] Fabian Hassler. *Show_document*. Lecture Notes. 2024. URL: https://www.quantuminfo.physik.rwth-aachen.de/global/show_document.asp?id=aaaaaaaacmgbnkz&utm_source=chatgpt.com (visited on 07/31/2026) (cit. on p. 51).

[Hau+24] Johannes Hauschild et al. “Tensor Network Python (TeNPy) Version 1”. In: *SciPost Phys. Codebases* (2024), p. 41. DOI: 10.21468/SciPostPhysCodeb.41 (cit. on p. 53).

[Hau26] Markus Hauru. *Mhauru/Abeliantensors*. May 16, 2026. URL: https://github.com/mhauru/abeliantensors (visited on 06/30/2026) (cit. on p. 53).

[HG24] Ludmil Hadjiivanov and Lachezar S. Georgiev. “Braiding Fibonacci Anyons”. In: *Journal of High Energy Physics* 2024.8 (Aug. 12, 2024), p. 84. DOI: 10.1007/JHEP08(2024)084. arXiv: 2404.01778 [hep-th]. (Visited on 08/13/2026) (cit. on p. 48).

[HJ12] Roger A. Horn and Charles R. Johnson. *Matrix Analysis*. 2nd ed. Cambridge: Cambridge University Press, 2012. ISBN: 978-1-139-02041-1. DOI: 10.1017/CBO9781139020411. (Visited on 06/30/2026) (cit. on pp. 15, 51).

[HS25] Simon H. Hille and Attila Szabó. *TeMFpy: A Python Library for Converting Fermionic Mean-Field States into Tensor Networks*. 2025. arXiv: 2510.05227 [cond-mat.str-el] (cit. on p. 53).

[Hub+] Claudius Hubig et al. *The SyTen Toolkit*. URL: https://syten.eu (cit. on p. 53).

[Hub17] Claudius Hubig. “Symmetry-Protected Tensor Networks”. PhD thesis. LMU München, 2017. URL: https://edoc.ub.uni-muenchen.de/21348/ (cit. on p. 53).

[Hub18] Claudius Hubig. “Global Symmetries in Tensor Networks”. 2018. URL: https://nextcloud.tfk.ph.tum.de/etn/wp-content/uploads/2018/11/201811_dresden.pdf (visited on 09/16/2026) (cit. on p. 53).

[IR19] Alberto Ibort and Miguel A. Rodriguez. *An Introduction to Groups, Groupoids and Their Representations*. Boca Raton: CRC Press, Oct. 28, 2019. 362 pp. ISBN: 978-1-315-23294-2. DOI: 10.1201/b22019 (cit. on pp. 12, 13, 34).

[Jav+24] Ali Javadi-Abhari et al. *Quantum Computing with Qiskit*. 2024. DOI: 10.48550/arXiv.2405.08810. arXiv: 2405.08810 [quant-ph] (cit. on p. 48).

[Jon85] Vaughan F. R. Jones. “A Polynomial Invariant for Knots via von Neumann Algebras”. In: *Bulletin of the American Mathematical Society* 12.1 (1985), pp. 103–111. DOI: 10.1090/S0273-0979-1985-15304-2. (Visited on 10/15/2026) (cit. on p. 46).

[JS91] André Joyal and Ross Street. "The Geometry of Tensor Calculus, I". In: *Advances in Mathematics* 88.1 (July 1, 1991), pp. 55–112. DOI: 10.1016/0001-8708(91)90003-P. (Visited on 09/23/2026) (cit. on pp. 27, 31).

[JVW90] F. Jaeger, D. L. Vertigan, and D. J. A. Welsh. "On the Computational Complexity of the Jones and Tutte Polynomials". In: *Mathematical Proceedings of the Cambridge Philosophical Society* 108.1 (July 1990), pp. 35–53. DOI: 10.1017/S0305004100068936. (Visited on 10/15/2026) (cit. on p. 46).

[KBS13] Vadym Kliuchnikov, Alex Bocharov, and Krysta M. Svore. *Asymptotically Optimal Topological Quantum Compiling*. arXiv.org. Oct. 15, 2013. DOI: 10.1103/PhysRevLett.112.140504. (Visited on 08/03/2026) (cit. on p. 48).

[KHC15] Ying-Jer Kao, Yun-Da Hsieh, and Pochung Chen. "Uni10: An Open-Source Library for Tensor Network Algorithms". In: *Journal of Physics: Conference Series* 640 (Sept. 28, 2015), p. 012040. DOI: 10.1088/1742-6596/640/1/012040. (Visited on 09/16/2026) (cit. on p. 53).

[Kir+23] Nico Kirchner et al. "Numerical Simulation of Non-Abelian Anyons". In: *Physical Review B* 107.19 (May 15, 2023), p. 195129. DOI: 10.1103/PhysRevB.107.195129. arXiv: 2206.14730 [cond-mat.str-el]. (Visited on 07/16/2026) (cit. on p. 5).

[Kit03] A. Yu Kitaev. "Fault-Tolerant Quantum Computation by Anyons". In: *Annals of Physics* 303.1 (Jan. 2003), pp. 2–30. DOI: 10.1016/S0003-4916(02)00018-0. arXiv: quant-ph/9707021. (Visited on 07/17/2026) (cit. on pp. 2, 5, 47).

[Kit06] Alexei Kitaev. "Anyons in an Exactly Solved Model and Beyond". In: *Annals of Physics* 321.1 (Jan. 2006), pp. 2–111. DOI: 10.1016/j.aop.2005.10.005. arXiv: cond-mat/0506438v3. (Visited on 07/14/2026) (cit. on pp. 2, 5, 27, 38).

[Kit97] A Yu Kitaev. "Quantum Computations: Algorithms and Error Correction". In: *Russian Mathematical Surveys* 52.6 (Dec. 31, 1997), pp. 1191–1249. DOI: 10.1070/RM1997v052n06ABEH002155. (Visited on 07/26/2026) (cit. on pp. 5, 46, 47).

[KMS07] Samuel A. Kutin, David Petrie Moulton, and Lawren M. Smithline. *Computation at a Distance*. Jan. 26, 2007. DOI: 10.48550/arXiv.quant-ph/0701194. arXiv: quant-ph/0701194. (Visited on 09/10/2026). Pre-published (cit. on p. 48).

[Kon+20] Liang Kong et al. "Algebraic Higher Symmetry and Categorical Symmetry: A Holographic and Entanglement View of Symmetry". In: *Physical Review Research* 2.4 (Oct. 15, 2020), p. 043086. DOI: 10.1103/PhysRevResearch.2.043086. (Visited on 07/05/2026) (cit. on p. 5).

[KTZ20] Liang Kong, Yin Tian, and Zhi-Hao Zhang. "Defects in the 3-Dimensional Toric Code Model Form a Braided Fusion 2-Category". In: *Journal of High Energy Physics* 2020.12 (Dec. 2020), p. 78. DOI: 10.1007/JHEP12(2020)078. arXiv: 2009.06564 [cond-mat.str-el]. (Visited on 10/15/2026) (cit. on p. 25).

[Kup25] Greg Kuperberg. *Breaking the Cubic Barrier in the Solovay-Kitaev Algorithm*. Oct. 8, 2025. DOI: 10.48550/arXiv.2306.13158. arXiv: 2306.13158 [quant-ph]. (Visited on 10/14/2026). Pre-published (cit. on p. 46).

[KZ22] Liang Kong and Zhi-Hao Zhang. *An Invitation to Topological Orders and Category Theory*. May 31, 2022. DOI: 10.48550/arXiv.2205.05565. arXiv: 2205.05565 [cond-mat.str-el]. (Visited on 07/02/2026). Pre-published (cit. on pp. 2, 5, 30, 41).

[KZ23] Liang Kong and Hao Zheng. "Categorical Computation". In: *Frontiers of Physics* 18.2 (Apr. 2023), p. 21302. DOI: 10.1007/s11467-022-1251-5. arXiv: 2102.04814 [quant-ph]. (Visited on 07/01/2026) (cit. on p. 5).

[LDV24] Laurens Lootens, Clement Delcamp, and Frank Verstraete. "Dualities in One-Dimensional Quantum Lattice Models: Topological Sectors". In: *PRX Quantum* 5.1 (Mar. 6, 2024), p. 010338. DOI: 10.1103/PRXQuantum.5.010338. arXiv: 2211.03777 [quant-ph]. (Visited on 06/08/2026) (cit. on p. 44).

[Lee26] Seung-Sup Lee. *Ssblee/Telum.Jl*. June 30, 2026. URL: https://github.com/ssblee/Telum.jl (visited on 07/15/2026) (cit. on p. 53).

[Liu+22] Yu-Jie Liu et al. “Methods for Simulating String-Net States and Anyons on a Digital Quantum Computer”. In: *PRX Quantum* 3.4 (Nov. 7, 2022), p. 040315. DOI: 10.1103/PRXQuantum.3.040315. (Visited on 07/16/2026) (cit. on p. 5).

[LM77] J. M. Leinaas and J. Myrheim. “On the Theory of Identical Particles”. In: *Il Nuovo Cimento B (1971-1996)* 37.1 (Jan. 1, 1977), pp. 1–23. DOI: 10.1007/BF02727953. (Visited on 07/19/2026) (cit. on pp. 4, 23).

[Mac78] Saunders Mac Lane. *Categories for the Working Mathematician*. Vol. 5. Graduate Texts in Mathematics. New York, NY: Springer, 1978. ISBN: 978-1-4419-3123-8 978-1-4757-4721-8. DOI: 10.1007/978-1-4757-4721-8. (Visited on 12/19/2024) (cit. on pp. 27, 30).

[McM19] Nathan McMahon. *Qnla/SymLibrary*. Mar. 4, 2019. URL: https://github.com/qnla/SymLibrary (visited on 07/05/2026) (cit. on p. 53).

[Men26] Christian B. Mendl. *Qc-Tum/Chemtensor*. qc-tum, May 28, 2026. URL: https://github.com/qc-tum/chemtensor (visited on 06/30/2026) (cit. on p. 53).

[MG02] I. P. McCulloch and M. Gulácsi. “The Non-Abelian Density Matrix Renormalization Group Algorithm”. In: *Europhysics Letters* 57.6 (Mar. 1, 2002), p. 852. DOI: 10.1209/epl/i2002-00393-0. (Visited on 07/16/2026) (cit. on pp. 4, 15, 53).

[MMS17] A. Mironov, A. Morozov, and A. Sleptsov. “On $6j$-Symbols for Symmetric Representations of $U_q(\mathfrak{su}_N)$”. In: *JETP Letters* 106.10 (Nov. 2017), pp. 630–636. DOI: 10.1134/S0021364017220040. arXiv: 1709.02290 [hep-th]. (Visited on 07/15/2026) (cit. on p. 15).

[Moc03] Carlos Mochon. “Anyons from Nonsolvable Finite Groups Are Sufficient for Universal Quantum Computation”. In: *Physical Review A* 67.2 (2003). DOI: 10.1103/PhysRevA.67.022315 (cit. on p. 5).

[Moc04] Carlos Mochon. “Anyon Computers with Smaller Groups”. In: *Physical Review A* 69.3 (Mar. 11, 2004), p. 032306. DOI: 10.1103/PhysRevA.69.032306. arXiv: quant-ph/0306063. (Visited on 07/01/2026) (cit. on p. 5).

[Mos25] Milo Moses. “Fusion Categories”. 2025. URL: https://milomoses.info/fusion-talk.pdf (visited on 06/16/2026) (cit. on p. 38).

[MS08] Igor L. Markov and Yaoyun Shi. “Simulating Quantum Computation by Contracting Tensor Networks”. In: *SIAM Journal on Computing* 38.3 (Jan. 2008), pp. 963–981. DOI: 10.1137/050644756. arXiv: quant-ph/0511069. (Visited on 10/12/2026) (cit. on pp. 9, 46).

[MSB20] Nathan A. McMahon, Sukhbinder Singh, and Gavin K. Brennen. “A Holographic Duality from Lifted Tensor Networks”. In: *npj Quantum Information* 6.1 (Apr. 24, 2020), p. 36. DOI: 10.1038/s41534-020-0255-7. (Visited on 07/05/2026) (cit. on p. 53).

[MSS24] David Jaz Myers, Hisham Sati, and Urs Schreiber. “Topological Quantum Gates in Homotopy Type Theory”. In: *Communications in Mathematical Physics* 405.7 (July 2024), p. 172. DOI: 10.1007/s00220-024-05020-8. arXiv: 2303.02382 [quant-ph]. (Visited on 07/17/2026) (cit. on p. 5).

[Nay+08] Chetan Nayak et al. “Non-Abelian Anyons and Topological Quantum Computation”. In: *Reviews of Modern Physics* 80.3 (Sept. 12, 2008), pp. 1083–1159. DOI: 10.1103/RevModPhys.80.1083. arXiv: 0707.1889 [cond-mat.str-el]. (Visited on 07/17/2026) (cit. on p. 5).

[Neo16] Klil H. Neori. *Identical Particles in Quantum Mechanics: Operational and Topological Considerations*. Mar. 20, 2016. DOI: 10.48550/arXiv.1603.06282. arXiv: 1603.06282 [quant-ph]. (Visited on 10/23/2026). Pre-published (cit. on p. 24).

[nLa26] nLab authors. *Spherical Braid Group*. July 2026. URL: https://ncatlab.org/nlab/show/spherical+braid+group (cit. on p. 25).

[OR95] Stellan Ostlund and Stefan Rommer. “Thermodynamic Limit of the Density Matrix Renormalization for the Spin-1 Heisenberg Chain”. In: *Physical Review Letters* 75.19 (Nov. 6, 1995), pp. 3537–3540. DOI: 10.1103/PhysRevLett.75.3537. arXiv: cond-mat/9503107. (Visited on 07/15/2026) (cit. on p. 9).

[Orú14] Román Orús. “Advances on Tensor Network Theory: Symmetries, Fermions, Entanglement, and Holography”. In: *The European Physical Journal B* 87.11 (Nov. 26, 2014), p. 280. DOI: 10.1140/epjb/e2014-50502-9. (Visited on 05/06/2026) (cit. on p. 2).

[Pen71] Roger Penrose. “Applications of Negative Dimensional Tensors”. In: (1971). Ed. by D.J.A. Welsh, pp. 221–224. URL: https://ncatlab.org/nlab/files/PenroseNegativeDimensionalTensors.pdf (visited on 06/30/2026) (cit. on pp. 9, 27).

[Pfe+10] Robert N. C. Pfeifer et al. “Simulation of Anyons with Tensor Network Algorithms”. In: *Physical Review B* 82.11 (Sept. 29, 2010), p. 115126. DOI: 10.1103/PhysRevB.82.115126. arXiv: 1006.3532 [cond-mat.str-el]. (Visited on 06/16/2026) (cit. on pp. 2, 4, 27, 44, 45).

[Pfe12] Robert N. C. Pfeifer. *Simulation of Anyons Using Symmetric Tensor Network Algorithms*. Feb. 10, 2012. DOI: 10.48550/arXiv.1202.1522. arXiv: 1202.1522 [cond-mat]. (Visited on 05/11/2026). Pre-published (cit. on pp. 2, 4, 27, 44, 45).

[Pfe14] Robert N. C. Pfeifer. “Measures of Entanglement in Non-Abelian Anyonic Systems”. In: *Physical Review B* 89.3 (Jan. 6, 2014), p. 035105. DOI: 10.1103/PhysRevB.89.035105. (Visited on 07/05/2026) (cit. on pp. 4, 44).

[PS15] Robert N. C. Pfeifer and Sukhwinder Singh. “Finite Density Matrix Renormalisation Group Algorithm for Anyonic Systems”. In: *Physical Review B* 92.11 (Sept. 16, 2015), p. 115135. DOI: 10.1103/PhysRevB.92.115135. arXiv: 1505.00100 [cond-mat.str-el]. (Visited on 10/12/2026) (cit. on p. 44).

[Ram+25] Marek M. Rams et al. “YASTN: Yet Another Symmetric Tensor Networks; A Python Library for Abelian Symmetric Tensor Network Calculations”. In: *SciPost Phys. Codebases* (2025), p. 52. DOI: 10.21468/SciPostPhysCodeb.52 (cit. on p. 53).

[Rie] Emily Riehl. “Category Theory in Context”. In: () (cit. on p. 27).

[Rie25] Jeremias Rieser. *Jkr11/Exactsynth-Hs*. Nov. 20, 2025. URL: https://github.com/jkr11/exactsynth-hs (visited on 08/03/2026) (cit. on p. 48).

[Rob+19] Chase Roberts et al. *TensorNetwork: A Library for Physics and Machine Learning*. 2019. arXiv: 1905.01330 [physics.comp-ph] (cit. on p. 53).

[RSW09] Eric Rowell, Richard Stong, and Zhenghan Wang. *On Classification of Modular Tensor Categories*. Nov. 9, 2009. DOI: 10.48550/arXiv.0712.1377. arXiv: 0712.1377 [math.QA]. (Visited on 09/23/2026). Pre-published (cit. on p. 27).

[RT91] N. Reshetikhin and V. G. Turaev. “Invariants of 3-Manifolds via Link Polynomials and Quantum Groups”. In: *Inventiones mathematicae* 103.1 (Dec. 1, 1991), pp. 547–597. DOI: 10.1007/BF01239527. (Visited on 07/15/2026) (cit. on pp. 5, 27).

[RW17] Eric C. Rowell and Zhenghan Wang. *Mathematics of Topological Quantum Computing*. Dec. 5, 2017. DOI: 10.48550/arXiv.1705.06206. arXiv: 1705.06206 [math.QA]. (Visited on 09/23/2026). Pre-published (cit. on p. 27).

[Sch+20] Philipp Schmoll et al. “A Programming Guide for Tensor Networks with Global $SU(2)$ Symmetry”. In: *Annals of Physics* 419 (Aug. 1, 2020), p. 168232. DOI: 10.1016/j.aop.2020.168232. (Visited on 05/07/2026) (cit. on pp. 2, 4, 27).

[Sch05] Issai Schur. “Neue Begruendungen der Theorie der Gruppencharaktere”. In: *Sitzungsberichte der Königlich Preussischen Akademie der Wissenschaften zu Berlin* (Januar bis Juni 1905), pp. 406–433. URL: http://archive.org/details/bub_gb_KwUoAAAAYAAJ (visited on 07/15/2026) (cit. on pp. 13, 34).

[Sch20] Philipp Schmoll. “Tensor Network Simulations with Global SU(2) Symmetry”. Mainz: Johannes Gutenberg University, July 2020. 150 pp. URL: https://openscience.ub.uni-mainz.de/server/api/core/bitstreams/d9d9ca4e-46c8-4a8e-a831-7611475d2504/content (visited on 05/08/2026) (cit. on p. 4).

[Sel11] P. Selinger. "A Survey of Graphical Languages for Monoidal Categories". In: *New Structures for Physics*. Ed. by Bob Coecke. Berlin, Heidelberg: Springer, 2011, pp. 289–355. ISBN: 978-3-642-12821-9. DOI: 10.1007/978-3-642-12821-9_4. arXiv: 0908.3347v1 [math.CT] (cit. on pp. 5, 27, 30).

[Sel16] Giuseppe Sellaroli. *Non-Compact Groups, Tensor Operators and Applications to Quantum Gravity*. Sept. 25, 2016. DOI: 10.48550/arXiv.1609.07795. arXiv: 1609.07795 [math-ph]. (Visited on 10/24/2026). Pre-published (cit. on p. 21).

[Ser77] Jean-Pierre Serre. *Linear Representations of Finite Groups*. Vol. 42. Graduate Texts in Mathematics. New York, NY: Springer, 1977. ISBN: 978-1-4684-9460-0 978-1-4684-9458-7. DOI: 10.1007/978-1-4684-9458-7. (Visited on 10/24/2026) (cit. on pp. 12, 13).

[Sim+06] S. H. Simon et al. "Topological Quantum Computing with Only One Mobile Quasiparticle". In: *Physical Review Letters* 96.7 (Feb. 23, 2006), p. 070503. DOI: 10.1103/PhysRevLett.96.070503. arXiv: quant-ph/0509175. (Visited on 07/19/2026) (cit. on pp. 5, 48).

[Sim23] Steven H. Simon. *Topological Quantum*. Oxford University Press, Sept. 29, 2023. ISBN: 978-0-19-888672-3. DOI: 10.1093/oso/9780198886723.001.0001. (Visited on 06/24/2026) (cit. on pp. 5, 25, 27).

[Sin+14] Sukhwinder Singh et al. "Matrix Product States for Anyonic Systems and Efficient Simulation of Dynamics". In: *Physical Review B* 89.7 (Feb. 13, 2014), p. 075112. DOI: 10.1103/PhysRevB.89.075112. (Visited on 07/08/2026) (cit. on p. 44).

[SM98] G. Sierra and M. A. Martin-Delgado. *The Density Matrix Renormalization Group, Quantum Groups and Conformal Field Theory*. Version 3. Nov. 30, 1998. DOI: 10.48550/arXiv.cond-mat/9811170. arXiv: cond-mat/9811170. (Visited on 07/16/2026). Pre-published (cit. on p. 4).

[SMB18] Sukhwinder Singh, Nathan A. McMahon, and Gavin K. Brennen. "Holographic Spin Networks from Tensor Network States". In: *Physical Review D* 97.2 (Jan. 26, 2018), p. 026013. DOI: 10.1103/PhysRevD.97.026013. arXiv: 1702.00392 [cond-mat.str-el]. (Visited on 05/11/2026) (cit. on p. 9).

[SO20] Philipp Schmoll and Roman Orus. "Benchmarking Global $SU(2)$ Symmetry in 2d Tensor Network Algorithms". In: *Physical Review B* 102.24 (Dec. 1, 2020), p. 241101. DOI: 10.1103/PhysRevB.102.241101. arXiv: 2005.02748 [cond-mat]. (Visited on 05/07/2026) (cit. on pp. 4, 22).

[SPV10] Sukhwinder Singh, Robert N. C. Pfeifer, and Guifre Vidal. "Tensor Network Decompositions in the Presence of a Global Symmetry". In: *Physical Review A* 82.5 (Nov. 8, 2010), p. 050301. DOI: 10.1103/PhysRevA.82.050301. arXiv: 0907.2994 [cond-mat.str-el]. (Visited on 05/11/2026) (cit. on pp. 2, 4, 20, 21, 53).

[SPV11] Sukhwinder Singh, Robert N. C. Pfeifer, and Guifre Vidal. "Tensor Network States and Algorithms in the Presence of a Global U(1) Symmetry". In: *Physical Review B* 83.11 (Mar. 15, 2011), p. 115125. DOI: 10.1103/PhysRevB.83.115125. arXiv: 1008.4774v1 [cond-mat.str-el]. (Visited on 05/06/2026) (cit. on p. 4).

[SS22] Hisham Sati and Urs Schreiber. "Differential Cohomotopy Implies Intersecting Brane Observables via Configuration Spaces and Chord Diagrams". In: *Advances in Theoretical and Mathematical Physics* 26.4 (2022), pp. 957–1051. DOI: 10.4310/ATMP.2022.v26.n4.a4. arXiv: 1912.10425 [hep-th]. (Visited on 09/08/2026) (cit. on p. 23).

[SS23] Hisham Sati and Urs Schreiber. "Anyonic Topological Order in Twisted Equivariant Differential (TED) K-Theory". In: *Reviews in Mathematical Physics* 35.03 (Apr. 2023), p. 2350001. DOI: 10.1142/S0129055X23500010. arXiv: 2206.13563 [hep-th]. (Visited on 09/09/2026) (cit. on p. 26).

[SS26] Hisham Sati and Urs Schreiber. "Entanglement of Sections: The Pushout of Entangled and Parameterized Quantum Information". In: *Quantum Studies: Mathematics and Foundations* 13.2 (June 2026), p. 23. DOI: 10.1007/s40509-026-00397-8. arXiv: 2309.07245 [quant-ph]. (Visited on 07/19/2026) (cit. on p. 5).

[SV12] Sukhwinder Singh and Guifre Vidal. “Tensor Network States and Algorithms in the Presence of a Global SU(2) Symmetry”. In: *Physical Review B* 86.19 (Nov. 9, 2012), p. 195114. DOI: 10.1103/PhysRevB.86.195114. arXiv: 1208.3919v1 [cond-mat.str-el]. (Visited on 05/11/2026) (cit. on pp. 2, 4, 15, 21, 39).

[SWD11] Mehdi Saeedi, Robert Wille, and Rolf Drechsler. “Synthesis of Quantum Circuits for Linear Nearest Neighbor Architectures”. In: *Quantum Information Processing* 10.3 (June 2011), pp. 355–377. DOI: 10.1007/s11128-010-0201-2. arXiv: 1110.6412 [quant-ph]. (Visited on 09/10/2026) (cit. on p. 48).

[Tre+08] Simon Trebst et al. “A Short Introduction to Fibonacci Anyon Models”. In: *Progress of Theoretical Physics Supplement* 176 (June 1, 2008), pp. 384–407. DOI: 10.1143/PTPS.176.384. arXiv: 0902.3275v1 [cond-mat.stat-mech]. (Visited on 06/28/2026) (cit. on pp. 4, 45).

[Unf24] Jakob Unfried. “Advancing Tensor Network Simulations -”. PhD thesis. Technische Universität München, 2024, p. 208. URL: https://mediatum.ub.tum.de/1755273 (cit. on pp. 2, 4, 27, 45, 53).

[Vak25] Ravi Vakil. *The Rising Sea: Foundations of Algebraic Geomtery*. Princeton University Press, Oct. 21, 2025. 688 pp. ISBN: 978-0-691-26867-5. URL: https://press.princeton.edu/books/paperback/9780691268675/the-rising-sea (visited on 06/24/2026) (cit. on p. 27).

[Val] Sachin J Valera. “TOPOLOGICAL QUANTUM TELEPORTATION AND SUPERDENSE CODING – WITHOUT BRAIDING”. In: () (cit. on pp. 26, 27).

[Val21a] Sachin J. Valera. “Fusion Structure from Exchange Symmetry in (2+1)-Dimensions”. In: *Annals of Physics* 429 (June 2021), p. 168471. DOI: 10.1016/j.aop.2021.168471. arXiv: 2004.06282 [quant-ph]. (Visited on 09/09/2026) (cit. on pp. 26, 27).

[Val21b] Sachin Jayesh Valera. “Topological Quantum and Skein-Theoretic Aspects of Braided Fusion Categories”. PhD thesis. Norway: University of Bergen, 2021. ISBN: 978-82-308-5158-6 978-82-308-6990-1. URL: 11250/2770245 (visited on 06/08/2026) (cit. on p. 36).

[VC04] F. Verstraete and J. I. Cirac. *Renormalization Algorithms for Quantum-Many Body Systems in Two and Higher Dimensions*. July 2, 2004. DOI: 10.48550/arXiv.cond-mat/0407066. arXiv: cond-mat/0407066. (Visited on 07/15/2026). Pre-published (cit. on p. 9).

[Vid08] G. Vidal. “A Class of Quantum Many-Body States That Can Be Efficiently Simulated”. In: *Physical Review Letters* 101.11 (Sept. 12, 2008), p. 110501. DOI: 10.1103/PhysRevLett.101.110501. arXiv: quant-ph/0610099. (Visited on 07/15/2026) (cit. on p. 9).

[VVW18] Gert Vercleyen, Frank Verstraete, and Universiteit Gent. Faculteit Wetenschappen. “The Mathematical Structure of Tensor Networks”. MA thesis. Master of Science in de wiskunde, 2018. URL: https://lib.ugent.be/fulltxt/RUG01/002/479/664/RUG01-002479664_2018_0001_AC.pdf (cit. on p. 5).

[Wei12a] Andreas Weichselbaum. “Non-Abelian Symmetries in Tensor Networks: A Quantum Symmetry Space Approach”. In: *Annals of Physics* 327.12 (Dec. 1, 2012), pp. 2972–3047. DOI: 10.1016/j.aop.2012.07.009. (Visited on 05/07/2026) (cit. on pp. 4, 53).

[Wei12b] Andreas Weichselbaum. “Tensor Networks and the Numerical Renormalization Group”. In: *Physical Review B* 86.24 (Dec. 20, 2012), p. 245124. DOI: 10.1103/PhysRevB.86.245124. (Visited on 06/30/2026) (cit. on p. 2).

[Wei20] Andreas Weichselbaum. “X-Symbols for Non-Abelian Symmetries in Tensor Networks”. In: *Physical Review Research* 2.2 (June 23, 2020), p. 023385. DOI: 10.1103/PhysRevResearch.2.023385. (Visited on 06/30/2026) (cit. on p. 4).

[Wei24] Andreas Weichselbaum. “QSpace - An Open-Source Tensor Library for Abelian and Non-Abelian Symmetries”. In: *SciPost Physics Codebases* (Nov. 14, 2024), p. 40. DOI: 10.21468/SciPostPhysCodeb.40. arXiv: 2405.06632v1 [cond-mat.str-el]. (Visited on 05/08/2026) (cit. on p. 53).

[Wen+25] Maurice Wenig et al. *The Syntax and Semantics of Einsum*. Oct. 20, 2025. DOI: 10.48550/arXiv.2509.20020. arXiv: 2509.20020 [cs.PL]. (Visited on 10/10/2026). Pre-published (cit. on p. 9).

[Whi92] Steven R. White. “Density Matrix Formulation for Quantum Renormalization Groups”. In: *Physical Review Letters* 69.19 (Nov. 9, 1992), pp. 2863–2866. DOI: 10.1103/PhysRevLett.69.2863. (Visited on 07/15/2026) (cit. on p. 9).

[Whi93] Steven R. White. “Density-Matrix Algorithms for Quantum Renormalization Groups”. In: *Physical Review B* 48.14 (Oct. 1, 1993), pp. 10345–10356. DOI: 10.1103/PhysRevB.48.10345. (Visited on 07/15/2026) (cit. on p. 9).

[Wil18] Jenny Wilson. “The Geometry and Topology of Braid Groups”. In: RTG Geometry–Topology Summer School. University of Chicago, 2018 (cit. on p. 24).

[Wil90] Frank Wilczek. *Fractional Statistics and Anyon Superconductivity*. WORLD SCIENTIFIC, 1990. DOI: 10.1142/0961. eprint: https://www.worldscientific.com/doi/pdf/10.1142/0961 (cit. on pp. 4, 23).

[Wit89] Edward Witten. “Quantum Field Theory and the Jones Polynomial”. In: *Communications in Mathematical Physics* 121.3 (Sept. 1, 1989), pp. 351–399. DOI: 10.1007/BF01217730. (Visited on 10/15/2026) (cit. on p. 46).

[Wou+14] Sebastian Wouters et al. “CheMPS2: A Free Open-Source Spin-Adapted Implementation of the Density Matrix Renormalization Group for Ab Initio Quantum Chemistry”. In: *Computer Physics Communications* 185.6 (June 2014), pp. 1501–1514. DOI: 10.1016/j.cpc.2014.01.019. arXiv: 1312.2415 [cond-mat.str-el]. (Visited on 09/16/2026) (cit. on p. 53).

[Wu+25] Kai-Hsin Wu et al. “The Cytnx Library for Tensor Networks”. In: *SciPost Physics Codebases* (Mar. 13, 2025), p. 53. DOI: 10.21468/SciPostPhysCodeb.53. arXiv: 2401.01921 [cs.MS]. (Visited on 09/16/2026) (cit. on p. 53).

[Wu84] Yong-Shi Wu. “General Theory for Quantum Statistics in Two Dimensions”. In: *Physical Review Letters* 52.24 (June 11, 1984), pp. 2103–2106. DOI: 10.1103/PhysRevLett.52.2103. (Visited on 09/08/2026) (cit. on p. 23).

[Yos26] Atis Yosprakob. *Ayosprakob/Grassmanntn*. June 7, 2026. URL: https://github.com/ayosprakob/grassmanntn (visited on 06/30/2026) (cit. on p. 53).

[Zha+23] Huanchen Zhai et al. “Block2: A Comprehensive Open Source Framework to Develop and Apply State-of-the-Art DMRG Algorithms in Electronic Structure and Beyond”. In: *The Journal of Chemical Physics* 159.23 (Dec. 18, 2023), p. 234801. DOI: 10.1063/5.0180424. (Visited on 09/17/2026) (cit. on p. 53).